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Topology Optimization of Flow Configurations Operating in the Turbulent Regime

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Preface

This thesis marks the culmination of my Master of Science in Mechanical Engineering at KU Leuven. The journey of exploring computational fluid dynamics and mathematical optimization has been both intensely challenging and deeply rewarding. Bridging the gap between theoretical mathematics and practical engineering applications required countless hours of coding, debugging, and analyzing flow fields, but seeing the optimizer successfully carve out organic, aerodynamic fluid channels made the entire process worthwhile.

I would like to express my sincere gratitude to my supervisors, Prof. Niels Horsten and Prof. Maarten Blommaert, for providing me with the opportunity to work on this fascinating topic. Their academic guidance, critical insights, and continuous support were instrumental in shaping the direction and rigor of this research. A very special thanks goes to my assistant-supervisor, Esteban Foronda Obando, for his day-to-day patience, technical expertise, and willingness to help me navigate the complex numerical hurdles within FEniCS.

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Tiebert Lefebure

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Abstract

As compact electronic, automotive, and energy systems become more power-dense, thermal-management hardware increasingly depends on low-loss internal-flow paths. This thesis addresses the hydraulic part of that problem: turbulent flow-only topology optimization for manifolds, bends, U-bends, and diffusers before conjugate heat transfer is added. A density-based FEniCS framework is developed using Brinkman penalization, wall-resolved Spalart-Allmaras (SA) turbulence modeling, a relaxed Eikonal wall-distance calculation, and a frozen-turbulence adjoint in which the eddy-viscosity and wall-distance fields are updated in the forward solve but fixed during sensitivity assembly.

The work first defines the modeling limits of SA for curved and separated internal flows, then verifies the custom FEniCS-SA solver against Ansys Fluent for a periodic turbulent channel and a two-dimensional U-bend. These cases show qualitative and profile-level agreement sufficient for use as the forward solver in the optimization framework, while retaining the known limitations of a two-dimensional, one-equation eddy-viscosity model.

The framework is then evaluated on turbulent pipe-bend, U-bend, and diffuser benchmarks. For the completed pipe-bend verification, frozen sensitivities agree with central finite differences to a maximum coordinate discrepancy of 5.92×10^{-5}, and the directional Taylor check shows approximately second-order remainder convergence. Ansys re-analysis of the extracted pipe-bend design shows moderate closure sensitivity: replacing Ansys-SA by SST $k - \omega$ increases pressure drop by about 4.5% and viscous dissipation by about 8.5%. Under identical SST $k - \omega$ operating conditions, the turbulent design reduces pressure drop by approximately 72% and viscous dissipation by approximately 47% relative to the laminar-optimized reference. The U-bend is retained as the companion 180° return-flow benchmark and is interpreted using the same validation standard. The pressure-outlet diffuser gives a more cautionary result: the turbulent SA design improves on the laminar reference under Ansys-SA, but the ranking reverses under SST $k - \omega$, showing that separated diffuser performance is strongly closure-sensitive.

The main conclusion is that frozen-SA topology optimization can produce credible low-loss turbulent internal-flow designs when its assumptions are explicit and final geometries are independently re-analyzed. The main remaining limitations are the frozen-adjoint approximation, two-dimensional suppression of secondary-flow physics, and sensitivity of final performance to geometry extraction.

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List of Abbreviations and Symbols

Abbreviations

AMG	Algebraic Multigrid
BiCGSTAB	Bi-Conjugate Gradient Stabilized method
CAD	Computer-Aided Design
CEV	Constant Eddy Viscosity
CFD	Computational Fluid Dynamics
CG1	Continuous Galerkin (Linear Basis Functions)
CG2	Continuous Galerkin (Quadratic Basis Functions)
CHT	Conjugate Heat Transfer
DG0	Discontinuous Galerkin (Piecewise Constant Elements)
DNS	Direct Numerical Simulation
DOF	Degree of Freedom
FEM	Finite Element Method
FD	Finite Difference
FVM	Finite Volume Method
GMRES	Generalized Minimal Residual method
IPCS	Incremental Pressure Correction Scheme
ILU	Incomplete LU factorization
MMA	Method of Moving Asymptotes
MUMPS	Multifrontal Massively Parallel Sparse direct Solver
NS-SA	Navier-Stokes-Spalart-Allmaras coupling
PDE	Partial Differential Equation
RAMP	Rational Approximation of Material Properties
RANS	Reynolds-Averaged Navier-Stokes
SA	Spalart-Allmaras
SIMP	Solid Isotropic Material with Penalization
SNES	Scalable Nonlinear Equations Solver
SUPG	Streamline-Upwind/Petrov–Galerkin stabilization
SST $k-\omega$	Shear Stress Transport $k-\omega$ model
TMR	NASA Turbulence Modeling Resource
TO	Topology Optimization
UFL	Unified Form Language
VMFL	Ansys Fluent verification benchmark case
XFEM	Extended Finite Element Method

Symbols

Fluid Dynamics & Turbulence

D, D_h	Pipe diameter and hydraulic diameter [m]
De	Dean number [-]
H, L	Channel height and characteristic length [m]
k	Turbulent kinetic energy [m^2/s^2]
$p, \bar{p}, p'$	Instantaneous, mean, and fluctuating static pressure [Pa]
ΔP	Macroscopic total pressure drop [Pa]
R_c	Radius of curvature [m]
Re	Bulk Reynolds number [-]
$S, \tilde{S}$	Scalar vorticity magnitude and modified SA vorticity magnitude [1/s]
t	Time / Pseudo-time [s]
τ_w	Wall shear stress [Pa]
$\mathbf{u}, \bar{u}_i$	Velocity vector / Mean velocity component [m/s]
u_i'	Fluctuating velocity component [m/s]
$U_{\text{ref}}, U_{\text{bulk}}, U_{\text{max}}$	Reference, bulk, and maximum velocity [m/s]
u^+	Non-dimensional mean velocity [-]
u_τ	Friction velocity [m/s]
y, d_w	Distance to the nearest wall [m]
y^+	Non-dimensional wall distance [-]
C_f	Skin-friction coefficient [-]
c_{b1}, c_{b2}, c_{v1}	Spalart-Allmaras production and viscosity-ratio constants [-]
c_{w1}, c_{w2}, c_{w3}	Spalart-Allmaras wall-destruction constants [-]
f_{v1}, f_{v2}, f_w	Spalart-Allmaras damping and wall-destruction functions [-]
$r_{\text{SA}}, g_{\text{SA}}$	Spalart-Allmaras wall-destruction auxiliary variables [-]
δ_{ij}	Kronecker delta [-]
μ, μ_t	Dynamic and turbulent eddy viscosity [Pa·s]
ν, ν_t	Kinematic and turbulent eddy kinematic viscosity [m^2/s]
$\tilde{\nu}$	Spalart-Allmaras working variable [m^2/s]
ρ	Fluid density [kg/m^3]
ω	Scalar vorticity in two-dimensional flow [1/s]
Ω_{ij}	Vorticity tensor [1/s]
χ	Spalart-Allmaras viscosity ratio, $\tilde{\nu}/\nu$ [-]
κ	von Kármán constant [-]
σ	Spalart-Allmaras turbulent Prandtl number [-]

Numerical Implementation (FEniCS)

G, G_0	Relaxed Eikonal field variable and boundary value [1/m]
h, h_{avg}	Finite element cell diameter and average cell size [m]
p_k	Kinematic pressure [$\mathrm{m}^2/\mathrm{s}^2$]
s	Arclength coordinate along a profile line [m]
e_i	Pointwise profile error or design-space basis vector, depending on context [-]
E_∞, E_{L_2}	Maximum and relative L_2 profile discrepancies [-]
C_{SUPG}	User-selected SUPG stabilization factor [-]
Δh	Finite-difference design perturbation [-]
$\Delta t_{\tilde{\nu}}$	Pseudo-time step used for optional SA transport damping [s]
$S_i^{\mathrm{FD}}, S_i^{\mathrm{fr}}$	Finite-difference and frozen-adjoint sensitivity of sampled design variable i
$\varepsilon_i^{\mathrm{fr}}$	Relative frozen-adjoint sensitivity discrepancy [-]
$\|\| \cdot \|\|_{L_2}$	L_2 (Euclidean) norm [-]
$[\![\cdot]\!]$	Jump operator across an internal element facet [-]
$\Gamma_{\mathrm{in}}, \Gamma_{\mathrm{out}}, \Gamma_D$	Inlet, outlet, and profile-extraction boundaries
$\mathbf{n}$	Outward-pointing normal vector [-]
$\mathcal{R}_{\tilde{\nu}}$	Strong residual of the Picard-linearized SA transport equation [-]
T_h, V_h, Q_h	Discrete turbulence, velocity, and pressure finite-element spaces
$\tau_{\tilde{\nu}}$	SUPG stabilization parameter for SA working-variable transport [s]
v, ξ	Test functions for pressure and turbulence equations [-]
$\alpha_u, \alpha_{\tilde{\nu}}$	Relaxation factors for velocity and turbulence [-]
Δt	Pseudo-time step [s]
ϵ_{tol}	Convergence tolerance [-]
ϕ	General scalar field variable [-]
σ_w	Eikonal relaxation constant [-]

Topology Optimization

$\alpha(\gamma)$	Continuous Brinkman penalization term / inverse permeability
$\alpha_{solid}, \alpha_{fluid}$	Maximum solid and minimum fluid Brinkman penalties
α_{dg}	Artificial diffusion scaling factor for DG0 filter [-]
β	Heaviside projection steepness parameter [-]
γ, γ_{raw}	Continuous and raw design variable (1 for fluid, 0 for solid) [-]
$\tilde{\gamma}, \bar{\gamma}$	Filtered and Heaviside-projected design variables [-]
η	Heaviside projection threshold [-]
Π_β	Heaviside projection operator [-]
$\mathcal{H}$	Helmholtz filter operator [-]
M	Active design mask [-]
J	Generic objective function used in the optimization problem [problem dependent]
J_D	Dissipation objective, evaluated from viscous dissipation [W, or W/m for two-dimensional unit-depth cases]
J_p	Average inlet-pressure objective [Pa]
Φ_D	Sharp-geometry viscous dissipation metric [W, or W/m for two-dimensional unit-depth cases]
q	RAMP penalization tuning parameter [-]
r	Continuous Helmholtz filter radius [m]
$r_V^{(k)}$	Volume residual at optimization iteration k [-]
$v_f^{(k)}$	Realized fluid volume fraction at optimization iteration k [-]
$\mathbf{s}, \mathbf{s}^*$	Monolithic forward state and frozen-flow adjoint state vectors
V_f, V_{domain}	Physical fluid volume and total design domain volume [m^3]
V_{mask}	Volume-mask measure [m^3]
V_{max}	Maximum allowable fluid volume fraction constraint [-]
$\Omega, \partial\Omega, \Gamma_{int}$	Computational domain, domain boundary, and internal element facets
Ω_D, Ω_V	Objective-integration and volume-integration regions

Chapter 1

Introduction

1.1 Background and Motivation

Energy systems increasingly rely on compact flow devices that must move heat, mass, or working fluids through limited space with minimal loss. Examples include liquid-cooled electronics, battery and power-electronics thermal management, fuel-cell and electrolyzer flow fields, compact heat exchangers, and coolant manifolds in automotive and aerospace applications. In these systems, the thermal design is often constrained before any heat-transfer equation is considered: the fluid must first be distributed, collected, redirected, or expanded without creating an excessive pressure drop. A cooling plate or heat exchanger with excellent local heat-transfer potential can still be impractical if the manifold requires too much pumping power or if the flow distribution is poor.

This thesis is positioned within that broader energy and thermal-management context, but its actual framework is deliberately narrower. It does not solve the energy equation, does not perform conjugate heat-transfer optimization, and does not optimize temperature directly. Instead, it studies the flow-only configurations that make those applications possible: manifolds, bends, U-bends, collectors, and diffusers whose performance is governed by pressure loss, viscous dissipation, separation, and turbulence-model behavior. These flow configurations are a necessary stepping stone toward turbulent heat-transfer optimization because reliable thermal design requires a credible and stable prediction of the hydraulic losses first.

Mathematical optimization offers a way to design such internal-flow layouts without prescribing the final channel topology in advance. Among these strategies, topology optimization has emerged as a powerful computational design tool for fluid and thermal-management problems [24]. In a fully coupled thermal problem, the eventual objective may involve heat-transfer rate, maximum temperature, or thermal resistance. In the flow-only problem treated here, the relevant objectives are instead pressure drop and dissipated mechanical energy. This restriction is not merely a simplification for convenience: turbulent density-based topology optimization remains challenging even before the energy equation is added, because the flow solver, turbulence model, wall treatment, geometry representation, and adjoint sensitivity calculation all interact with the evolving topology.

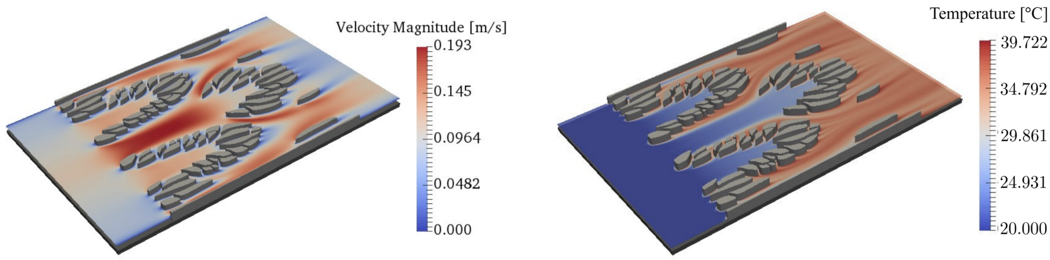


Figure 1.1: Topology-optimized heat sink from Barroca et al. [11], showing the optimized fin layout with velocity magnitude (left) and temperature (right).

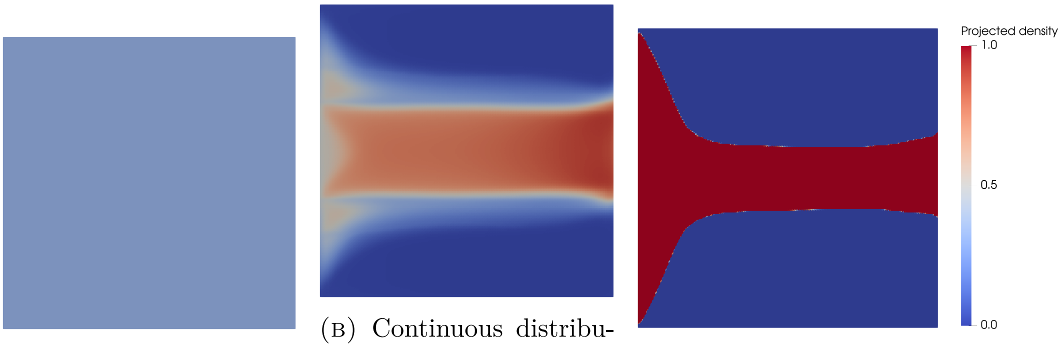
The central gap addressed in this thesis is therefore that density-based fluid topology optimization is far more mature for laminar flows than for turbulent internal flows. Laminar assumptions are attractive because they simplify both the governing equations and the associated adjoint sensitivities, but they are often not representative of compact engineering flow paths operating at industrial Reynolds numbers. In turbulent internal flows, the chosen Reynolds-Averaged Navier-Stokes (RANS) closure, near-wall treatment, and response to curvature or separation can influence the optimized topology itself. Before turbulent conjugate heat-transfer optimization can be treated as a reliable design tool, the underlying turbulent flow-only optimization problem must first be stabilized, verified, validated, and interpreted critically.

1.2 Conceptual Overview of Density-Based Topology Optimization

To understand the challenges of applying optimization to turbulent fluids, it is first necessary to distinguish topology optimization from traditional design methodologies. Historically, computational design relied on size optimization, which changes the thickness of predefined components, or shape optimization, which moves the boundary nodes of a predefined geometry. While effective, these methods are inherently limited by their starting topology. A shape optimization algorithm cannot spontaneously create a new hole or split a single fluid channel into two.

Density-based topology optimization, pioneered by Bendsoe and Kikuchi in 1988 for solid mechanics [13], approaches the problem differently. Instead of moving boundaries, the design space is discretized into a fixed grid of finite elements. The algorithm then determines the optimal distribution of material across this grid. In fluid problems, this is achieved by assigning a continuous design variable, $\gamma \in [0, 1]$, to every element in the domain. In the convention adopted for this thesis, $\gamma = 1$ represents a pure fluid channel, while $\gamma = 0$ represents a solid, impermeable structure.

Because gradient-based optimizers require continuous and differentiable functions, this binary choice is mathematically relaxed. The design variable is allowed to take intermediate "gray" values between 0 and 1. To give physical meaning to these intermediate states, the fluid equations are modified using a fictitious porous medium approach, where intermediate densities correspond to varying levels of flow permeability. The optimization algorithm iteratively updates the design field to minimize a specific objective function, such as pressure loss or dissipated power, while satisfying constraints such as a maximum allowable fluid volume fraction. In the conceptual figures in Figure 1.2, the displayed field should be interpreted as the physical, projected density used by the flow equations. Chapter 5 distinguishes this field from the raw optimizer variable and the filtered density.



(A) Initial projected field

(B) Continuous distribution

(C) Discrete topology

FIGURE 1.2: Iterative density-based topology optimization process: initialization with a uniform projected density field, intermediate continuous material distribution, and final discrete fluid topology. The full raw-filtered-projected density chain is introduced in Chapter 5.

1.3 The Evolution of Fluid Topology Optimization

While density-based topology optimization became a standard tool in structural engineering in the late 1990s, its application to fluid dynamics is relatively recent. The optimization of fluid manifolds remains an active and highly challenging area of research [24]. The genesis of fluid topology optimization is widely attributed to Borrvall and Petersson (2003), who successfully applied the technique to minimize power dissipation in creeping Stokes flow [14]. This was quickly followed by Gersborg-Hansen et al. (2005), who extended the framework to steady, laminar Navier-Stokes flows [26]. Over the past decade, the literature surrounding laminar flow optimization expanded extensively, adapting the methodology to standard incompressible Navier-Stokes equations [19, 4, 66].

However, the assumption of laminar flow significantly simplifies both the governing equations and the derivation of adjoint sensitivities required for gradient-based optimization. Unfortunately, this assumption prevents the application of these methodologies to many industrial and high-performance applications, where flow velocities are high and turbulence dominates the flow physics. Only recently has the computational mechanics community begun to tackle the severe non-linearities and numerical instabilities associated with optimizing geometries directly in the turbulent regime [63, 20, 64].

Extending topology optimization to the turbulent regime is highly non-trivial. It requires coupling optimization algorithms with Reynolds-Averaged Navier-Stokes (RANS) turbulence models. These models introduce steep velocity gradients, highly non-linear source terms, near-wall resolution requirements, and severe numerical instabilities during the iterative optimization process. To mitigate this, gradient-based turbulent optimizations frequently rely on the "frozen turbulence" assumption, where the turbulent viscosity field is updated in the forward solve but treated as fixed during sensitivity assembly. While this assumption provides necessary solver stability and has driven success in industrial adjoint shape optimization for automotive aerodynamics [43], advancing toward fully coupled turbulent sensitivities in topology optimization remains an ongoing mathematical challenge.

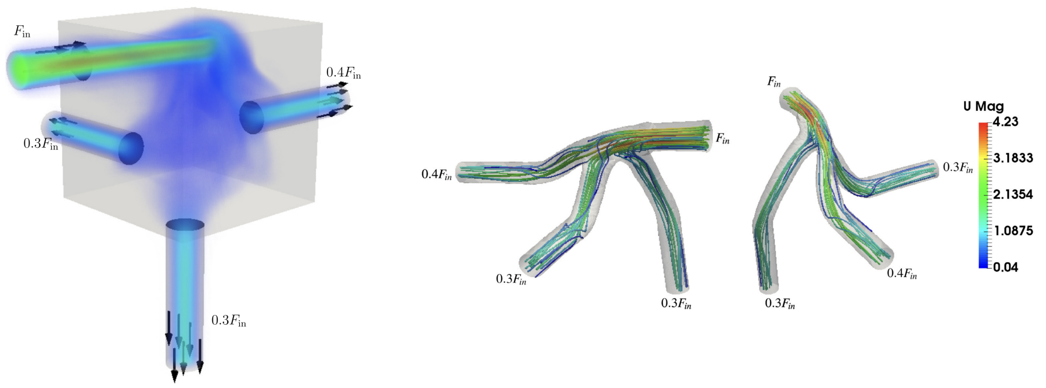


Figure 1.3: Topology optimized manifolds with turbulent flow. Adapted from Dilgen et al. [20].

Pioneering works by Yoon (2016) [63] and Dilgen et al. (2018) [20] have pushed the boundaries of turbulent frameworks. As seen in Figure 1.3, integrating turbulence models allows the optimizer to generate complex fluid manifolds tailored to guide turbulent flow distributions efficiently.

1.4 Objectives and Research Questions

The main objective of this Master's thesis is to extend density-based topology optimization to internal flow configurations operating in the turbulent regime. The phrase "flow configurations" in the title is intentional: the work targets the hydraulic part of energy and thermal-management devices, not the complete heat-transfer device. To achieve this goal, the thesis investigates the coupling of incompressible flow, RANS turbulence modeling, Brinkman topology penalization, and gradient-based optimization within an automated finite-element framework. By isolating the fluid mechanics from heat-transfer calculations, the thesis bounds its scope to establish a stable and verifiable baseline for future turbulent thermal design.

Rather than relying solely on black-box commercial CFD software, the core framework is implemented in **FEniCS**, an open-source computing platform for solving partial differential equations [5]. FEniCS and UFL provide the mathematical flexibility required to write finite-element weak forms directly, verify the one-equation Spalart-Allmaras (SA) turbulence model, and assemble the residual derivatives used by the frozen-turbulence adjoint.

At this stage, it is important to distinguish two different comparisons that appear later in the thesis. A comparison between laminar-optimized and turbulent-optimized layouts is useful as supporting design context, because it shows how inertia and turbulence alter the preferred topology. However, this comparison is not treated as the principal CFD validation step. The decisive assessment instead consists of re-analyzing the final SA-optimized geometries in Ansys Fluent, first with the same SA model and then with the SST $k-\omega$ closure.

Based on this methodology, this thesis seeks to answer the following research questions:

1. **How accurately does the Spalart-Allmaras turbulence model predict curved and separated internal turbulent flows, and where do its physical limitations appear?**

   Chapter 2 answers this question by combining turbulence-model theory, curved-pipe and separated-flow literature evidence, and Ansys Fluent comparisons on curved internal-flow benchmarks. The Dean number is used only as a rough physical scale for the bend cases, not as a universal validity threshold for SA.

2. **Can a custom FEniCS implementation of the Spalart-Allmaras turbulence model achieve quantitative agreement with Ansys Fluent for turbulent internal-flow benchmarks?**

   Chapters 3 and 4 answer this question by presenting the weak forms, implementing the standalone SA solver, and comparing it against Ansys Fluent on a turbulent channel and a U-bend benchmark.

3. **To what extent do SA-driven topology optimizations produce final designs whose predicted performance remains consistent under independent Ansys re-analysis with SA and SST $k-\omega$, and how do they compare with laminar-optimized designs under identical turbulent operating conditions?**

   Chapter 6 answers this question using the pipe-bend, U-bend, and diffuser benchmarks. It reports the FEniCS optimizations, sharp-geometry Ansys-SA validation, SST $k-\omega$ re-analysis, and laminar-versus-turbulent design comparisons. Appendix C contains the supporting frozen-sensitivity verification.

1.5 Outline of the Thesis

The remainder of this thesis is structured into the following six chapters:

- **Chapter 2 (Turbulence Modeling):** Provides the theoretical background on fluid dynamics and turbulence. It discusses the Navier-Stokes equations, RANS modeling, near-wall treatment, the physical characteristics of the Spalart-Allmaras model, and the curvature and separation limits relevant to this thesis.
- **Chapter 3 (Implementation of the Spalart-Allmaras Model in FEniCS):** Details the numerical implementation of the standalone SA solver in FEniCS, including weak forms, IPCS flow solves, Picard coupling, and wall-distance computation.
- **Chapter 4 (Verification of the Spalart-Allmaras Solver for Turbulent Internal Flows):** Verifies the standalone FEniCS-SA forward solver against Ansys Fluent on turbulent channel-flow and U-bend benchmarks.
- **Chapter 5 (Turbulent Topology Optimization Methodology):** Introduces the density-based optimization framework. It details Brinkman momentum penalization, topology penalties, reciprocal wall-distance penalization, filtering, projection, MMA, and the frozen-turbulence adjoint.
- **Chapter 6 (Turbulent Topology Optimization Results):** Presents the turbulent topology optimization results for the pipe-bend and U-bend benchmarks from Bayat et al. [12] and the diffuser benchmark from Yoon [63], and evaluates extracted designs with both SA and SST $k - \omega$ turbulence models. Appendix C documents the pipe-bend sensitivity verification and numerical support material.
- **Chapter 7 (Conclusion and Future Work):** Summarizes the main findings, answers the research questions conservatively, and outlines remaining limitations and future extensions toward three-dimensional and conjugate heat-transfer applications.

Chapter 2

Turbulence Modeling

2.1 Introduction to Fluid Dynamics and Turbulence

Fluid dynamics is governed by the Navier-Stokes equations, which describe the conservation of mass and momentum for a viscous continuum fluid [61]. In this introductory text, these equations are presented using index notation to clearly illustrate the individual tensor components. The indices i and j represent the three spatial dimensions, and a repeated index within a term implies summation over all spatial directions. In subsequent chapters dealing with finite element implementation, this notation shifts to standard vector calculus to align with the FEniCS Unified Form Language syntax.

The fundamental principle of mass conservation is expressed by the continuity equation. For a general compressible fluid, this is defined as:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_i)}{\partial x_i} = 0 \tag{2.1}$$

where ρ is the fluid density. For most liquid cooling applications and low-speed fluid manifolds, the fluid can be assumed incompressible. Under this assumption, ρ is constant in space and time, and the continuity equation reduces to

$$\frac{\partial u_i}{\partial x_i} = 0. \tag{2.2}$$

For an incompressible Newtonian fluid with constant dynamic viscosity μ, the conservation of momentum is expressed as:

$$\rho \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} \right) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial u_i}{\partial x_j} \right) \tag{2.3}$$

where p represents the static pressure. In principle, solving Equation 2.3 directly yields the exact flow field. This approach is known as Direct Numerical Simulation (DNS). However, as the Reynolds number (Re) increases, the flow transitions into a chaotic turbulent state characterized by a broad spectrum of spatial and temporal scales [45]. Resolving the smallest turbulent eddies requires extremely fine spatial grids. For industrial applications, including electronics cooling manifolds, DNS is computationally prohibitive. Engineers therefore rely on turbulence modeling to predict the averaged effects of these fluctuations without resolving the full turbulent spectrum [59, 6].

2.2 Reynolds-Averaged Navier-Stokes (RANS) Equations

The most common approach to turbulence modeling in engineering is the Reynolds-Averaged Navier-Stokes (RANS) framework, based on the theories originally introduced by Osborne Reynolds [45]. The core principle of RANS is the Reynolds decomposition, which separates instantaneous flow variables into a time-averaged mean component ($\bar{u}_i$, $\bar{p}$) and a fluctuating component (u_i', p'):

$$u_i = \bar{u}_i + u_i' \quad \text{and} \quad p = \bar{p} + p'. \tag{2.4}$$

Figure 2.1 illustrates this separation for velocity and pressure signals measured at a fixed point in a turbulent flow. The instantaneous signal contains rapid turbulent fluctuations, while the dashed mean line represents the slowly varying quantity retained by the RANS equations. By construction, the mean of the fluctuating component is zero, but products of fluctuations, such as $\overline{u_i' u_j'}$, remain non-zero and generate the Reynolds-stress terms.

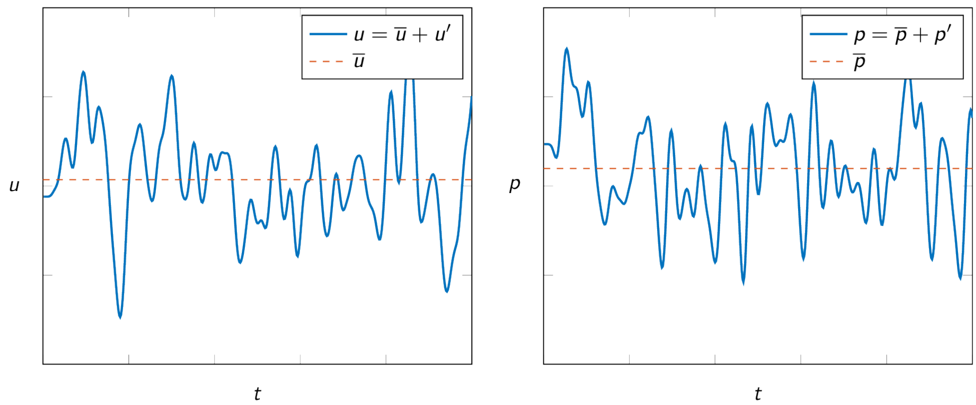


Figure 2.1: Reynolds decomposition of velocity and pressure signals into mean and fluctuating components. Adapted from the Fluid Mechanics on the Web Chapter 6 material based on White's viscous-flow treatment [41].

Substituting these decomposed variables into the standard Navier-Stokes equations and taking the time average yields the RANS momentum equations:

$$\rho \left(\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} \right) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial \bar{u}_i}{\partial x_j} - \rho \overline{u_i' u_j'} \right). \tag{2.5}$$

The averaging process introduces a new term, $-\rho\overline{u_i' u_j'}$, known as the Reynolds stress tensor. This tensor represents the momentum transfer caused by turbulent fluctuations. The introduction of this term creates a mathematical closure problem because the RANS equations now contain more unknowns than available physical equations [59].

2.2.1 The Boussinesq Hypothesis and the Isotropic Assumption

To close the system and solve for the mean flow, most RANS models rely on the Boussinesq eddy viscosity hypothesis [45]. This hypothesis assumes that turbulent momentum transfer is analogous to molecular momentum transfer. It relates the Reynolds stresses linearly to the mean velocity gradients using a scalar turbulent eddy viscosity (μ_t):

$$-\rho\overline{u_i' u_j'} = \mu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \frac{2}{3}\rho k \delta_{ij} \tag{2.6}$$

where k is the turbulent kinetic energy and δ_{ij} is the Kronecker delta. The primary task of any Boussinesq-based turbulence model is to calculate this local eddy viscosity.

It is crucial to highlight a fundamental physical limitation of Equation 2.6: the Boussinesq hypothesis assumes that μ_t is an isotropic scalar quantity. It maps all turbulence effects into one scalar multiplier aligned with the mean strain rate. While this assumption is often adequate for attached shear flows, it is less reliable when curvature, rotation, or separation create Reynolds-stress anisotropy. Curved ducts and bends are especially relevant here because centrifugal effects redistribute momentum across the cross-section and can create secondary motion that a scalar eddy-viscosity closure cannot represent exactly [48, 47].

2.3 Boundary Layer Theory and Near-Wall Treatment

Before selecting a specific turbulence equation to calculate μ_t, the physical structure of a turbulent boundary layer must be considered. The boundary layer is traditionally divided into three regions based on the non-dimensional wall distance (y^+) and non-dimensional velocity (u^+):

$$y^+ = \frac{y u_\tau}{\nu}, \quad u^+ = \frac{\bar{u}}{u_\tau} \tag{2.7}$$

where y is the distance from the wall, ν is the kinematic viscosity, and $u_\tau = \sqrt{\tau_w/\rho}$ is the friction velocity derived from the wall shear stress (τ_w) [45].

Figure 2.2 summarizes the wall-unit regions used when discussing RANS wall treatment. The important point for this thesis is not that one near-wall strategy is universally required for topology optimization, but that the chosen turbulence closure and wall treatment must remain stable while the solid-fluid boundary evolves.

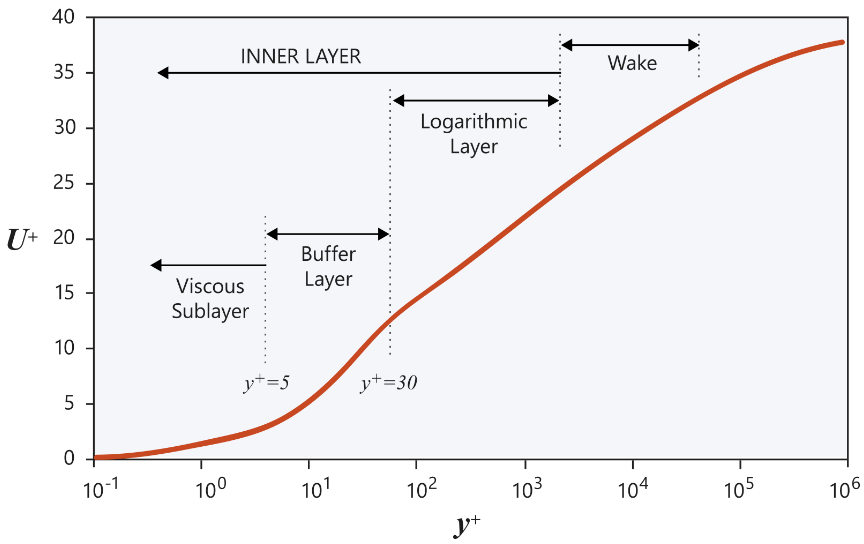


Figure 2.2: Near-wall turbulent boundary-layer regions in wall units. Adapted from Altair's AcuSolve near-wall modeling documentation [7].

1. **The viscous sublayer ($y^+ < 5$):** Immediately adjacent to the wall, turbulent fluctuations are damped by kinematic constraints. Viscous shear dominates and the velocity profile is approximately linear, $u^+ = y^+$. Low-Reynolds wall-resolved RANS calculations place the first cell around $y^+ \approx 1$ so this region is resolved rather than modeled.

2. **The buffer layer ($5 < y^+ < 30$):** This is a transition region where viscous and turbulent shear stresses are both important. It is generally an undesirable location for the first cell center because it is neither cleanly wall-resolved nor safely inside the log-law range used by wall functions.

3. **The log-law and outer regions ($y^+ > 30$):** Farther from the wall, turbulence dominates and the velocity follows an approximate logarithmic distribution controlled by the von Karman constant. High-Reynolds wall-function methods place the first cell in this range and model the inner layer empirically, while the outer wake region becomes increasingly sensitive to pressure-gradient and outer-flow effects [59, 52, 7].

High-Reynolds models, such as the standard $k - \epsilon$ model, use empirical wall functions to bridge the gap between the wall and the log-law region, typically requiring the first computational cell center to be placed at $y^+ > 30$. Wall functions are not fundamentally incompatible with turbulent topology optimization; Bayat et al. [12], for example, use a wall-function formulation with an implicit wall penalty. For this thesis, however, a wall-resolved $y^+ \approx 1$ strategy is more convenient because it avoids introducing a moving empirical wall-function interface into the FEniCS density-based formulation. The meshes used here are therefore generated as wall-resolved meshes, and the SA model is integrated through the viscous sublayer.

2.4 The Spalart-Allmaras One-Equation Model

While two-equation models such as $k-\epsilon$ or $k-\omega$ solve separate transport equations for turbulent kinetic energy and a turbulent length scale, they introduce highly non-linear source terms. These terms frequently cause numerical instabilities during the adjoint sensitivity analysis required for topology optimization. To achieve a trade-off between physical accuracy and computational efficiency, this thesis utilizes the Spalart-Allmaras (SA) model, originally developed in 1992 for aerodynamic flows with adverse pressure gradients [50]. The SA model solves a single transport equation for a working variable, $\tilde{\nu}$, which is subsequently linked to the turbulent kinematic eddy viscosity ($\nu_t = \mu_t/\rho$) [51].

The transport equation for the standard SA model is given by:

$$\frac{\partial \tilde{\nu}}{\partial t} + \bar{u}_j \frac{\partial \tilde{\nu}}{\partial x_j} = c_{b1}\tilde{S}\tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{d}\right)^2 + \frac{1}{\sigma}\left[\frac{\partial}{\partial x_j}\left((\nu+\tilde{\nu})\frac{\partial \tilde{\nu}}{\partial x_j}\right) + c_{b2}\left(\frac{\partial \tilde{\nu}}{\partial x_j}\right)^2\right]. \quad (2.8)$$

The terms on the right-hand side represent production, destruction, and diffusion of the turbulent working variable, respectively. The production term relies on a modified vorticity scalar, $\tilde{S}$, so turbulence generation in the SA model is fundamentally tied to shear. The destruction term is explicitly a function of d, the distance to the nearest wall. This makes the SA model a wall-resolved low-Reynolds model that integrates through the viscous sublayer without requiring empirical wall functions or a second length-scale equation [46].

At this stage of the thesis, the SA model is treated as a turbulence closure rather than an implementation object. Therefore, Chapter 2 only states the main transport equation and its physical role. Detailed constants, damping functions, linearization choices, and weak forms are deferred to Chapter 3, where the FEniCS implementation is discussed.

2.5 Evaluation of Curvature Effects and the Dean Number

Despite its computational advantages for topology optimization, the standard SA model relies on a single length scale tied to the nearest wall distance. It lacks an explicit rotation/curvature correction and remains constrained by the Boussinesq eddy-viscosity assumption. It can therefore struggle to capture the anisotropic physics of streamline curvature.

When fluid passes through a bend, centrifugal forces induce secondary motion perpendicular to the main flow direction. In curved-pipe literature, the dimensionless Dean number (De) is often used to report the combined Reynolds-number and curvature scale:

$$De = Re_d\sqrt{\frac{d}{2R_c}} \quad (2.9)$$

where d is the characteristic length of the cross-section, such as the pipe diameter D or hydraulic diameter D_h, R_c is the radius of curvature, and Re_d is the bulk

Reynolds number based on that same characteristic length [18]. A higher Dean number indicates a stronger cross-sectional influence of centrifugal forces.

The Dean number is academically relevant here because it separates curvature severity from Reynolds number alone. However, it should not be interpreted as a universal threshold for SA validity. The breakdown of an eddy-viscosity model depends on geometry, separation, mesh resolution, boundary conditions, and the validation quantity being compared. In this thesis, the Dean number is used only as a rough physical descriptor of the bend cases; the final judgment still comes from profile comparisons and independent CFD re-analysis.

For mildly curved flows, the SA model can perform relatively well. Apalowo [9] evaluated a 90° circular pipe bend with a Dean number of $De \approx 9,000$ at $Re_D \approx 34,000$. Under these moderate conditions, where $R_c = 7D$, the SA result in the bottom panel of Figure 2.3 follows the measured streamwise-velocity profile about as well as the two k-ϵ variants. This can be seen from the similar profile shape and from the reported error values, which are slightly lower for the SA panel than for the two panels above it. The comparison is therefore used as evidence that mild curvature does not automatically invalidate SA.

At stronger bend curvature and higher Dean number, the limitations of the isotropic Boussinesq assumption become more visible. Abuhatira et al. [2] evaluated turbulent flow through a 90° pipe bend with $R_c = 2.0D$ and $Re_D = 60,000$, corresponding to $De \approx 30,000$. Figure 2.4 shows the velocity profile just outside the bend, at the beginning of the straight outlet section. None of the RANS models perfectly follows the experimental data, including SST $k-\omega$, which is a useful warning in itself. The yellow SA curve is nevertheless the worst of the compared closures near the inner-wall side of the profile, where it smooths the velocity redistribution more strongly than the other models. This provides supporting evidence that strong curvature can expose the limitations of the standard SA closure, but it is not treated as proof of a universal failure threshold.

The literature therefore provides useful rough-scale evidence: SA can be acceptable for mild curvature and can deteriorate for stronger curvature. The thesis-specific evidence must come from the Ansys comparisons below, which evaluate the geometries and quantities most relevant to the present optimization study.

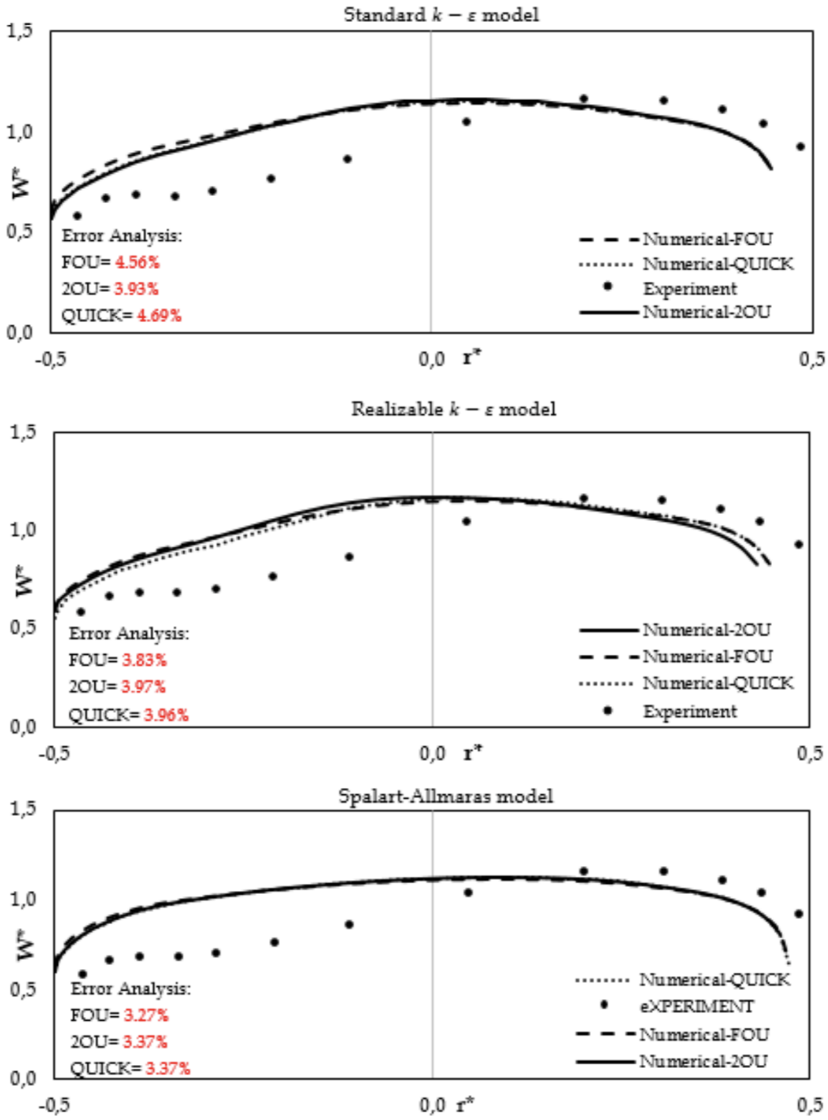


FIGURE 2.3: Streamwise velocity profiles in a 90° pipe bend ($R_c = 7D$, $Re_D \approx 34,000$, $De \approx 9,000$). The SA model is the bottom panel; its profile shape and reported error values are comparable to the two k-ϵ panels above it. Reprinted from Apalowo [9].

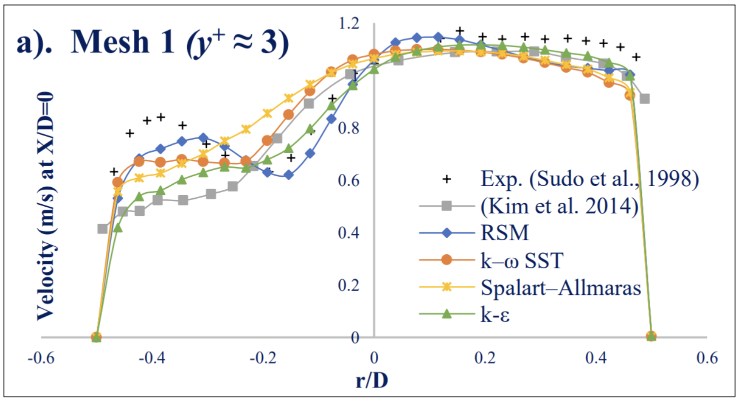


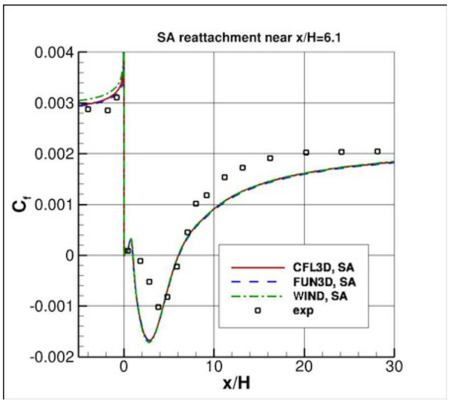
Figure 2.4: Turbulence model predictions just outside the 90° bend, at the beginning of the straight outlet section ($R_c = 2.0D$, $Re_D = 60,000$, $De \approx 30,000$). The SA result is the yellow curve and deviates most strongly from the experimental data, although the other RANS closures are also imperfect. Adapted from Abuhatira et al. [2].

2.6 Separated Flow Without Curvature

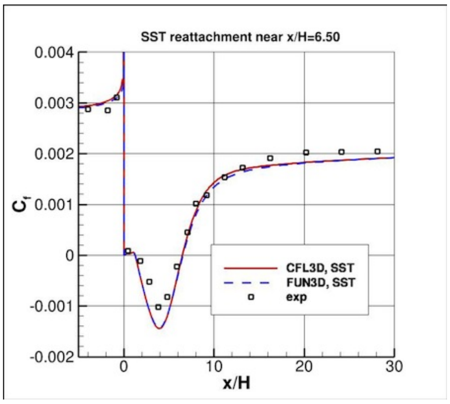
Curvature is not the only mechanism that can expose the limitations of a scalar eddy-viscosity model. A fully developed channel is an attached wall-bounded flow and is therefore not treated as a model-limitation case in this chapter; it is used later as an implementation-verification benchmark. A backward-facing step is more relevant here because it removes bend curvature while introducing a sudden expansion, separated shear layer, recirculation region, and downstream reattachment [22, 32]. This resembles the local separation that can occur in topology-optimized manifolds and diffuser-like designs even when the geometry is not dominated by pipe-bend curvature. Because there is no curvature radius, the Dean number is not meaningful for this case; the standard integral metric is the normalized reattachment length x_r/h, with h the step height.

The modeling difficulty is that separation breaks the near-wall local-equilibrium picture on which simple wall-resolved closures often behave well. Turbulence generated in the detached shear layer is transported downstream before it affects the mean flow, and reattachment depends on shear-layer growth, entrainment, pressure recovery, and stress redistribution. Linear eddy-viscosity models such as SA and SST $k-\omega$ can represent extra turbulent mixing through an effective scalar viscosity, but they cannot fully represent stress anisotropy or coherent unsteadiness during reattachment [45, 59]. For topology optimization, this matters because a locally separated design may appear favorable if the chosen closure overpredicts reattachment or underpredicts separation losses.

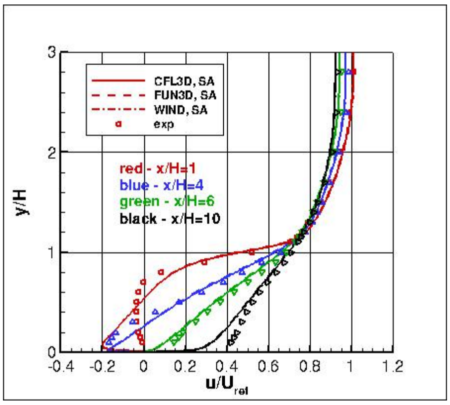
The Driver and Seegmiller backward-facing-step experiment is a canonical validation case for separated internal flow [21]. In the NASA Turbulence Modeling Resource comparison shown in Figure 2.5, the experimental reattachment length is $x_r/h = 6.26 \pm 0.10$. Representative SA computations predict reattachment near



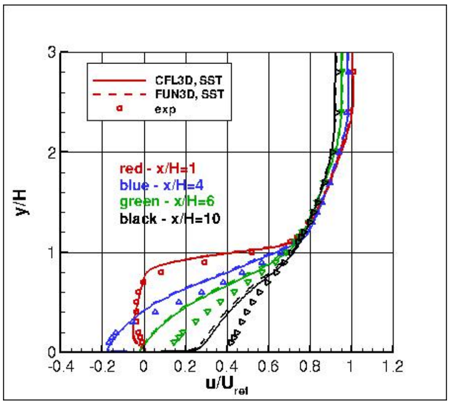
(A) SA skin friction.



(B) SST $k-\omega$ skin friction.



(C) SA velocity profiles.



(D) SST $k-\omega$ velocity profiles.

FIGURE 2.5: NASA Turbulence Modeling Resource comparison of SA and SST $k-\omega$ predictions with the Driver-Seegmiller backward-facing-step experiment at $Re_H \approx 36{,}000$ [21, 39]. The velocity-profile comparison indicates better separated-flow recovery for SST $k-\omega$ than for SA.

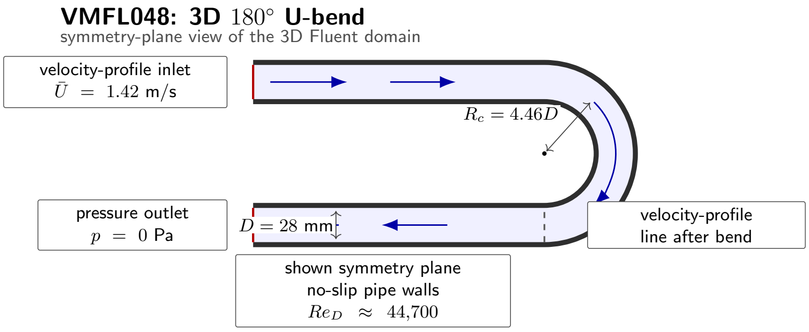
$x_r/h \approx 6.1$, while the SST $k-\omega$ results reported by NASA as SSTm predict reattachment near $x_r/h \approx 6.5$ [39]. Judged only by this integral metric, SA does not simply fail. The velocity profiles in panels (c) and (d), however, show the more relevant limitation: SST $k-\omega$ follows the measured separated-flow recovery more closely than SA. Since topology optimization is sensitive to local velocity recovery, wall shear, and loss generation, the backward-facing-step evidence supports a cautious conclusion: separated internal flows are strongly turbulence-model dependent, and SA-based predictions should be checked with an independent closure when local separation is present.

2.7 Benchmark Studies in Ansys Fluent

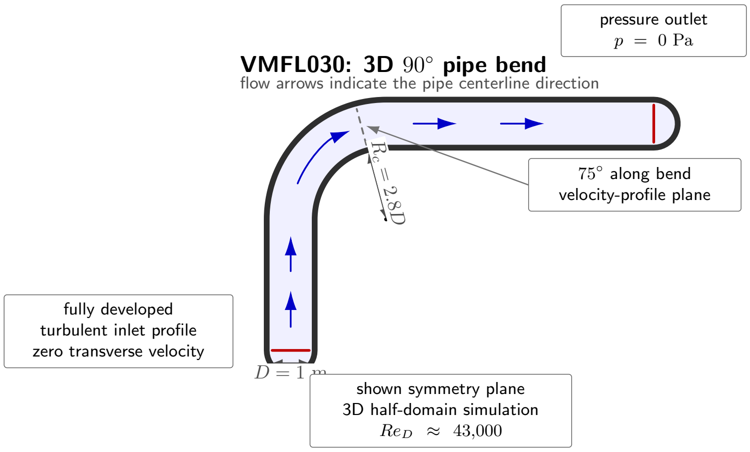
To investigate where the SA model's accuracy deteriorates for the present study, benchmark simulations were performed using **Ansys Fluent**. Three internal-flow benchmarks were considered. The first two are fully three-dimensional curved-flow cases: a 180° U-bend and a tighter 90° pipe bend. To ensure a standardized numerical baseline for these curved cases, the computational meshes, fluid properties, and boundary conditions were adopted from the official Ansys Fluid Dynamics Verification Manual [8]. Specifically, the 3D 90° bend corresponds to verification case VMFL030, while the 3D U-bend corresponds to case VMFL048. The third benchmark, the backward-facing step, is treated differently: it is not used as a new validation case, because the literature-supported validation evidence is already given in Section 2.6. It is instead used as a separated-flow model-sensitivity benchmark in Appendix A, comparing SA against SST $k-\omega$ [38] on the local backward-facing-step geometry used in the repository and in the corresponding thesis work [36]. The Fluent convergence monitors and final-window criteria are reported in Appendix A.1, so the main chapter can focus on the physical comparison between SA and SST $k-\omega$.

2.7.1 Mild Curvature: The U-Bend Case

The first benchmark evaluated a 3D U-bend geometry characterized by a moderate curvature ratio of $R_c \approx 4.46D$. The Fluent setup is summarized in Figure 2.6a: the VMFL048 case uses a velocity-profile inlet, a zero-gauge pressure outlet, no-slip pipe walls, and a symmetry boundary on one side of the 3D domain. Operating at a bulk Reynolds number of $Re_D \approx 44,700$, this geometry resulted in a Dean number of $De \approx 15,000$.

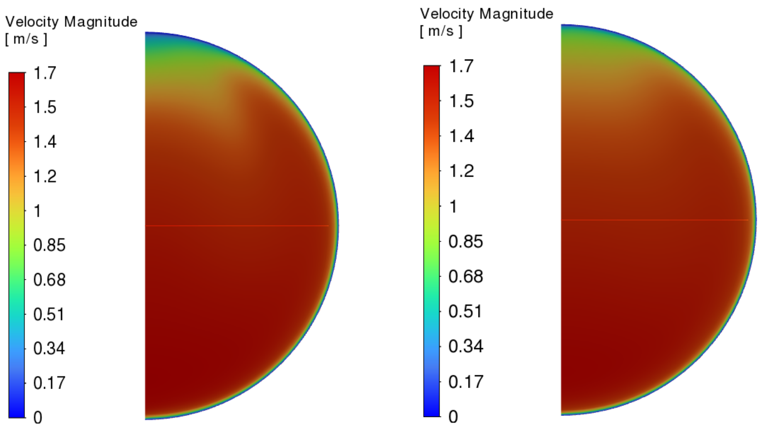


(A) VMFL048 3D U-bend.

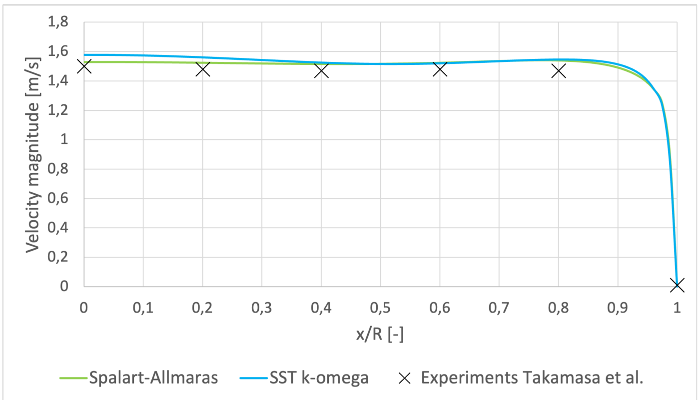


(B) VMFL030 3D 90° bend.

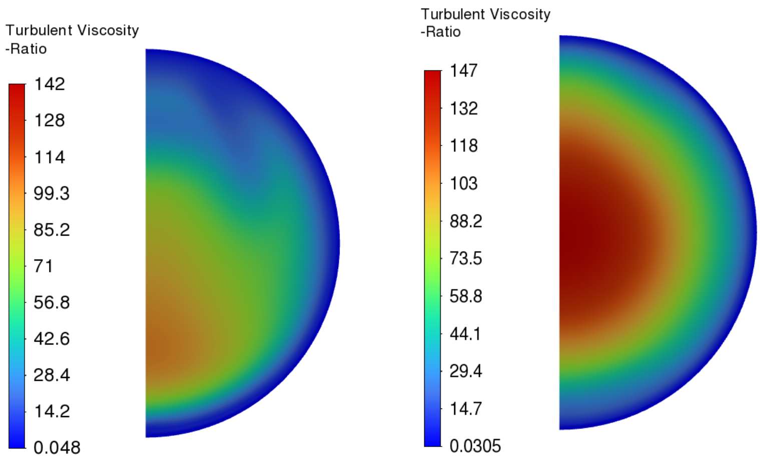
FIGURE 2.6: Geometry and boundary conditions of the two Ansys Fluent curved-flow benchmark cases adopted from the Ansys Fluid Dynamics Verification Manual [8].



(A) Velocity, SST $k - \omega$ (B) Velocity, SA



(C) Velocity profiles extracted at the bend exit



(D) μ_t/μ, SST $k - \omega$ (E) μ_t/μ, SA

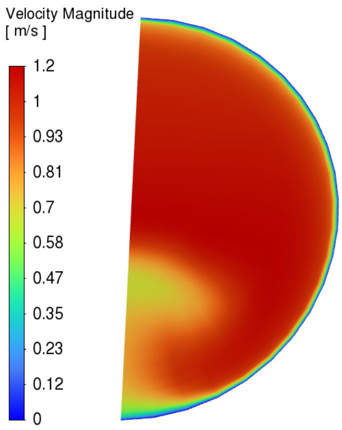
Figure 2.7: VMFL048 U-bend RANS comparison: velocity contours, extracted velocity profile, and turbulent-viscosity ratio contours.

Consistent with the moderate-curvature literature evidence, the SA model performed well under these conditions. As shown in Figure 2.7, the qualitative velocity contours of the SA model reproduce the bulk flow distribution of the SST $k-\omega$ model. In panels (a) and (b), the line drawn on the exit-plane contour is the radius along which the VMFL048 velocity profiles in panel (c) are extracted. The profile coordinate follows this radius from left to right, i.e. from the inner part of the pipe at $x/R = 0$ toward the pipe wall at $x/R = 1$. This links the one-dimensional profile comparison directly to the displayed exit-plane flow field. The quantitative velocity profiles agree well with the experimental data from Takamasa and Kondo [56].

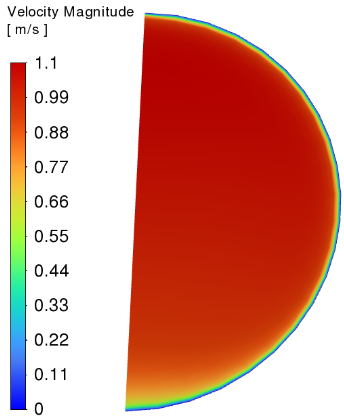
The turbulent-viscosity panels in Figure 2.7 clarify why the VMFL048 velocity agreement is good but not model-identical. Both closures damp μ_t/μ to low values at the no-slip wall and generate the largest eddy viscosity away from the wall where mean shear and turbulent mixing are strongest. The SA contour is more radially uniform than the SST $k-\omega$ contour and its scale reaches a slightly higher maximum, which is consistent with a mild tendency to overpredict eddy viscosity and smooth gradients. In this moderately curved U-bend, that extra diffusion remains spatially aligned with the main shear region, so it does not strongly distort the extracted velocity profile.

2.7.2 Strong Curvature: The $90°$ Bend Case

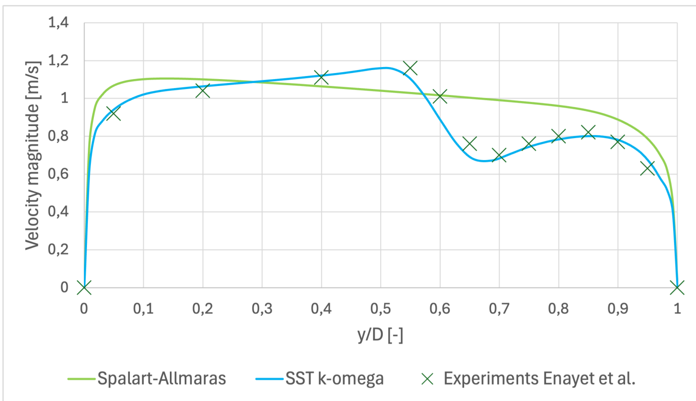
The second benchmark evaluated a tighter $90°$ pipe bend characterized by $R_c = 2.8D$. The corresponding VMFL030 setup is shown in Figure 2.6b, including the fully developed turbulent inlet profile, pressure outlet, no-slip pipe walls, half-domain symmetry, and the $75°$ velocity-profile plane used for comparison. Despite operating at a similar bulk Reynolds number of $Re_D \approx 43,000$, this tighter bend radius pushed the Dean number to $De \approx 18,000$. Because the Reynolds numbers are similar between the two benchmark cases, the comparison isolates the geometric effect of streamline curvature more clearly than a Reynolds-number-only comparison.



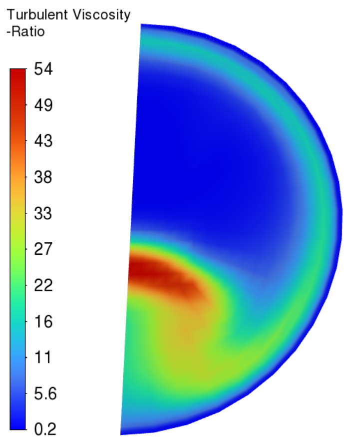
(A) Velocity, SST $k-\omega$



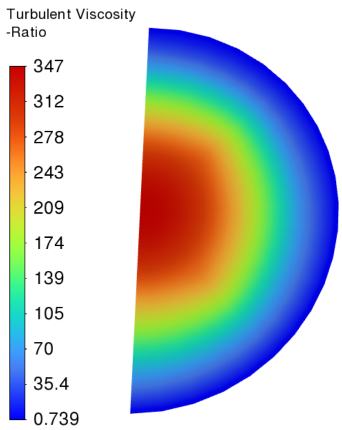
(B) Velocity, SA



(C) Velocity profiles extracted at 75° along the bend



(D) μ_t/μ, SST $k-\omega$



(E) μ_t/μ, SA

Figure 2.8: VMFL030 90° pipe-bend RANS comparison: velocity contours, extracted velocity profile, and turbulent-viscosity ratio contours.

As shown in Figure 2.8, the velocity profile at the 75° plane along the bend demonstrates clear deterioration of the SA prediction. The profile in panel (c) is extracted along the pipe diameter. In the contour views of panels (a) and (b), this diameter coincides with the left edge of the plotted cross-section, so an additional guide line cannot be drawn without hiding the contour boundary. The coordinate follows the diameter from top to bottom, with $y/D = 0$ at the top and $y/D = 1$ at the bottom. Driven by the tighter bend radius and stronger anisotropic centrifugal effects, the SA model over-predicts the turbulent viscosity. This causes it to over-smooth the velocity profile, failing to capture the sharp velocity gradients near the inner wall when compared against both the SST $k-\omega$ prediction and the experimental data from Enayet et al. [23].

The turbulent-viscosity panels in Figure 2.8 explain the poorer SA velocity prediction in the tighter bend. The SST $k-\omega$ result keeps the high μ_t/μ region more localized, with a low-viscosity core where the curvature-induced redistribution is not simply equivalent to local shear production. The SA result instead produces a broad high-viscosity region with a peak several times larger than the SST $k-\omega$ scale. Physically, the strong bend curvature drives secondary motion and Reynolds-stress anisotropy, while the standard SA closure still represents turbulence through a scalar eddy viscosity based on wall distance and a modified vorticity scale. In this case SA assigns too much eddy viscosity to too much of the bend cross-section, so the modeled turbulent diffusion smooths velocity gradients that remain visible in both the SST $k-\omega$ result and the experiment. This behavior is much more severe than in VMFL048: in the U-bend the SA overprediction is moderate and spatially consistent with the SST $k-\omega$ field, whereas in VMFL030 the eddy-viscosity magnitude and distribution are both wrong enough to wash out the measured velocity-gradient structure.

2.7.3 Sudden Expansion: The Backward-Facing Step Case

The backward-facing-step benchmark is retained as a separated-flow model-sensitivity check. Its setup, mesh-sensitivity table, Ansys contours, and downstream profiles are reported in Appendix A. The main argument relies on the literature comparison in Figure 2.5: SA can predict a plausible reattachment length, but SST $k-\omega$ follows the local velocity recovery more closely. The thesis-specific Fluent backstep results support the same caution without being treated as a separate experimental validation case.

2.8 Conclusion

This chapter answers the first research question regarding the predictive capabilities and physical limitations of the Spalart-Allmaras model for internal turbulent flows relevant to topology optimization. The SA model is attractive because it is a one-equation, wall-resolved closure that avoids empirical wall functions and remains substantially easier to stabilize than two-equation alternatives. The channel-like and mildly curved evidence supports its use as a practical forward model for wall-resolved internal-flow topology optimization.

The limitation is not simply high Reynolds number. The main issue is curvature- and separation-driven Reynolds-stress anisotropy under a scalar eddy-viscosity closure. Literature cases at $De \approx 9,000$ and $De \approx 30,000$ provide useful supporting evidence, while the thesis-specific Ansys comparisons give a rough operating scale for the present bend cases: the U-bend case at $De \approx 15,000$ remains credible, whereas the tighter $90°$ bend at $De \approx 18,000$ already shows clear SA deterioration for the selected profile comparison. The separated-flow evidence points in the same direction. In the Driver-Seegmiller validation data, SST $k - \omega$ follows the measured backward-facing-step velocity profiles more closely than SA, and the thesis-specific backward-facing-step case confirms that the predicted recirculation and recovery are visibly model dependent. These diagnostics are therefore treated as case-specific evidence of model sensitivity, not as universal thresholds.

Consequently, using SA as the turbulence model inside the FEniCS topology optimization loop is defensible for this thesis because of its numerical simplicity and compatibility with wall-resolved optimization. The resulting layouts should be interpreted as SA-optimized design candidates, not as closure-independent predictions by themselves. For attached or mildly curved cases, independent sharp-geometry re-analysis can show whether the SA-optimized layout remains robust. For designs containing strong curvature or local separation, quantitative performance claims must be conditioned on an independent closure check, such as SST $k - \omega$, because the separated-flow losses may be strongly model dependent.

Chapter 3

Implementation of the Spalart-Allmaras Model in FEniCS

3.1 Introduction

With the theoretical foundation of the Spalart-Allmaras (SA) model established in Chapter 2, this chapter details its numerical translation into a working FEniCS solver. The focus is methodological: the finite-element weak forms, Picard coupling strategy, pressure-correction flow solver, SA transport equation, and relaxed Eikonal wall-distance calculation are described before the solver is verified separately in Chapter 4.

The numerical simulations and eventual topology optimization in this thesis are built upon FEniCS, an open-source computing platform for solving partial differential equations (PDEs). Unlike commercial computational fluid dynamics (CFD) software such as Ansys, which typically relies on the Finite Volume Method (FVM), FEniCS uses the Finite Element Method (FEM) and requires the user to explicitly define the mathematical formulation of the problem [35]. FEniCS is useful in this thesis because the same weak-form language used to assemble the forward equations can also be used to assemble residual derivatives for the topology-optimization adjoint in Chapter 5.

To solve a PDE in FEniCS, the governing equations (the "strong form") must be mathematically transformed into a weak formulation, or variational form. FEniCS utilizes the Unified Form Language (UFL) to express these weak forms in Python code that closely mirrors the mathematical notation [33]. This high-level mathematical scripting is particularly advantageous for topology optimization, as it makes the connection between the implemented residuals and the sensitivity equations explicit.

For reproducibility, two project folders accompany this thesis. `Turbulence Models` contains the standalone SA solver and the verification setup files. `FluidTO` contains the topology-optimization code and the frozen-adjoint benchmark setup files. The appendices refer to these folders when tabulating the numerical setups.

3.1.1 Derivation of a Weak Formulation

Before diving into the complex turbulence equations, it is instructive to demonstrate how a continuous PDE is translated into a discrete FEniCS formulation. To illustrate this methodology, consider the Poisson equation for pressure, which is a fundamental component of incompressible fluid solvers. The strong form of the Poisson equation is given by:

$$-\nabla^2 p = f \quad \text{in}\, \Omega \tag{3.1}$$

where p is the pressure field, f is a source term derived from the velocity divergence, and Ω is the computational domain. To obtain the weak form, the equation is multiplied by a sufficiently smooth test function v and integrated over the entire domain Ω, following standard Galerkin finite element procedures [35, 33]:

$$-\int_\Omega (\nabla^2 p) v \, dx = \int_\Omega f v \, dx. \tag{3.2}$$

Applying integration by parts to the left-hand side yields:

$$\int_\Omega \nabla p \cdot \nabla v \, dx - \int_{\partial\Omega} (\nabla p \cdot \mathbf{n}) v \, ds = \int_\Omega f v \, dx, \tag{3.3}$$

where $\partial\Omega$ is the boundary of the domain and $\mathbf{n}$ is the outward-pointing normal vector. This integral formulation lowers the continuity requirements of the solution, requiring only first-order derivatives to be square-integrable. Neumann boundary conditions can also be substituted directly into the boundary integral. This variational approach forms the foundation of all fluid and turbulence equations implemented in this thesis.

3.2 Numerical Solution Strategy for Fluid Dynamics

Solving the full Reynolds-Averaged Navier-Stokes (RANS) equations is numerically challenging due to the non-linear convective terms, the saddle-point nature of the coupled velocity-pressure system, and the intense cross-coupling with the turbulence model. To achieve a stable and efficient steady-state solution in FEniCS, the verification solver in this chapter uses segregated Picard coupling between the velocity-pressure and SA fields.

3.2.1 Decoupling the Physics: The Picard Iteration Scheme

Because the momentum equation relies on the turbulent eddy viscosity (ν_t), and the turbulence transport equation relies on the convective velocity ($\mathbf{u}$), solving the entire fluid-turbulence system simultaneously in a massive monolithic matrix frequently leads to ill-conditioning and solver divergence.

To resolve this, the forward physics are decoupled using a **Picard iteration scheme** (successive substitution), a robust approach for segregated RANS solvers [25, 59]. During each Picard step, the flow solver advances using the eddy viscosity from the previous iteration. The newly updated velocity field is then passed to the Spalart-Allmaras solver to calculate a new eddy viscosity field. This alternating sequence forms the outer loop of the solver, iteratively updating the decoupled equations until the coupled physical system reaches a steady-state equilibrium.

3.2.2 Incremental Pressure Correction Scheme (IPCS)

Within the Picard loop, the Navier-Stokes equations themselves must be solved. Rather than solving for velocity and pressure simultaneously, the standalone verification solver utilizes an Incremental Pressure Correction Scheme (IPCS). This algorithmic splitting is based on the fractional-step projection methods originally pioneered by Chorin [17] and Temam [57]. The weak finite-element statements below follow standard Galerkin discretization practice used in FEniCS [35, 33]. To drive the non-linear convective momentum terms toward a stable steady state, a pseudo-time stepping approach (Δt) is employed for the flow variables. The IPCS algorithm splits the fluid equations into three sequential steps per pseudo-time step:

1. **Tentative velocity step:** The momentum equation is solved using the pressure field from the previous iteration (p^n) to compute an intermediate, non-divergence-free velocity field ($\mathbf{u}^*$). Because this is a RANS framework, the diffusion term utilizes the effective kinematic viscosity, $(\nu + \nu_t)$. Treating the convective term explicitly and the diffusive term implicitly, the semi-discrete equation is:

$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} + (\mathbf{u}^n \cdot \nabla)\mathbf{u}^* - \nabla \cdot (2(\nu + \nu_t^n)\boldsymbol{\varepsilon}(\mathbf{u}^*)) = -\nabla p^n + \mathbf{f}, \tag{3.4}$$

   where $\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}\left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right)$ is the symmetric rate-of-strain tensor.

2. **Pressure correction step:** A Poisson equation is solved to find the pressure variation that enforces the continuity constraint ($\nabla \cdot \mathbf{u}^{n+1} = 0$):

$$\nabla^2 p^{n+1} = \nabla^2 p^n + \frac{1}{\Delta t}\nabla \cdot \mathbf{u}^*. \tag{3.5}$$

3. **Velocity update step:** The tentative velocity is corrected using the gradient of the pressure difference to yield a mass-conserving velocity field:

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla (p^{n+1} - p^n). \tag{3.6}$$

In finite-element form, the three IPCS subproblems are written on a velocity space V_h and pressure space Q_h. For all test functions $\mathbf{v} \in V_h$ and $q \in Q_h$, the tentative velocity step solves

$$\begin{aligned} \int_\Omega \frac{1}{\Delta t}\mathbf{u}^* \cdot \mathbf{v}\, dx + \int_\Omega ((\mathbf{u}^n \cdot \nabla)\mathbf{u}^*) \cdot \mathbf{v}\, dx + \int_\Omega 2(\nu + \nu_t^n)\boldsymbol{\varepsilon}(\mathbf{u}^*) : \boldsymbol{\varepsilon}(\mathbf{v})\, dx \\ = \int_\Omega \frac{1}{\Delta t}\mathbf{u}^n \cdot \mathbf{v}\, dx - \int_\Omega \nabla p^n \cdot \mathbf{v}\, dx + \int_\Omega \mathbf{f} \cdot \mathbf{v}\, dx. \end{aligned} \tag{3.7}$$

The pressure correction is then obtained from

$$\int_\Omega \nabla p^{n+1} \cdot \nabla q \, dx = \int_\Omega \nabla p^n \cdot \nabla q \, dx - \int_\Omega \frac{1}{\Delta t} (\nabla \cdot \mathbf{u}^*) q \, dx, \tag{3.8}$$

and the corrected velocity is recovered through

$$\int_\Omega \mathbf{u}^{n+1} \cdot \mathbf{v} \, dx = \int_\Omega \mathbf{u}^* \cdot \mathbf{v} \, dx - \int_\Omega \Delta t \, \nabla (p^{n+1} - p^n) \cdot \mathbf{v} \, dx. \tag{3.9}$$

In the forward-flow verification scripts, pressure is treated as a kinematic pressure and the viscous operator is written in the symmetric strain-rate form. No Streamline-Upwind/Petrov–Galerkin (SUPG) term is added to the momentum equation in the verification solver. SUPG is a residual-based stabilization that adds numerical diffusion along streamlines to damp nonphysical oscillations in advection-dominated transport. In this implementation, it is used only for the SA working-variable equation described below. The topology-optimization-specific Brinkman and density-penalty terms are introduced later in Chapter 5, after the forward SA equation has been defined.

The simulation is deemed converged to a steady state when the L_2 norm of the residual difference between successive Picard iterations falls below a defined tolerance limit (ϵ_{tol}) for both the fluid and turbulence fields.

3.3 Detailed Formulation of the Spalart-Allmaras Model

As concluded in Chapter 2, the Spalart-Allmaras (SA) one-equation model was selected for its balance of physical accuracy and numerical stability. The SA transport equation is solved as a steady Picard problem within the outer flow loop. In selected convection-dominated verification runs, the linearized SA update is additionally stabilized by the SUPG and pseudo-time terms described below. The later frozen topology-optimization configurations keep the SA pseudo-time damping disabled as a practical choice: pseudo-time stabilization can improve robustness in difficult forward solves, but it adds a tuning parameter and would need to be included consistently in the residual and adjoint if used during optimization. The steady transport equation for the continuous nonlinear working variable $\tilde{\nu}$ is given by:

$$\mathbf{u} \cdot \nabla \tilde{\nu} = c_{b1} \tilde{S} \tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{y} \right)^2 + \frac{1}{\sigma} \left[\nabla \cdot ((\nu + \tilde{\nu}) \nabla \tilde{\nu}) + c_{b2} (\nabla \tilde{\nu})^2 \right] \tag{3.10}$$

where ν is the molecular kinematic viscosity, $\sigma = 2/3$ is the turbulent Prandtl number, c_{b1} and c_{w1} are calibrated closure constants, $\tilde{S}$ is a modified vorticity magnitude, f_w is the wall-damping function, and y is the physical distance to the nearest solid boundary [51].

3.3.1 Picard-Linearized Weak SA Transport Form

The FEniCS implementation does not solve Eq. 3.10 as a fully nonlinear scalar problem. Instead, it uses the outer Picard state to freeze the nonlinear SA closure terms and then solves a linearized weak problem for the next working-variable field. The closure functions used in this weak form are defined in the following subsection.

Let $\tilde{\nu}^k$ denote the previous Picard value of the SA working variable, while $\tilde{\nu}^{k+1}$ is the unknown being solved. The modified vorticity $\tilde{S}^k$, wall function f_w^k, diffusion coefficient, and explicit source terms are evaluated from $\tilde{\nu}^k$ and the current velocity field. The scalar finite-element space for the transported turbulence variable is denoted by T_h, analogous to the velocity space V_h and pressure space Q_h used in the IPCS weak forms. The Picard-linearized SA step is: find $\tilde{\nu}^{k+1} \in T_h$ such that, for all scalar test functions $\xi \in T_h$,

$$\begin{aligned} &\int_\Omega (\mathbf{u} \cdot \nabla \tilde{\nu}^{k+1})\xi \, dx + \frac{1}{\sigma} \int_\Omega (\nu + \tilde{\nu}^k) \nabla \tilde{\nu}^{k+1} \cdot \nabla \xi \, dx \\ &\quad + \int_\Omega c_{w1} f_w^k \frac{\tilde{\nu}^k}{y^2} \tilde{\nu}^{k+1} \xi \, dx \\ &= \int_\Omega \left[c_{b1} \tilde{S}^k \tilde{\nu}^k + \frac{c_{b2}}{\sigma} \nabla \tilde{\nu}^k \cdot \nabla \tilde{\nu}^k \right] \xi \, dx. \end{aligned} \tag{3.11}$$

The left-hand side contains the advection term $(\mathbf{u} \cdot \nabla \tilde{\nu}^{k+1})$, implicit diffusion, and implicit wall destruction. Production and cross-diffusion remain on the right-hand side as explicit Picard sources. After solving Eq. 3.11, the working variable is under-relaxed and bounded before the turbulent eddy viscosity ν_t is recomputed from the standard SA algebraic relation.

When pseudo-time stabilization is enabled for the SA solve, as in the steady U-bend verification configuration, the left-hand side of Eq. 3.11 is augmented by

$$\int_\Omega \frac{\tilde{\nu}^{k+1} - \tilde{\nu}^k}{\Delta t_{\tilde{\nu}}} \xi \, dx. \tag{3.12}$$

This is a numerical damping term for the Picard update of the transported working variable; it does not alter the steady SA closure being targeted after convergence. It is therefore treated as optional forward-solver stabilization, not as a defining part of the topology-optimization formulation.

For convection-dominated internal-flow cases, the implementation may augment the SA transport solve with SUPG stabilization [16]. This stabilization is applied to the transported working variable $\tilde{\nu}$ only; it is not a modification of the RANS momentum equation and it is not an additional turbulence-model source term. With element size h and local speed $|\mathbf{u}|$, the added streamline contribution has the form

$$F_{\mathrm{SUPG},\tilde{\nu}} = \int_\Omega \tau_{\tilde{\nu}} (\mathbf{u} \cdot \nabla \xi) \, \mathcal{R}_{\tilde{\nu}} \, dx, \qquad \tau_{\tilde{\nu}} = C_{\mathrm{SUPG}} \frac{h}{2|\mathbf{u}|}, \tag{3.13}$$

where $\mathcal{R}_{\tilde{\nu}}$ is the strong residual corresponding to Eq. 3.11: the pointwise advection, diffusion, wall-destruction reaction, and explicit production and cross-diffusion sources are collected on one side before being projected along the streamline test direction. When Eq. 3.12 is active, its pseudo-time contribution is included in the same residual. The coefficient C_{SUPG} is a user-selected stabilization factor. In the verification benchmarks and in the turbulent topology-optimization runs, Eq. 3.13 is used to suppress element-scale oscillations in $\tilde{\nu}$ while preserving the same underlying SA closure.

3.3.2 Closure Terms and Calibration Constants

The Spalart-Allmaras model relies on several interconnected closure functions to accurately capture boundary layer physics. The production term relies on a modified vorticity magnitude, $\tilde{S}$, which prevents the destruction of turbulence in regions where the vorticity goes to zero, while maintaining correct scaling near the wall [51]. The implementation uses the vorticity-based SA definition. First,

$$S = \sqrt{2\Omega_{ij}\Omega_{ij}} \quad \text{and} \quad \bar{S} = \frac{\tilde{\nu}}{\kappa^2 y^2} f_{v2}. \tag{3.14}$$

Here S is the scalar magnitude of the vorticity tensor, κ is the von Karman constant, and y is the distance to the nearest wall. During nonlinear iterations, the code evaluates the standard protected form of $\tilde{S}$ recommended in the SA reference implementation [46] so that the production term remains well defined when $S + \bar{S}$ becomes small. Small numerical floors are added in the implementation to prevent division by zero. The function f_{v2} is a near-wall modifier:

$$f_{v2} = 1 - \frac{\chi}{1 + \chi f_{v1}} \quad \text{where} \quad \chi = \frac{\tilde{\nu}}{\nu} \quad \text{and} \quad f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3}. \tag{3.15}$$

Similarly, the wall destruction term is modulated by the non-dimensional function f_w, which accelerates the decay of turbulent viscosity inside the viscous sublayer:

$$f_w = g \left(\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6} \right)^{1/6} \tag{3.16}$$

where the intermediate variables g and r are evaluated as:

$$g = r + c_{w2}(r^6 - r) \quad \text{and} \quad r = \min\left(\frac{\tilde{\nu}}{\tilde{S}\kappa^2 y^2}, 10 \right). \tag{3.17}$$

The upper bound of 10 on r ensures numerical stability during solver initialization when $\tilde{S}$ may momentarily approach zero.

The original formulation by Spalart and Allmaras [51] includes additional geometric trip functions (f_{t1} and f_{t2}) designed to manually trigger the transition from laminar to turbulent flow at specific user-defined coordinates. Following the topology optimization methodology established by Yoon [63], these transition functions are omitted from this thesis. Because the topological boundaries are unknown a priori and evolve during optimization, tracking discrete trip points is mathematically infeasible; therefore, the model assumes a fully turbulent flow regime.

To close the system, the standard empirical calibration constants formulated by Spalart and Allmaras are utilized throughout the FEniCS implementation [46]:

$$\begin{aligned} c_{b1} &= 0.1355 & \sigma &= 2/3 & c_{b2} &= 0.622 \\ \kappa &= 0.41 & c_{w2} &= 0.3 & c_{w3} &= 2.0 \\ c_{v1} &= 7.1 & c_{v2} &= 0.7 & c_{v3} &= 0.9 \\ c_{w1} &= \frac{c_{b1}}{\kappa^2} + \frac{1+c_{b2}}{\sigma}. \end{aligned}$$

3.3.3 Wall-Distance Calculation via the Relaxed Eikonal Equation

A major computational challenge in the SA model is the calculation of the wall distance y. In a standard static CFD simulation, y is purely geometric and can be pre-computed. However, to ensure this verification solver translates into the topology optimization code, where solid boundaries evolve continuously, the implementation uses a PDE-based approach.

The true geometric wall distance satisfies the standard Eikonal equation, $|\nabla y| = 1$. However, this formulation contains singularities at the medial axes of the domain. To guarantee continuous differentiability, a relaxed reciprocal-distance equation proposed by Yoon [63] is solved:

$$\nabla G \cdot \nabla G + \sigma_w G(\nabla \cdot \nabla G) = (1 + 2\sigma_w)G^4. \tag{3.18}$$

Here, G is an intermediate reciprocal-distance field variable. The parameter σ_w is a relaxation constant, typically set to 0.1, that controls the smoothness of the field. The PDE is subjected to a Dirichlet boundary condition at the physical walls, $G = G_0$, where G_0 is an arbitrarily large constant. Once G is solved over the domain, the physical wall distance y is recovered algebraically using the inverse relationship:

$$y = \frac{1}{G} - \frac{1}{G_0}. \tag{3.19}$$

The code applies small numerical floors and non-negativity safeguards during this recovery to prevent division-by-zero and roundoff artifacts, but the mathematical equation used in the thesis remains the Yoon formulation above.

This system is implemented as a nonlinear weak residual for G. Following the Galerkin statement of Yoon's relaxed wall-distance equation [63], the PDE is multiplied by a scalar test function and the $\sigma_w G\nabla^2 G$ term is integrated by parts, so that the residual contains only first derivatives of G:

$$(1-\sigma_w)\int_\Omega |\nabla G|^2 z\,dx - \sigma_w \int_\Omega G\nabla G \cdot \nabla z\,dx - (1+2\sigma_w)\int_\Omega G^4 z\,dx = 0. \tag{3.20}$$

3.4 Picard-IPCS Coupling Algorithm

The solver coupling is independent of the particular verification geometry, so the iteration scheme is part of the implementation rather than the benchmark definition.

For fixed-wall verification cases, the relaxed Eikonal problem is solved first using Eq. 3.20, giving the wall-distance field y used by the SA closure. The coupled RANS-SA state is then advanced by the outer Picard loop summarized in Figure 3.1.

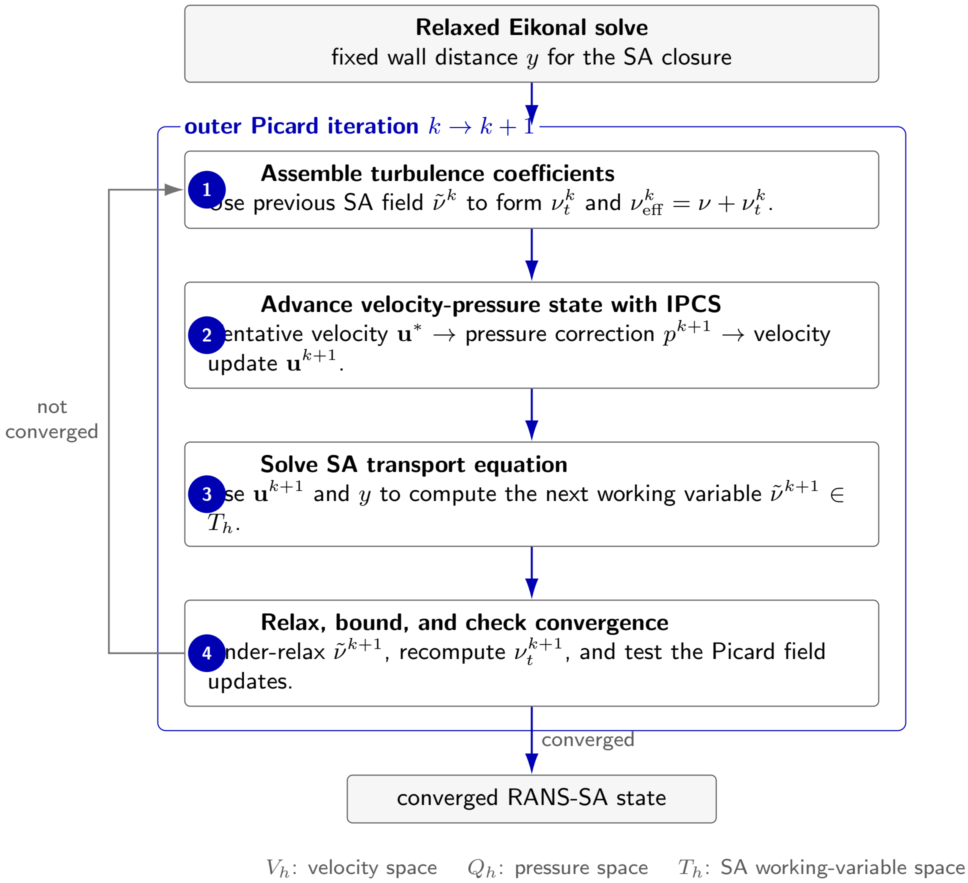


Figure 3.1: Nested Picard-IPCS coupling used by the FEniCS RANS-SA solver. The eddy viscosity ν_t is assembled from the previous SA working variable $\tilde{\nu}$, the velocity-pressure pair is advanced by the IPCS weak problems, and the updated velocity then drives the next SA transport solve.

Inside Picard iteration k, the eddy viscosity ν_t^k is assembled from the previous SA working variable $\tilde{\nu}^k$ and inserted into the effective viscosity $\nu + \nu_t^k$ of the tentative velocity problem. The velocity-pressure pair is then advanced through the IPCS weak problems in Eqs. 3.7–3.9. The resulting velocity field $\mathbf{u}^{k+1}$ is used in the SA weak transport solve, Eq. 3.11, optionally with the pseudo-time term in Eq. 3.12 and the SUPG term in Eq. 3.13. After the SA solve, $\tilde{\nu}^{k+1}$ is under-relaxed and bounded, ν_t^{k+1} is recomputed, and the field updates are checked against the prescribed Picard tolerance.

The pressure-drop and profile-error definitions are intentionally not part of this algorithmic section. They depend on the benchmark geometry and boundary conditions, so they are introduced with the channel and U-bend verification cases in Chapter 4.

3.5 Conclusion

This chapter has established the numerical formulation of the FEniCS-SA forward solver used for the verification benchmarks. The key ingredients are the weak-form finite-element implementation, Picard decoupling of the velocity-pressure and turbulence fields, IPCS pseudo-time stepping for the incompressible flow solve, the steady SA transport equation, and the relaxed Eikonal wall-distance formulation.

Separating these methods from their verification keeps the thesis logic clean: this chapter explains how the forward SA solver is constructed, Chapter 4 verifies its turbulent-flow behavior against Ansys Fluent benchmark cases, and Chapter 5 extends the verified forward solver to density-based topology optimization and frozen adjoint sensitivities.

Chapter 4

Verification of the Spalart-Allmaras Solver for Turbulent Internal Flows

4.1 Introduction

Chapter 3 introduced the numerical implementation of the Spalart-Allmaras (SA) model in FEniCS. Before using this solver inside the topology optimization framework, the forward fluid solver is verified against independent Ansys Fluent calculations on two benchmark flows. The first benchmark is a fully developed turbulent channel, which isolates wall damping, turbulence production, and Picard coupling in a simple geometry. The second benchmark is a curved U-bend, which tests the same implementation under strong convection, streamline curvature, and separation.

This chapter keeps the essential benchmark definitions in the main text: geometry type, boundary conditions, Reynolds number, wall resolution, and the comparison metrics used to judge agreement. The appendices retain the exhaustive mesh hierarchies, solver tolerances, and full configuration tables. This separation keeps the verification argument readable while still making the numerical setup reproducible. The purpose is to verify the forward-flow solver developed in Chapter 3, because this is the turbulent-flow core on which the topology optimization framework is built. Chapter 5 later adds Brinkman penalization, topology-dependent SA and wall-distance penalties, and a frozen-adjoint sensitivity residual, but those additions build on the same SA transport equation, wall-distance computation, and segregated flow-coupling strategy verified here.

All mesh-independence relative differences referenced from Appendix B follow the appendix convention: coarse–medium differences are normalized by the coarse value, while medium–fine differences are normalized by the fine value.

4.2 Channel Verification Benchmark

To verify the fundamental correctness of the custom fluid and turbulence solvers, the code was first tested on a standard 2D turbulent channel flow. To cleanly isolate and verify turbulence production, wall damping, and Picard coupling without the artifacts of a developing boundary layer, this benchmark was formulated as a pressure-driven periodic channel rather than a conventional inlet-outlet duct.

The velocity field ($\mathbf{u}$) and Spalart-Allmaras working variable ($\tilde{\nu}$) were constrained periodically between the streamwise boundaries, while the top and bottom boundaries were treated as standard no-slip walls ($\mathbf{u} = 0, \tilde{\nu} = 0$). Flow was driven entirely by a fixed pressure drop from the upstream plane ($p = 2$ Pa) to the downstream plane ($p = 0$ Pa). Based on a reference bulk velocity of $U_{\text{ref}} = 18.5$ m/s and using the full channel height ($H = 2.0$ m) as the characteristic length scale, the solver was targeted to resolve a fully turbulent flow regime at a channel Reynolds number of $Re_H \approx 20{,}300$. The complete definition of the geometric, physical, and boundary-condition parameters is provided in Table B.1 of Section B.1.1 in Appendix B.

The same finite-element implementation choices are used for both verification benchmarks: the momentum equation is assembled with the symmetric effective-viscosity stress form, while SUPG stabilization is restricted to the SA working-variable transport equation.

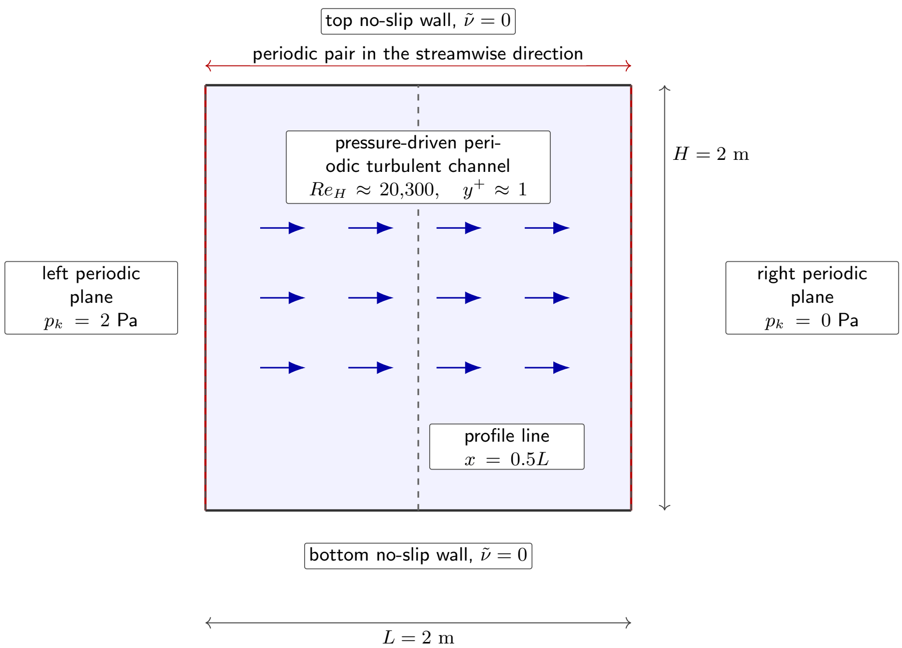


FIGURE 4.1: Schematic of the pressure-driven periodic channel verification setup. The vertical dashed line marks the mid-plane profile extraction location used for the FEniCS–Ansys comparison.

4.2.1 Mesh Resolution Requirements

Because the SA formulation is a low-Reynolds model that does not use empirical wall functions, it requires mathematical integration directly through the viscous sublayer. To satisfy this strict boundary-layer requirement, a custom structured inflation mesh was generated for the channel flow. Because the domain is a simple periodic rectangle, this anisotropic grid was constructed programmatically using NumPy arrays and converted into a FEniCS-compatible format via the `meshio` library. Based on the target bulk Reynolds number, this approach enforced a physical first-cell height of approximately 1.7×10^{-3} m, aiming for a non-dimensional wall distance of $y^{+} \approx 1$ across the entire domain.

To verify that the numerical results were not dominated by spatial discretization, a grid-independence study was conducted. The full spatial verification is documented in Section B.1.3 of Appendix B. The mesh hierarchy is detailed in Table B.3, and visual confirmation of convergence is provided by the velocity profiles in Figure B.1.

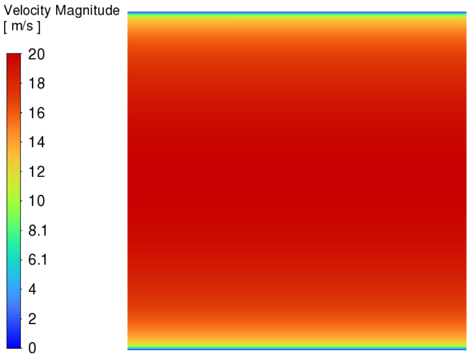
4.2.2 FEniCS vs. Ansys Comparison

Prior to this comparative analysis, the Ansys baseline itself was verified for grid independence to ensure a robust reference (see Table B.4 and Figure B.2 in Section B.1.3 of Appendix B). The channel flow was then simulated in FEniCS using the decoupled Picard iteration method and compared against the Ansys Fluent baseline.

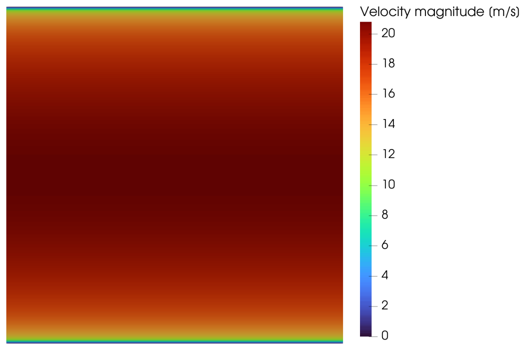
As shown in Figure 4.2, the qualitative velocity contours of the custom FEniCS implementation show good agreement with the Ansys Fluent baseline. Both solvers produce a fully developed, streamwise-uniform channel field with zero velocity at the walls and a broad high-speed core around the centerline. The remaining visual differences are confined mainly to the thickness of the near-wall color bands and the exact width of the central high-velocity region, which are expected to be sensitive to mesh resolution, plotting scale, and the different finite-element and finite-volume discretizations. For the quantitative comparison, velocity profiles are extracted from both solvers across the middle of the channel, i.e. at $x = 0.5L$, with y varied from the lower wall to the upper wall. The custom solver resolves the logarithmic region of the turbulent boundary layer and predicts the turbulent diffusion toward the channel center.

Figure 4.3 shows the corresponding modified turbulent viscosity field. The two contours have the same physical structure: $\tilde{\nu}$ is suppressed at the no-slip walls by the SA boundary condition and wall-destruction term, then rises rapidly through the buffer/log region toward an almost uniform core value. The FEniCS field is slightly smoother in the wall-normal transition, but it preserves the same symmetry and wall-to-core trend as the Ansys reference. This agreement is important because $\tilde{\nu}$, rather than velocity alone, controls the eddy-viscosity field used by the momentum equation.

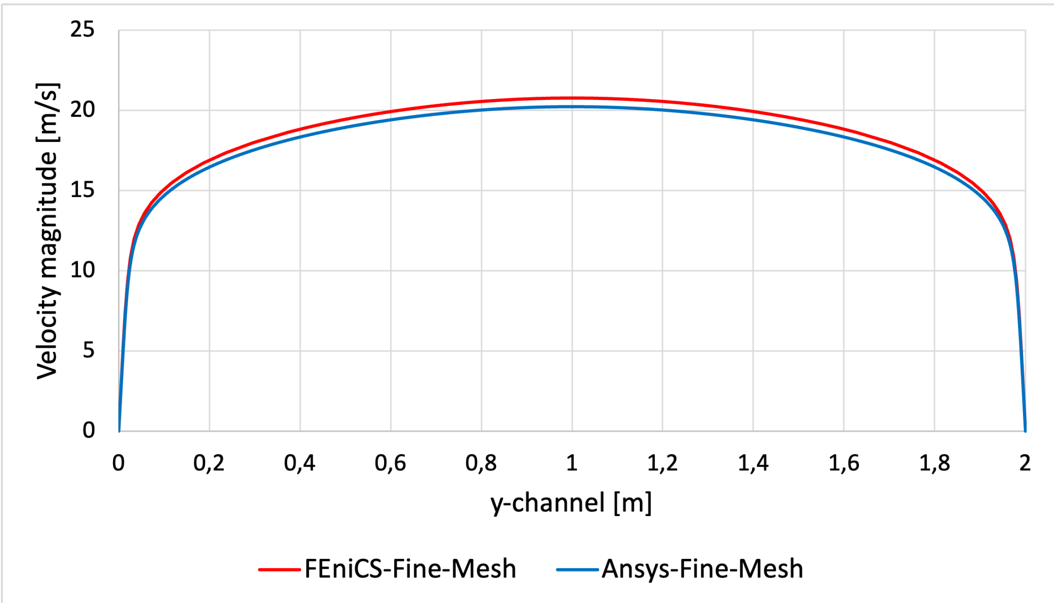
Because the channel is driven by a prescribed pressure difference, $\Delta p = 2.0$ Pa is identical by construction and is not a useful comparison metric. The retained global



(A) Ansys Fluent SA contour for u

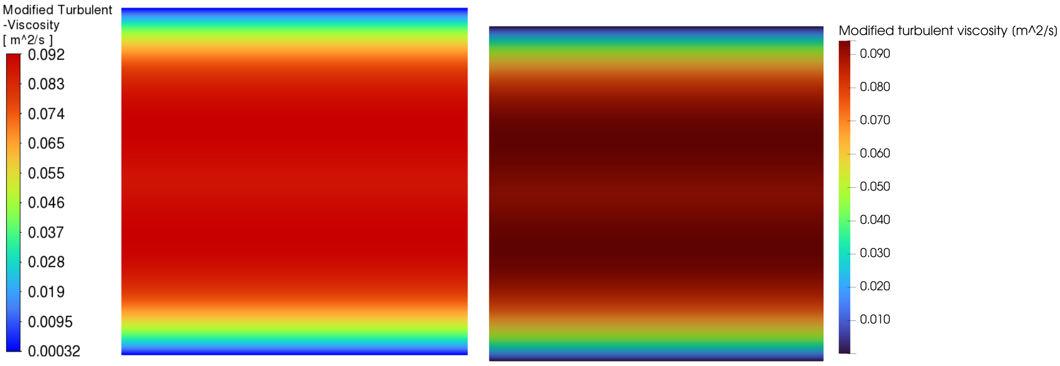


(B) FEniCS SA contour for u



(C) Velocity profiles extracted at the streamwise mid-plane of the channel

FIGURE 4.2: Verification of the 2D channel flow ($Re_H \approx 20{,}300$). The comparison includes qualitative velocity contours from Ansys Fluent and FEniCS, plus quantitative velocity profiles extracted at the streamwise mid-plane, $x = 0.5L$, while varying the wall-normal coordinate y.



(A) Ansys Fluent SA contour for $\tilde{\nu}$ (B) FEniCS SA contour for $\tilde{\nu}$

Figure 4.3: Comparison of the modified turbulent viscosity $\tilde{\nu}$, i.e. the Spalart-Allmaras transported working variable, for the 2D channel benchmark. Both solvers drive $\tilde{\nu}$ to zero at the no-slip walls and recover a nearly streamwise-uniform turbulent core.

metric is therefore the bulk velocity,

$$U_{\text{bulk}} = \frac{1}{H} \int_0^H u_x(0.5L, y)\, \mathrm{d}y, \tag{4.1}$$

and the corresponding plane-channel skin-friction coefficient,

$$C_f = \frac{\tau_w}{\frac{1}{2}\rho U_{\text{bulk}}^2} = \frac{\Delta p\, H}{\rho L U_{\text{bulk}}^2}. \tag{4.2}$$

Here u_x is the streamwise velocity component, H is the channel height, L is the streamwise periodic length, τ_w is the mean wall shear stress, ρ is the fluid density, and Δp is the imposed pressure difference. When the solver pressure is treated as kinematic pressure, the same expression is evaluated consistently with $\rho = 1$. Since Δp is fixed, C_f is not independent of U_{bulk}, but it is still useful because it converts the imposed forcing into the wall friction implied by the predicted flow rate.

The two error metrics are cross-code discrepancies, not separate errors for each solver. The exported fine-mesh profiles provide velocity magnitude along the channel diameter; for this fully developed channel this is equivalent to the streamwise velocity within the exported precision. After interpolating the FEniCS profile onto the Ansys line coordinates y_i, the pointwise difference is

$$e_i = u_{x,\text{FEniCS}}(y_i) - u_{x,\text{Ansys}}(y_i). \tag{4.3}$$

The maximum pointwise velocity discrepancy is normalized by the maximum Ansys profile velocity to avoid singular relative errors at the no-slip walls,

$$E_\infty = \frac{\max_i |e_i|}{\max_i |u_{x,\text{Ansys}}(y_i)|} \times 100\%, \tag{4.4}$$

and the relative profile error is evaluated as a line-integrated norm,

$$E_{L_2} = \left(\frac{\int_0^H |u_{x,\text{FEniCS}} - u_{x,\text{Ansys}}|^2 \, \mathrm{d}y}{\int_0^H |u_{x,\text{Ansys}}|^2 \, \mathrm{d}y} \right)^{1/2} \times 100\%. \tag{4.5}$$

These metrics are useful compact supplements when exported FEniCS and Ansys profile arrays are available. They should not be computed from visually digitized plots, because plot digitization and axis scaling would introduce an uncertainty unrelated to the solver comparison. For the fine-mesh channel profiles, the FEniCS data are sampled at 1001 uniformly spaced diameter coordinates, while the Ansys data are sampled at 190 exported diameter coordinates. Three duplicate Ansys coordinates have identical velocity values and are collapsed before integration. The FEniCS profile is then linearly interpolated onto the 187 unique Ansys coordinates, and Eq. (4.5) is evaluated with a trapezoidal line integral over those coordinates.

TABLE 4.1: Macroscopic flow metrics and fine-mesh profile discrepancies for the 2D channel benchmark. The profile-error norms are evaluated from the exported Ansys and FEniCS diameter profiles, with the FEniCS profile interpolated onto the unique Ansys coordinates.

Metric	**Ansys**	**FEniCS**	**Discrepancy**
Bulk velocity (U_{bulk}) [m/s]	17.977	18.458	2.67%
Friction coefficient (C_f)	6.188×10^{-3}	5.870×10^{-3}	5.14%
Maximum profile discrepancy (E_∞)	fine-mesh velocity profile		2.66%
Relative profile norm (E_{L_2})	fine-mesh velocity profile		2.64%

This agreement supports the verification that the core Navier-Stokes solver, relaxed Eikonal wall-distance field, and SA production/destruction terms behave correctly in a canonical wall-bounded turbulent flow. The main-text parameters are sufficient to interpret the result, while Appendix B provides the detailed mesh and solver settings needed to reproduce it.

4.3 U-Bend Verification Benchmark

Following the channel flow verification, the solver was tested on the highly convective separating U-bend flow evaluated in Chapter 2. This case is closer to the curved internal-flow topologies studied later in the thesis because the flow experiences strong streamline curvature and pressure-gradient-driven redistribution.

The verification used a 2D geometry corresponding to the symmetry mid-plane of the full 3D VMFL048 case. Since the FEniCS domain is mathematically a 2D planar approximation, it cannot resolve true 3D Dean secondary vortices. Instead, this benchmark validates the 2D symmetry-plane approximation and the solver's ability to capture curvature-induced separation and momentum redistribution. The straight inlet and outlet legs are shorter than in the original VMFL048 Ansys mesh to reduce meshing and computational effort; both the FEniCS and Ansys verification runs use this same short-leg U-bend geometry.

To drive the flow, a uniform prescribed velocity ($\mathbf{u}_{in} = 1.42$ m/s) was imposed at the inlet alongside a fully turbulent boundary value ($\tilde{\nu}_{in} = 9.34 \times 10^{-5}$ m^2/s), while a zero relative static pressure ($p = 0$ Pa) was assigned to the outlet. While the domain is 2D, the 3D physical pipe diameter ($D = 0.028$ m) and radius of curvature ($R_c = 0.125$ m) were retained as the characteristic length scales. This preserves the target Reynolds number ($Re_D \approx 44{,}700$) and Dean number ($De \approx 15{,}000$) of the 3D Ansys reference case. The physical setup and boundary conditions are listed in Table B.5 of Section B.2.1 in Appendix B.

The U-bend is significantly more convection dominated than the periodic channel benchmark, but it uses the same SA-transport SUPG stabilization described in Chapter 3. This stabilization is applied only to the transported working variable $\tilde{\nu}$; no additional SUPG term is introduced in the momentum equation.

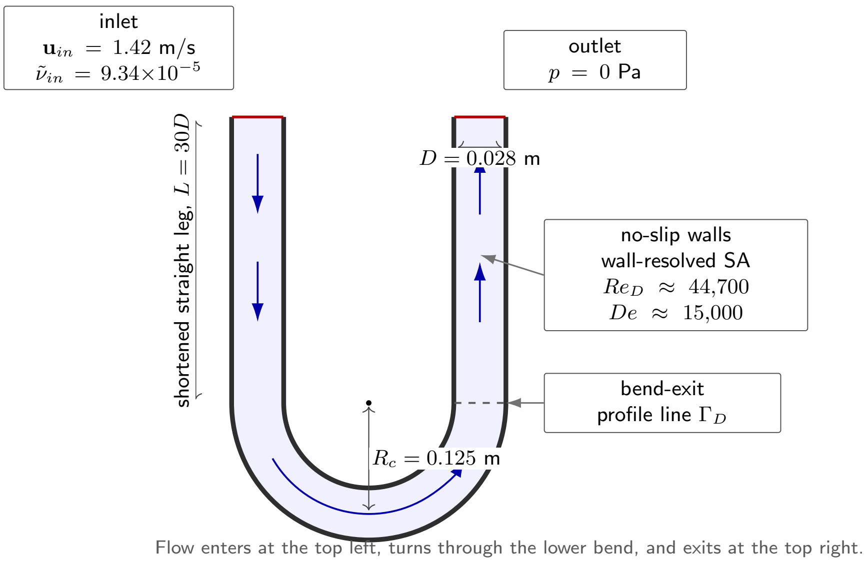


Figure 4.4: Schematic of the shortened 2D U-bend verification setup based on the VMFL048 operating point from the Ansys verification manual [8]. The profile line Γ_D is used for the bend-exit velocity comparison.

4.3.1 Mesh Resolution Requirements

The U-bend geometry features curved inner and outer arcs, complex corner transitions, and anisotropic boundary-layer inflation around non-orthogonal walls. To manage this geometric complexity while satisfying the $y^+ \approx 1$ resolution requirement, a custom mesh was generated using Gmsh [27]. The first element height adjacent to the wall was controlled to 1.20×10^{-5} m across the entire domain. The independence studies are documented in Section B.2.3 of Appendix B: the inflation parameters and element densities of the Gmsh hierarchy are listed in Table B.7, and the extracted velocity profiles confirming grid convergence at the bend exit are illustrated in Figure B.3.

4.3.2 FEniCS vs. Ansys Comparison

Having validated the spatial resolution of the FEniCS mesh, an equivalent grid-independence study was executed for the commercial Ansys Fluent baseline (detailed in Table B.8 and Figure B.4 of Section B.2.3 in Appendix B). With both solvers operating on spatially converged grids, an independent Ansys Fluent simulation was executed on the shortened 2D domain.

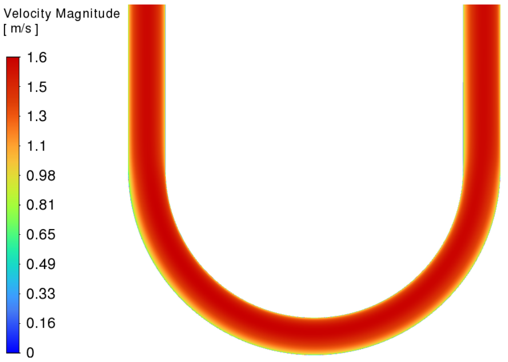
As shown in Figure 4.5, the global flow fields and separation regions observed in the qualitative velocity contours demonstrate visual agreement. Both solvers recover the acceleration around the inner part of the bend, the momentum redistribution toward the outer radius, and the low-speed regions associated with the no-slip boundary layers. The most relevant differences are local rather than topological: small shifts in the separated or low-momentum region and in the near-wall velocity gradients are expected because the FEniCS case is a 2D finite-element approximation of a curved internal flow, whereas the Ansys reference uses a finite-volume formulation on its own mesh. To provide a more robust assessment, velocity data are extracted along a cross-sectional line probe positioned at the bend exit, spanning the full pipe diameter $D = 0.028$ m from the inner wall to the outer wall. The quantitative velocity profile computed by the custom FEniCS implementation traces the commercial Ansys results closely, adequately capturing the boundary-layer growth and the momentum shift toward the outer wall.

Figure 4.6 provides the companion check on the SA transport variable. The similarity sought here is not pointwise identity of every contour band, but the same physical organization: low $\tilde{\nu}$ at all walls, elevated $\tilde{\nu}$ in the bend core and downstream leg, and smooth convection of the turbulent working variable through the curved section. Differences in peak value or band thickness indicate local discrepancies in modeled turbulence production, destruction, and numerical diffusion, but the shared spatial pattern verifies that the wall-distance field and SA transport equation remain coupled to the U-bend flow in the same qualitative way as the Ansys baseline.

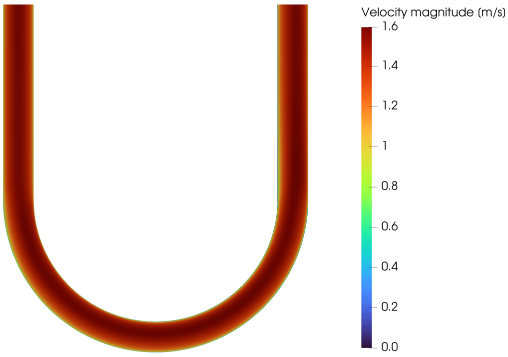
For the U-bend, the inlet velocity is prescribed rather than produced by a fixed pressure gradient, so pressure drop remains a meaningful global comparison metric. A channel-style friction coefficient is not retained because the curved geometry does not have a single fully developed wall-shear relation equivalent to the periodic channel. In FEniCS, the pressure-drop diagnostic is evaluated from boundary-measure-weighted mean pressures on the inlet and outlet facets,

$$\Delta p_{\mathrm{FEniCS}} = \rho \left[\frac{1}{|\Gamma_{\mathrm{in}}|} \int_{\Gamma_{\mathrm{in}}} p_k \, \mathrm{d}\Gamma - \frac{1}{|\Gamma_{\mathrm{out}}|} \int_{\Gamma_{\mathrm{out}}} p_k \, \mathrm{d}\Gamma \right], \tag{4.6}$$

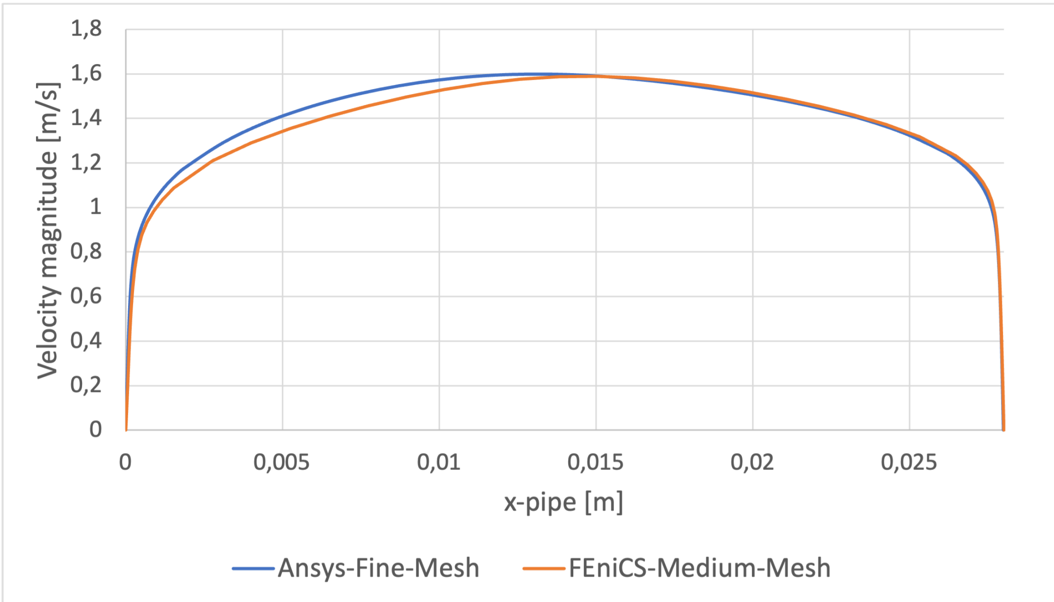
where p_k is the kinematic pressure solved by the IPCS formulation, $\rho = 997$ kg/m^3, and $|\Gamma| = \int_\Gamma 1 \, \mathrm{d}\Gamma$. The inlet and outlet integrals correspond to the marked FEniCS facets `ds(2)` and `ds(3)`, respectively, so the reported value is the area-averaged inlet static pressure minus the area-averaged outlet static pressure, converted to pascals.



(A) Ansys Fluent SA contour for u

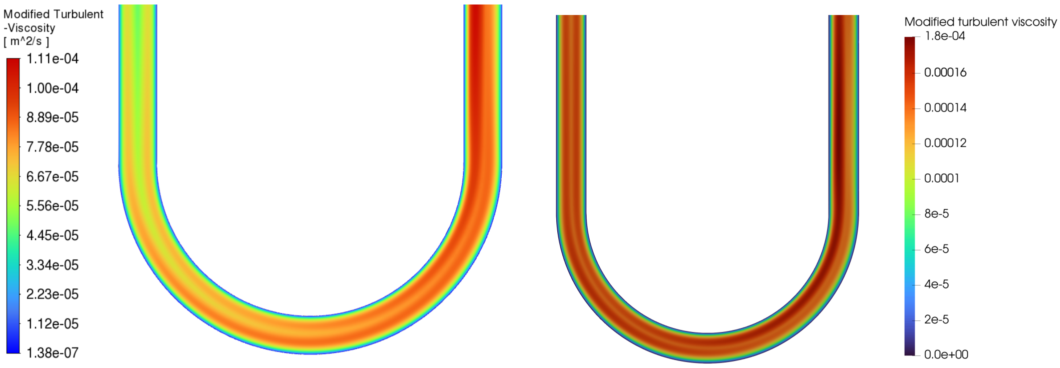


(B) FEniCS SA contour for u



(C) Velocity profiles extracted at the bend exit

Figure 4.5: Verification of the 2D U-bend flow ($Re_D \approx 44{,}700$, $De \approx 15{,}000$). The comparison includes qualitative velocity contours from Ansys Fluent and FEniCS, showing separation characteristics, plus quantitative velocity profiles extracted at the bend exit across the pipe diameter $D = 0.028$ m.



(A) Ansys Fluent SA contour for $\tilde{\nu}$ (B) FEniCS SA contour for $\tilde{\nu}$

FIGURE 4.6: Comparison of the modified turbulent viscosity $\tilde{\nu}$ for the 2D U-bend benchmark. The fields are expected to vanish at the no-slip walls and increase in the curved core where shear, convection, and adverse pressure-gradient effects transport the SA working variable through the bend.

If coordinate-matched bend-exit profiles are exported, the same local comparison metrics used for the channel are computed as

$$E_\infty = \frac{\max_i \left||\mathbf{u}_{\text{FEniCS}}(s_i)| - |\mathbf{u}_{\text{Ansys}}(s_i)|\right|}{\max_i |\mathbf{u}_{\text{Ansys}}(s_i)|} \times 100\%, \tag{4.7}$$

and

$$E_{L_2} = \left(\frac{\int_{\Gamma_D} \left(|\mathbf{u}_{\text{FEniCS}}| - |\mathbf{u}_{\text{Ansys}}|\right)^2 \, \mathrm{d}s}{\int_{\Gamma_D} |\mathbf{u}_{\text{Ansys}}|^2 \, \mathrm{d}s} \right)^{1/2} \times 100\%, \tag{4.8}$$

where Γ_D denotes the line segment across the bend-exit diameter and s is the corresponding arclength coordinate. As in the channel case, these profile norms are useful only when evaluated from the exported profile arrays rather than from visual plot digitization. Since the FEniCS formulation solves for kinematic pressure, any retained pressure-drop value must be converted to physical static pressure using $\rho = 997$ kg/m^3 before comparison; equivalently, the fine Ansys pressure drop of 792.995 Pa corresponds to $\Delta p/\rho = 0.7954$ m^2/s^2.

TABLE 4.2: Macroscopic pressure-drop comparison and exported bend-exit profile discrepancies for the 2D U-bend benchmark. The profile-error norms use the fine Ansys profile and the medium FEniCS profile, with the FEniCS profile interpolated onto the unique Ansys diameter coordinates.

Metric	**Ansys**	**FEniCS**	**Discrepancy**
Pressure drop (Δp) [Pa]	792.995	815.130	2.79%
Maximum profile discrepancy (E_∞)	bend-exit velocity profile		8.50%
Relative profile norm (E_{L_2})	bend-exit velocity profile		2.75%

4.4 Conclusion

The two verification cases answer the second research question by establishing the practical operating range of the custom FEniCS-SA solver before it is embedded in the topology optimization loop. The periodic channel confirms that the solver reproduces canonical wall-bounded turbulent behavior, including the near-wall damping and production balance that are central to the SA model. On the fine mesh, the channel bulk velocity differs from the Ansys reference by 2.67%, while the exported diameter profile gives $E_\infty = 2.66\%$ and $E_{L_2} = 2.64\%$. These numbers do not imply pointwise equivalence with a commercial CFD code, but they show that the solver captures the dominant velocity scale and wall-bounded profile shape with sufficient accuracy for use as a forward-flow model in topology optimization. The U-bend then checks the same implementation in a curved, separation-prone internal flow, where streamline curvature and adverse pressure gradients make the forward-flow prediction more demanding than in the channel.

These results should therefore be interpreted as a targeted verification of the forward solver, not as a claim that the custom implementation perfectly aligns with Ansys Fluent. The known limitations of a 2D SA model remain: secondary Dean vortices, anisotropic Reynolds stresses, and full three-dimensional separation physics are outside the present solver scope. Within that scope, however, the implemented SA equations, wall-distance calculation, and segregated coupling strategy produce stable and physically plausible turbulent internal-flow fields that are accurate enough to support the pressure-drop and dissipation objectives used in the subsequent topology optimization campaigns. The separate sensitivity checks in Appendix C then verify the additional frozen-adjoint derivative used by MMA.

Chapter 5

Turbulent Topology Optimization Methodology

5.1 Introduction

Building upon the Spalart-Allmaras (SA) fluid dynamics solver [50, 51] developed in Chapter 3 and verified in Chapter 4, this chapter introduces the mathematical framework required to perform Topology Optimization (TO) in the turbulent regime. The present chapter extends the verified forward solver by embedding the same SA transport model and reciprocal wall-distance machinery in a density-based design loop and differentiating the resulting residuals for gradient-based optimization.

The core objective of this methodology is to algorithmically determine the optimal distribution of fluid channels within a given design domain to reduce hydraulic or mechanical energy loss under a prescribed flow condition. This chapter details the density-based fictitious-domain formulation used to represent solid and fluid regions on a fixed computational mesh [14, 26]. It then explains the turbulent-flow ingredients retained for the remainder of the thesis: Brinkman momentum resistance, topology penalties for the SA and wall-distance equations, the reciprocal penalized wall-distance formulation, frozen-turbulence adjoint sensitivities, active-domain filtering, Heaviside projection, and MMA updates.

The topology-optimization code is therefore not a separate turbulence solver. It uses the verified Chapter 3 forward-flow machinery as its starting point and augments it with design-dependent physics. During the optimization loop, IPCS solves are used as robust warm starts and Picard updates. For the adjoint step, the final frozen-viscosity state is assembled as a steady weak residual with Brinkman resistance, dynamic viscosity, and the symmetric strain tensor. This residual is the differentiable state equation used for design sensitivities, while the underlying SA transport and wall-distance updates remain the same forward-model components verified in Chapter 4.

5.2 The Fictitious Domain Method and Brinkman Penalization

In standard shape optimization, the solid-fluid boundary is explicitly defined by the mesh, and the nodes are physically moved to improve the design. In density-based topology optimization, however, the computational domain (Ω) remains strictly fixed. To create solid walls and fluid channels within this fixed grid, the Fictitious Domain Method is employed.

Originally applied to Stokes flow by Borrvall and Petersson [14] and extended to the Navier-Stokes equations by Gersborg-Hansen et al. [26], this method assigns a continuous design variable to the spatial coordinates in the domain. In the convention used here, a physical topology value of $\bar{\gamma} = 1$ represents a pure **fluid** channel, while $\bar{\gamma} = 0$ represents a **solid**, impermeable structure. To force the fluid velocity to zero inside solid regions without mathematically removing the finite elements, the fluid is treated as a fictitious porous medium.

This flow resistance is quantified by an inverse permeability term, or Brinkman penalty function, denoted as $\alpha(\bar{\gamma})$. To ensure the optimization algorithm can transition smoothly between fluid and solid states, α must be continuous and differentiable. The formulation uses a convex Rational Approximation of Material Properties (RAMP) interpolation [53] scaled between a maximum solid penalty (α_{solid}) and a minimum fluid penalty (α_{fluid}):

$$\alpha(\bar{\gamma}) = \alpha_{solid} + (\alpha_{fluid} - \alpha_{solid}) \frac{\bar{\gamma}(1+q)}{\bar{\gamma} + q} \tag{5.1}$$

where α_{solid} is a sufficiently large penalty factor that stagnates the flow when $\bar{\gamma} = 0$, and q is a tuning parameter controlling the convexity of the penalization curve. Equation 5.1 is evaluated using the projected effective density $\bar{\gamma}$, rather than the raw optimizer variable γ, because the physical flow equations are formulated on the filtered, projected, and domain-bounded topology. For visualization, the Brinkman resistance is normalized as

$$\hat{\alpha}(\bar{\gamma}) = \frac{\alpha(\bar{\gamma}) - \alpha_{fluid}}{\alpha_{solid} - \alpha_{fluid}},$$

so that $\hat{\alpha} = 1$ in a fully solid cell and $\hat{\alpha} = 0$ in a fully fluid cell. If $\alpha_{fluid} = 0$, this reduces to $\alpha(\bar{\gamma})/\alpha_{solid}$.

Figure 5.1 shows the normalized resistance curve for increasing values of q. Small values of q create a very steep drop in resistance immediately away from the solid end ($\bar{\gamma} = 0$), which makes the curve almost flat near the fluid end ($\bar{\gamma} = 1$). Larger values of q spread the transition more evenly across intermediate densities. During continuation, the RAMP parameter is adjusted together with the projection steepness to control how aggressively intermediate densities are penalized while keeping the interpolation differentiable. Equivalent Brinkman interpolation curves are sometimes written with a reciprocal penalty parameter, e.g. $\hat{\alpha} = (1 - \bar{\gamma})/(1 + Q\bar{\gamma})$ for $\alpha_{fluid} = 0$, which corresponds to $Q = 1/q$ in the present convention. Unlike traditional structural optimization, which often relies on the SIMP power law [13], fluid optimization generally avoids SIMP for momentum because its derivative can become ineffective near the solid limit. The RAMP formulation retains a useful gradient across the active design domain [53, 55].

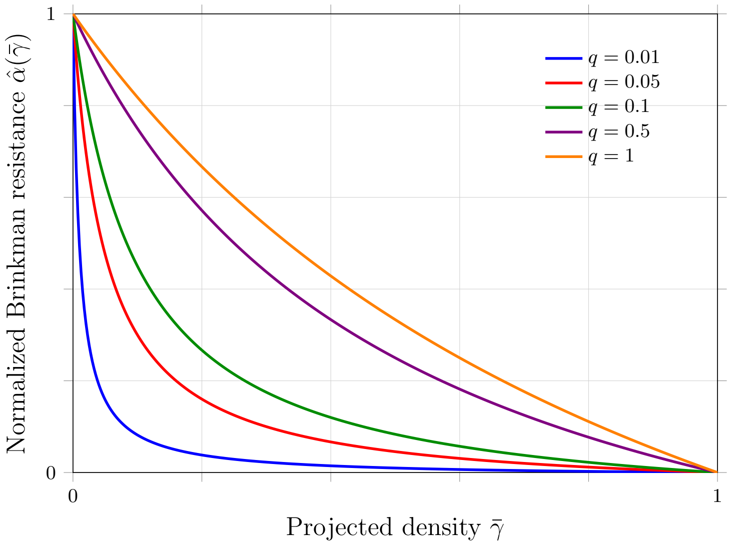


FIGURE 5.1: Normalized RAMP Brinkman resistance from Eq. 5.1 for $q = \{0.01, 0.05, 0.1, 0.5, 1.0\}$, adapted from Stolpe and Svanberg [53].

5.2.1 The Density Chain and Domain Bounding

While conceptually represented by the physical topology field in the governing equations, the layout is managed through a distinct chain of density representations to handle design and non-design domains. To explicitly avoid mathematical confusion with the fluid mass density (ρ) utilized in the Reynolds-Averaged Navier-Stokes equations, this thesis exclusively utilizes the γ notation for all topological variables:

- **Raw design density** (γ): the baseline parameter manipulated by the optimizer.
- **Filtered density** ($\tilde{\gamma}$): the spatially smoothed design field.
- **Projected effective density** ($\bar{\gamma}$): the bounded physical topology field evaluated by the governing equations.

The raw design density, γ, is defined as a Discontinuous Galerkin (DG0) [29] function over the entire mesh. The optimization algorithm strictly manipulates the degrees of freedom corresponding to the actively optimized design space, while passive non-design cells are subsequently restored to their prescribed bounding values.

This raw variable is subjected to a spatial PDE filter [34, 15] to yield a smoothed field $\tilde{\gamma}$. To recover sharp boundaries and strictly preserve required non-design regions, the filtered density is then projected and mapped into the bounded effective density ($\bar{\gamma}$) [28, 60]:

$$\bar{\gamma} = \gamma_{\text{lower}} + (\gamma_{\text{upper}} - \gamma_{\text{lower}})\Pi_{\beta}(\tilde{\gamma}) \tag{5.2}$$

where Π_β is the projection operator, and γ_{lower} and γ_{upper} define the absolute permissible bounds for each cell. This bounding guarantees that predefined passive-fluid features, such as non-design inlet and outlet extensions ($\gamma_{\text{lower}} = \gamma_{\text{upper}} = 1$), remain fixed without interfering with the mathematical behavior of the active optimization space.

5.2.2 Design and Non-Design Domains

The computational mesh is partitioned into active design cells and passive fluid non-design cells. The active design domain contains the cells whose raw densities are updated by MMA. Fluid non-design domains are fixed as permanent fluid, typically to preserve inlet and outlet extensions, buffer channels, or measurement regions.

These passive fluid cells still participate in the forward flow, turbulence, and wall-distance solves. They are excluded only from the optimization masks: the volume constraint is evaluated over the prescribed volume mask Ω_V, and any domain-integrated post-processing metric is evaluated over its prescribed integration mask. Consequently, prescribed inlet/outlet buffers do not count toward the optimized fluid fraction and do not directly bias reported domain-integrated metrics.

This active/passive partition is used by the active-mask-normalized filtering step introduced in Section 5.7, where the masking convention is shown together with the Helmholtz filtering chain.

5.3 Penalization of the Governing Equations

To optimize turbulent manifolds, the respective penalties must be integrated into the flow, turbulence, and wall-distance equations so that the physics remain consistent as solid boundaries evolve. All physical penalties use the projected effective density, $\bar{\gamma}$.

5.3.1 Modified Reynolds-Averaged Navier-Stokes Equations

The primary application of the Brinkman penalty is the modification of the steady-state, incompressible RANS momentum equation. The RAMP-interpolated inverse permeability $\alpha(\bar{\gamma})$ is added as a linear sink term to the momentum equation:

$$\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nabla \cdot \left[(\mu + \mu_t)\left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T\right)\right] - \alpha(\bar{\gamma})\mathbf{u}. \tag{5.3}$$

The topology-optimization residual retains the full symmetric rate-of-strain tensor. Since the turbulent eddy viscosity (μ_t) varies across the spatial domain [45], the effective viscosity is not constant and the transposed gradient term contributes to the momentum balance. In regions where the optimizer places solid material ($\bar{\gamma} \to 0$), the large magnitude of $\alpha_{solid}\mathbf{u}$ dominates the local balance and drives the velocity toward zero.

The final steady flow state used by the frozen adjoint is assembled as a weak residual. With $\mu_{\text{eff}} = \mu + \mu_t^{fr}$ and test functions $(\mathbf{v}, q)$, the implemented residual is: find $(\mathbf{u}, p)$ such that

$$R_{\text{NS}}((\mathbf{u},p),(\mathbf{v},q)) = \int_\Omega \rho(\mathbf{u}\cdot\nabla)\mathbf{u}\cdot\mathbf{v}\,dx + \frac{1}{2}\int_\Omega \mu_{\text{eff}}\left(\nabla\mathbf{u}+\nabla\mathbf{u}^T\right):\left(\nabla\mathbf{v}+\nabla\mathbf{v}^T\right)\,dx \\ + \int_\Omega \nabla p\cdot\mathbf{v}\,dx + \int_\Omega q\,\nabla\cdot\mathbf{u}\,dx + \int_\Omega \alpha(\bar{\gamma})\mathbf{u}\cdot\mathbf{v}\,dx = 0. \tag{5.4}$$

Equation 5.4 is the differentiable state equation used for the velocity-pressure adjoint. The forward Picard updates may use IPCS warm starts for robustness, but the final sensitivity solve differentiates this steady weak residual.

5.3.2 Modified Spalart-Allmaras Transport Equation

As the Fictitious Domain Method forces velocity to zero in solid regions, turbulence generation inside those regions should theoretically cease. However, numerical diffusion can allow the SA working variable $(\tilde{\nu})$ to bleed into solid regions. This unphysical behavior destabilizes the solver and distorts the adjoint sensitivity analysis.

To enforce a zero-turbulence state inside solid structures, a corresponding penalty sink term is added to the Spalart-Allmaras transport equation [63, 20]:

$$\mathbf{u}\cdot\nabla\tilde{\nu} = c_{b1}\tilde{S}\tilde{\nu} - c_{w1}f_w\left(\frac{\tilde{\nu}}{y}\right)^2 + \frac{1}{\sigma}\left[\nabla\cdot((\nu+\tilde{\nu})\nabla\tilde{\nu}) + c_{b2}(\nabla\tilde{\nu})^2\right] \\ - \alpha_{\tilde{\nu}}(1-\bar{\gamma})^{N_{\tilde{\nu}}}\tilde{\nu}. \tag{5.5}$$

The last term is the topology penalty used to suppress the SA working variable in solid material. Because $\bar{\gamma} = 1$ denotes fluid and $\bar{\gamma} = 0$ denotes solid, the penalty is applied to the local solid fraction $(1 - \bar{\gamma})$. The coefficient $\alpha_{\tilde{\nu}}$ sets the damping strength and the exponent $N_{\tilde{\nu}}$ controls how sharply the damping grows as a cell becomes solid. In the final benchmark computations these two parameters are held constant throughout the optimization.

In the Picard-linearized finite-element solve, the topology penalty enters as an additional reaction term in the weak SA equation. Using the previous Picard field $\tilde{\nu}^k$ to evaluate $\tilde{S}^k$, f_w^k, and the explicit source terms, the topology-aware SA step solves for $\tilde{\nu}^{k+1}$ from

$$\int_\Omega(\mathbf{u}\cdot\nabla\tilde{\nu}^{k+1})\xi\,dx + \frac{1}{\sigma}\int_\Omega(\nu+\tilde{\nu}^k)\nabla\tilde{\nu}^{k+1}\cdot\nabla\xi\,dx \\ + \int_\Omega c_{w1}f_w^k\frac{\tilde{\nu}^k}{y^2}\tilde{\nu}^{k+1}\xi\,dx + \int_\Omega \alpha_{\tilde{\nu}}(1-\bar{\gamma})^{N_{\tilde{\nu}}}\tilde{\nu}^{k+1}\xi\,dx \\ = \int_\Omega\left[c_{b1}\tilde{S}^k\tilde{\nu}^k + \frac{c_{b2}}{\sigma}\nabla\tilde{\nu}^k\cdot\nabla\tilde{\nu}^k\right]\xi\,dx. \tag{5.6}$$

This weak form makes the artificial solid damping explicit: as $\bar{\gamma}$ approaches zero, the reaction term forces the SA working variable toward zero in the same cells where the Brinkman term suppresses velocity.

Because the evolving topology can create sharp local changes in velocity, wall distance, and turbulence damping, the topology-optimization implementation keeps SUPG stabilization enabled for the SA working-variable transport equation. As in the verification solver, this stabilization is restricted to the $\tilde{\nu}$ transport solve and is not added to the momentum equation. Its role is numerical: it suppresses element-scale oscillations and overshoots in the convected SA working variable during Picard updates, while leaving the Spalart-Allmaras closure terms and the frozen-turbulence adjoint residual unchanged.

5.3.3 Reciprocal Penalized Wall-Distance Equation

The Spalart-Allmaras model is highly sensitive to the wall distance, y. While differential equations for wall distance computation are well-established [58], topology optimization introduces a unique challenge. The location of the wall is not only a static mesh boundary, but also the dynamically evolving $\bar{\gamma} = 0$ contour. Therefore, the relaxed Eikonal equation must be penalized to dynamically recognize fictitious solid as a physical boundary [63]:

$$\nabla G \cdot \nabla G + \sigma_w G(\nabla \cdot \nabla G) - (1 + 2\sigma_w)G^4 - \alpha_G(1 - \bar{\gamma})^{N_G}(G - G_0) = 0. \tag{5.7}$$

The wall-distance equation uses the same power-law topology concept. The coefficient α_G sets the strength of the artificial wall-distance forcing and the exponent N_G controls its growth through gray material. In fluid regions the term vanishes, while in topology-created solid regions it pulls G toward G_0. The algebraic recovery function

$$y = \frac{1}{G} - \frac{1}{G_0} \tag{5.8}$$

then collapses the SA wall distance toward zero inside topology-created solid regions while retaining the physical wall distance in fluid regions. As in the standalone solver, small numerical floors and non-negativity safeguards are used in the implementation to avoid division-by-zero artifacts, but they are not part of the mathematical model statement.

The corresponding nonlinear weak residual is obtained by multiplying Eq. 5.7 by a scalar test function z and integrating the $\sigma_w G\nabla^2 G$ contribution by parts:

$$\begin{aligned} R_G(G, z) =& (1 - \sigma_w) \int_\Omega |\nabla G|^2 z \, dx - \sigma_w \int_\Omega G\nabla G \cdot \nabla z \, dx - (1 + 2\sigma_w) \int_\Omega G^4 z \, dx \\ & - \int_\Omega \alpha_G(1 - \bar{\gamma})^{N_G}(G - G_0)z \, dx = 0. \end{aligned} \tag{5.9}$$

The last term is what turns topology-created solid into an artificial wall-distance boundary in the continuous-density setting.

5.4 Boundary Conditions and Turbulence Initialization

To stabilize the topology optimization and preserve physical interpretation, boundary constraints are managed across both fluid and turbulence variables. Non-design fluid buffer regions are mandated at inlets and outlets to isolate the active optimization domain from artificial numerical shear. The framework handles pressure outlets, optional velocity-component constraints, and pressure pin controls to fully constrain the discrete PDE problem.

At the domain inlets, the prescribed turbulence intensity and turbulence length scale are converted into an intermediate eddy-viscosity ratio. This ratio defines the inlet value for the Spalart-Allmaras working variable $\tilde{\nu}$. Throughout the computation, numerical floors, ceilings, and clipping functions are imposed to keep the working variable within stable physical limits during intermediate design cycles.

5.5 Sensitivity Analysis and the Frozen Turbulence Assumption

With the forward state equations fully defined and penalized, the topology optimization algorithm requires the gradient of the objective function with respect to the design variable. Because evaluating finite differences for every design cell is computationally impossible, an adjoint method is required [43, 44].

The implementation used in this thesis is best described as a **frozen-turbulence weak-form residual adjoint of the final steady flow problem**. It differentiates the finite-element residual assembled in UFL for the final frozen-viscosity state. It is not a tape-based differentiation of the entire IPCS/Picard forward algorithm, and it is not a fully coupled turbulence adjoint.

5.5.1 Reduced Adjoint Formulation

Let $\mathbf{s} = (\mathbf{u}, p)$ denote the final velocity-pressure state used for the design derivative. Under the frozen-turbulence approximation, the eddy viscosity ν_t^{fr} and reciprocal wall-distance field G^{fr} are treated as fixed coefficients during the sensitivity calculation. The final flow residual is written as

$$R(\mathbf{s}, \bar{\gamma}; \nu_t^{fr}, G^{fr}) = 0. \tag{5.10}$$

In the implementation, this abstract residual is the weak RANS-Brinkman form in Eq. 5.4. The dependence on $\bar{\gamma}$ therefore enters directly through the Brinkman interpolation $\alpha(\bar{\gamma})$ and, for the dissipation objective, through the objective's Brinkman drag contribution. The dependence of ν_t^{fr} and G^{fr} on the design is not differentiated in the adjoint step. For an objective $J(\mathbf{s}, \bar{\gamma})$, the reduced Lagrangian is

$$\mathcal{L}(\mathbf{s}, \bar{\gamma}, \mathbf{s}^*) = J(\mathbf{s}, \bar{\gamma}) + \mathbf{s}^* R(\mathbf{s}, \bar{\gamma}; \nu_t^{fr}, G^{fr}), \tag{5.11}$$

where $\mathbf{s}^*$ is the adjoint state. In finite-element notation, the product $\mathbf{s}^* R$ represents the assembled residual weighted by the adjoint velocity and pressure test functions, not a pointwise multiplication. The adjoint equation is obtained by eliminating the dependence of the total derivative on the forward-state variation:

$$\left(\frac{\partial R}{\partial \mathbf{s}}\right)^T \mathbf{s}^* = -\left(\frac{\partial J}{\partial \mathbf{s}}\right)^T. \tag{5.12}$$

The design derivative with respect to the projected physical density then becomes

$$\frac{dJ}{d\bar{\gamma}} = \frac{\partial J}{\partial \bar{\gamma}} + \mathbf{s}^* \frac{\partial R}{\partial \bar{\gamma}}. \tag{5.13}$$

This derivative is subsequently chained through the Heaviside projection and the transpose of the active-mask filter before being passed to MMA. The same transpose-filter machinery is used for the volume-constraint sensitivity.

5.5.2 Mechanism of Frozen Turbulence

The frozen turbulence assumption simplifies the sensitivity analysis by decoupling the turbulence transport from the design derivative. During the adjoint formulation, the turbulent eddy viscosity field (ν_t) generated by the forward SA solver is assumed to be locally independent of infinitesimal changes in the physical topology:

$$\frac{\partial \nu_t}{\partial \bar{\gamma}} \approx 0. \tag{5.14}$$

The reciprocal wall-distance field used by the SA model is treated analogously during sensitivity assembly. By treating ν_t and G as frozen spatially varying coefficients, the adjoints of the SA transport equation and the penalized wall-distance equation do not need to be solved.

This does *not* mean that turbulence is frozen in the forward physics. The forward loop still updates the wall distance, SA working variable, and eddy viscosity as the design evolves. The approximation is only in the derivative: the optimizer ignores the infinitesimal derivative of future turbulence and wall-distance fields with respect to the current topology change.

5.5.3 Justification and Limitations

A fully coupled turbulent adjoint would require adjoint equations for the SA working variable, the eddy-viscosity relation, the topology-dependent SA penalty, the reciprocal wall-distance equation, the wall-distance penalty, and the clipping or continuation devices used for numerical robustness. For evolving-wall topology optimization, this creates a very stiff coupled derivative problem. Exact SA adjoints are possible and have been developed for shape optimization and related applications [67, 37], but they are substantially more fragile than frozen-viscosity derivatives in the present topology setting.

The primary advantage of the frozen turbulence approach is therefore practical robustness. It reduces implementation complexity, lowers the cost per design iteration, and avoids differentiating the most nonlinear turbulence and wall-distance source terms. This is why frozen turbulence, or Constant Eddy Viscosity (CEV), remains common in industrial adjoint workflows [43, 49]. Wu et al. [62] show that the CEV assumption can still achieve useful optimization goals in benign flows, while also documenting that sensitivity accuracy and adjoint convergence can deteriorate in more challenging regimes.

The price of this robustness is that the gradients are approximate reduced derivatives. The finite-difference verification reported in Appendix C therefore validates only the implemented frozen derivative: the perturbed checks hold $\tilde{\nu}$ and G fixed in the same way as the adjoint. It should not be interpreted as validation of a fully coupled turbulence sensitivity.

5.6 Optimization Problem Formulation

Using the frozen-turbulence sensitivities described above, the topology optimization task is formulated as a mathematical programming problem: find a spatial density distribution that minimizes a performance metric subject to physical and geometric constraints.

5.6.1 Objective Formulation

The distinction between active and passive domains introduced in Section 5.2 is essential for the objective definition. Dissipation-type objectives are common in density-based flow topology optimization because they penalize mechanical energy loss directly. In Brinkman-penalized formulations, the corresponding power-loss objective is naturally written as the sum of physical viscous dissipation and Darcy/Brinkman drag over the optimized design domain [63, 20]:

$$J_D(\mathbf{u}, \bar{\gamma}) = \int_{\Omega_D} \left[\frac{1}{2} \mu_{\text{eff}} \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T \right) : \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T \right) + \alpha(\bar{\gamma})\, \mathbf{u} \cdot \mathbf{u} \right] d\Omega. \tag{5.15}$$

Here Ω_D denotes the optimized design domain used for objective integration, excluding fixed non-design inlet and outlet extensions, and $\mu_{\text{eff}} = \mu + \mu_t$ under the frozen-turbulence flow solve.

For completeness, the implementation also supports an average inlet-pressure objective. Because the outlet pressure is fixed to zero in the pressure-outlet cases, this pressure can be used as a proxy for the pressure loss required to drive the imposed flow rate through the topology:

$$J_p(p) = \frac{1}{|\Gamma_{\text{in}}|} \int_{\Gamma_{\text{in}}} p \, d\Gamma. \tag{5.16}$$

Here Γ_{in} is the inlet boundary of the fixed non-design extension, not an internal design-domain pressure plane; the division by $|\Gamma_{\text{in}}|$ is required so that the objective is an average pressure. Preliminary pressure-objective runs used this functional, but the final turbulent optimization framework uses J_D from Eq. 5.15 for the pipe-bend, U-bend, and diffuser campaigns. This choice follows the physical interpretation that pressure loss is proportional to mechanical energy dissipation in steady internal flow, while the domain-integrated J_D provides a less localized design signal by summing the objective contribution over all design cells. The code keeps J_p as a diagnostic and fallback objective, but pressure-drop results are interpreted alongside the dissipation objective rather than replacing it.

5.6.2 Volume Constraint

Because the goal is to design a targeted manifold or pipe system, a strict volume constraint must be imposed. The volume fraction constraint is evaluated only over the prescribed volume mask Ω_V, rather than over the entire computational mesh:

$$g(\bar{\gamma}) = \frac{\int_{\Omega_V} \bar{\gamma}\, d\Omega}{V_{\text{mask}}} - V_{\max} \leq 0 \tag{5.17}$$

where V_{mask} is the total volume of the masked region and $V_{\max}$ is the maximum allowable fluid volume fraction inside that region.

5.6.3 The Complete Continuous Optimization Problem

Combining the selected objective function, the governing equations, and the constraints, the continuous topology optimization problem for turbulent flow is formally stated as:

$$\begin{aligned} \underset{\gamma(\mathbf{x})\in[0,1]}{\text{minimize}} \quad & J(\mathbf{u}, p, \bar{\gamma}) \\ \text{subject to} \quad & \text{Modified RANS weak residual (Eq. 5.4)} \\ & \text{Modified Spalart-Allmaras weak form (Eq. 5.6)} \\ & \text{Penalized wall-distance weak residual (Eq. 5.9)} \\ & g(\bar{\gamma}) \leq 0. \end{aligned} \tag{5.18}$$

The Method of Moving Asymptotes (MMA) [55] is used to solve this constrained minimization problem. MMA builds a sequence of local convex approximations from the current objective value, constraint value, and their gradients. The moving asymptotes and move limits restrict how far each active design variable may change in one iteration, which prevents unstable density jumps in a highly nonlinear flow problem. In this implementation, MMA updates only the active raw design degrees of freedom; passive cells are restored to their prescribed bounds after filtering and projection.

5.7 Density Filtering and Projection

A well-known mathematical complication in density-based topology optimization is the lack of existence of a classical solution. If the design space is left unconstrained, the optimizer tends to create infinitely thin structures or checkerboard patterns. To guarantee a mesh-independent and manufacturable design, a minimum length scale must be enforced [15, 34].

5.7.1 The Discontinuous Galerkin Helmholtz Filter

To prevent checkerboarding, a density filter is applied to the raw design variable generated by the optimizer. Because a continuous nodal definition causes the Brinkman penalty to interpolate smoothly across boundary elements, the density field is discretized using piecewise constant elements (DG0) [10, 29]. By assigning a single density value to each element, the structural boundaries remain sharp at element edges.

Unlike an explicit cone-kernel density filter, the Helmholtz filter does not prescribe a direct weighted average over neighboring elements. Instead, it defines the filtered density as the solution of a smoothing problem: the filtered field must remain close to the raw optimizer field while jumps and rapid spatial oscillations are penalized. In a continuous finite-element discretization this smoothing is usually produced by a gradient term. In the present DG0 discretization, however, the density is constant inside each element and therefore has no element-interior gradient. Neighboring cells are coupled only through their shared facets, so the continuous Helmholtz filter is replaced by an interior-penalty form on the internal facets (Γ_{int}):

$$\int_{\Gamma_{int}} r^2 \frac{\alpha_{dg}}{h_{avg}} [\![\tilde{\gamma}]\!] \cdot [\![v]\!] \, dS + \int_{\Omega} \tilde{\gamma} v \, dx = \int_{\Omega} \gamma v \, dx \tag{5.19}$$

where v is the test function, $[\![\cdot]\!]$ denotes the jump operator across a facet, h_{avg} is the average cell size, and $r = R/(2\sqrt{3})$ is the scaled filter radius. The first term penalizes differences between adjacent element densities, while the second term keeps the filtered field anchored to the raw design. Increasing r strengthens the facet coupling and therefore spreads the influence of each raw cell over a wider neighborhood. The filter can thus be interpreted as an implicit, mesh-based averaging operation whose effective kernel is given by the response of Eq. 5.19 to a localized density perturbation.

Active-Mask Normalization

To handle boundary interactions near designated inlet and outlet buffers, this filtering is active-mask-normalized. If the filter were applied blindly across the entire mesh, fixed passive cells could dilute the active design field near non-design regions. The implementation therefore filters only the active contribution and divides by the filtered active mask:

$$\tilde{\gamma} = \frac{\mathcal{H}(M\gamma)}{\mathcal{H}(M)} \tag{5.20}$$

where $\mathcal{H}$ represents the Helmholtz PDE solution operator and M represents the isolated active design indicator mask. The numerator smooths the active raw design field, while the denominator renormalizes the local averaging footprint. As a result, inlet extensions, outlet buffers, and other passive regions do not artificially pull the nearby active design toward their prescribed density values. During sensitivity analysis, the transpose of this normalized filter maps objective and volume gradients back into the active DG0 design space.

In practice, the Helmholtz filter is applied to the active design contribution and normalized by the filtered active mask, so fixed passive-fluid cells do not dilute nearby design densities. After projection, the passive-fluid bounds are restored before the physical flow equations are assembled. The same chain is traversed in reverse when objective and volume sensitivities are mapped back to the active raw density variables.

The same Helmholtz operator is used in the reverse sensitivity chain. If $s_{\bar{\gamma}}$ denotes a cellwise derivative with respect to the bounded physical density, then the projection and bound mapping give

$$s_{\tilde{\gamma}} = s_{\bar{\gamma}}(\gamma_{\text{upper}} - \gamma_{\text{lower}})\frac{\partial \Pi_\beta}{\partial \tilde{\gamma}}. \tag{5.21}$$

The transpose of the active-mask-normalized filter then maps this quantity back to the raw active design space:

$$s_\gamma = M\,\mathcal{H}^{-T}\left(\frac{s_{\tilde{\gamma}}}{\mathcal{H}(M)}\right). \tag{5.22}$$

Because the Helmholtz bilinear form in Eq. 5.19 is symmetric, $\mathcal{H}^{-T}$ is evaluated by another Helmholtz solve with the transpose-filter right-hand side. Passive cells are set to zero in s_γ because MMA updates only the active design variables.

5.7.2 Heaviside Projection

While the Helmholtz filter imposes a minimum length scale, it smooths the design field and creates a gray transition zone between solid and fluid. To recover a sharper topology, a continuous Heaviside projection is applied to the filtered variable [28, 60]:

$$\Pi_\beta(\tilde{\gamma}) = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{\gamma} - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))} \tag{5.23}$$

where $\eta \in [0, 1]$ is the projection threshold, and β is a steepness parameter. The threshold η defines the filtered density around which the projection switches between topology states. In the convention used in this thesis, filtered values below the threshold are projected toward $\bar{\gamma} = 0$ and therefore become solid-like, whereas values above the threshold are projected toward $\bar{\gamma} = 1$ and become fluid-like. Increasing β sharpens this transition; in the limit $\beta \to \infty$, the function approaches a strict Heaviside step.

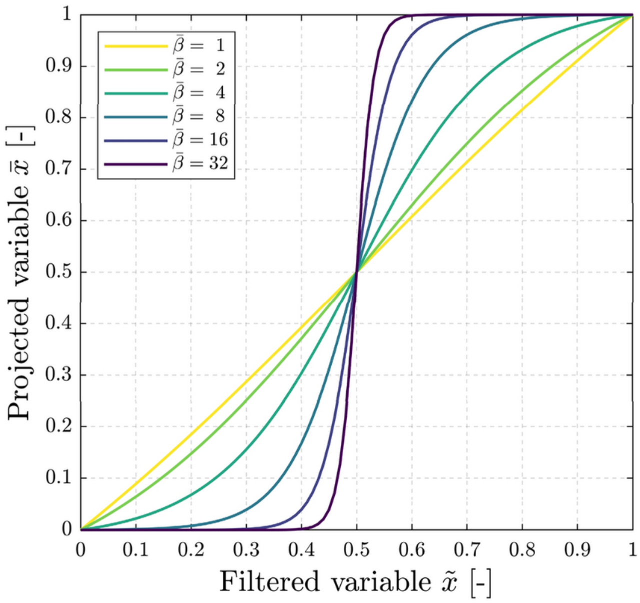


FIGURE 5.2: Heaviside projection from Eq. 5.23 at $\eta = 0.5$ for $\beta = \{1, 2, 4, 8, 16, 32\}$, reproduced from Ooms et al. [42].

5.8 Algorithmic Workflow and Solver Implementation

Executing the optimization requires a robust iterative algorithm. The solver combines density processing, wall-distance updates, Picard-decoupled SA turbulence updates, a final frozen-viscosity state solve, adjoint sensitivity assembly, transpose-filter mapping, volume-gradient assembly, and MMA updates.

Figure 5.3 summarizes the full optimization workflow. The raw active density is first filtered, projected, and restored into the bounded physical density field used by the governing equations. The forward loop then updates the wall-distance, velocity-pressure, and SA turbulence fields until an accepted turbulent state is obtained. Only after this forward update is complete is the frozen-viscosity residual assembled for the final state and adjoint solve. The resulting objective and volume sensitivities are mapped back through the projection slope and transpose filter before MMA updates the active raw design variables.

5.8.1 SA Numerical Controls and Hybrid Forward Flow Recovery

During the optimization loop, the flow solver advances the momentum field using the eddy viscosity from the previous iteration, and the updated velocity is then passed back to the SA solver. During volatile intermediate design steps, the framework uses Stokes-Brinkman warm starts and convection continuation to reintroduce inertial physics gradually. If convergence stalls, the SNES-based final solve can step through recovery schedules with altered tolerances and line-search methods.

The optimization architecture supports non-converged acceptance during difficult intermediate steps, allowing the algorithm to maintain topological momentum.

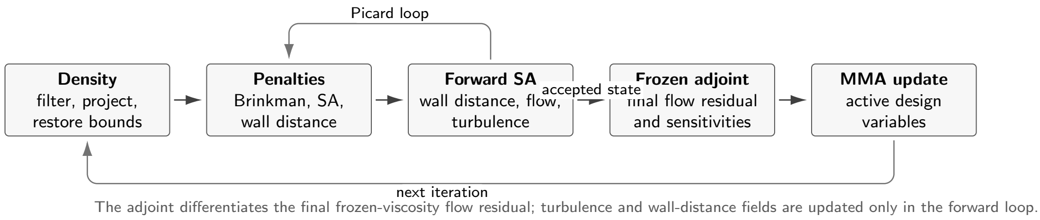


Figure 5.3: Algorithmic path of the density-based frozen-turbulence topology optimization loop. The forward turbulence fields are updated before the final frozen-viscosity adjoint sensitivity is assembled and passed to MMA.

Prior to adjoint assembly, stricter flow solves can be executed. When used, these strict solves confirm that the sensitivities and frozen turbulent fields are derived from a more equilibrated state. Otherwise, relying on accepted residual criteria is a pragmatic compromise required for highly nonlinear intermediate designs.

5.8.2 Algorithm and Continuation Strategy

To encourage convergence to a physically meaningful binary design, the algorithm iterates over a continuation schedule. The projection steepness (β) is increased as the optimization progresses, while the Brinkman RAMP parameter (q) is advanced according to the prescribed penalty schedule and MMA move limits are tightened. The SA and wall-distance power penalties do not use this continuation in the final benchmark runs; their amplitudes and exponents remain fixed while the local penalty magnitude changes through the evolving topology field.

Within each continuation stage, one MMA iteration consists of the following mathematical sequence. First, the raw active design field is filtered with Eq. 5.19, normalized by the active-mask denominator $\mathcal{H}(M)$, projected with Eq. 5.23, and restored to the prescribed passive-cell bounds. The resulting $\bar{\gamma}$ defines the Brinkman resistance $\alpha(\bar{\gamma})$, the SA damping coefficient in Eq. 5.6, and the wall-distance forcing in Eq. 5.9.

Second, the forward turbulence state is updated. The reciprocal wall-distance residual in Eq. 5.9 is solved for the current topology, the velocity-pressure equations are advanced with the current eddy viscosity, and the penalized SA weak form in Eq. 5.6 updates $\tilde{\nu}$ and therefore μ_t. These Picard updates continue until the configured residual criteria are accepted for the current design step.

Third, the final frozen-viscosity flow state is solved with Eq. 5.4. The adjoint equation from Section 5.5 is then assembled from the same weak residual and the selected objective; in the final turbulent benchmark campaigns this is J_D from Eq. 5.15. The design derivative is obtained from Eq. 5.13, chained through the projection derivative in Eq. 5.21, and mapped back to the active raw variables through the transpose filter in Eq. 5.22. MMA finally updates only those active raw density degrees of freedom, subject to the volume constraint in Eq. 5.17 and the current move limit.

5.9 Conclusion

This chapter has established the mathematical and algorithmic framework used for the turbulent topology optimization results in this thesis. By employing the Fictitious Domain Method with Brinkman momentum resistance [14, 26], the evolving structural topology is integrated into a fixed computational grid. The SA working-variable and reciprocal wall-distance equations are coupled to the topology through power-law penalties, while the DG0 Helmholtz interior penalty filter [34, 15] and continuous Heaviside projection [28, 60] enforce a minimum length scale and reduce mesh-dependent numerical artifacts.

The optimization is driven by a frozen-turbulence weak-form residual adjoint [62, 49]. The forward turbulence fields are updated during the optimization loop; only their derivatives are frozen during sensitivity assembly. This assumption is defensible as a stable engineering approximation, but the thesis treats it as an approximation whose limitations must be checked through finite-difference sensitivity verification and independent CFD validation. Embedded within a Picard-decoupled flow architecture using hybrid IPCS/SNES recovery schemes, this methodology transforms the verified fluid dynamics engine from Chapter 3 into the automated design tool evaluated in Chapter 6.

Chapter 6

Turbulent Topology Optimization Results

6.1 Introduction

Building upon the FEniCS-SA solver verified in Chapter 4 and the topology optimization methodology developed in Chapter 5, this chapter evaluates the turbulent topology optimization solver on the pipe-bend and U-bend internal-flow benchmarks from Bayat et al. [12] and the diffuser benchmark from Yoon [63]. The Bayat cases use reciprocal wall-distance penalization, topology-dependent SA and wall-distance reaction terms, an average inlet-pressure objective, and a frozen-turbulence adjoint. The diffuser benchmark uses the same frozen-SA topology optimization machinery, but it minimizes the dissipation objective J_D.

The optimized pipe-bend, U-bend, and diffuser layouts are interpreted through their projected density fields, velocity fields, and independent sharp-geometry CFD validation. The frozen-sensitivity verification, FEniCS convergence histories, numerical settings, and Ansys grid-independence studies are documented in Appendix C. For the completed pipe-bend sensitivity check, five central finite-difference samples agree with the frozen-adjoint derivative to a maximum relative discrepancy of 5.92×10^{-5}, and the directional Taylor check recovers approximately second-order remainder decay. This verifies the derivative used by MMA for the frozen-turbulence J_p setup, while leaving the fully coupled turbulence and wall-distance derivative outside the claim.

The benchmark sections below introduce the geometry, boundary conditions, objective, and validation metrics needed to read the results without consulting the appendices first. Appendix C retains the full turbulent and laminar numerical settings, FEniCS convergence histories, and Ansys grid-independence studies for the extracted designs, grouped by benchmark case.

All mesh-independence relative differences referenced from Appendix C follow the appendix convention: coarse–medium differences are normalized by the coarse value, while medium–fine differences are normalized by the fine value.

6.2 Benchmark Cases and Evaluation Metrics

The three benchmarks cover curved internal flows and a diffuser with a prescribed outlet contraction. The pipe bend targets the $V_f = 0.25$, $Re = 5{,}000$ case from Bayat et al. [12], while the U-bend targets their $V_f = 0.27$, $Re = 5{,}000$ case with a 180° flow reversal. The diffuser targets the $Re = 3{,}000$ turbulent diffuser from Yoon [63], where the original solid mass usage of 70% corresponds to a maximum fluid fraction of $V_f = 0.30$ in the present density convention. Table 6.1 summarizes the operating point and analyses reported for each case.

TABLE 6.1: Benchmark cases evaluated in Chapter 6. Detailed physical and numerical parameters are listed in Appendix C.

Case	V_f	Re	**Analyses reported**
Pipe bend	0.25	5,000	FEniCS optimization; Ansys-SA and Ansys-SST $k-\omega$ CFD validation; laminar-design comparison
U-bend	0.27	5,000	FEniCS optimization; Ansys-SA and Ansys-SST $k-\omega$ CFD validation; laminar-design comparison
Diffuser	0.30	3,000	Velocity-outlet baseline; pressure-outlet FEniCS optimization with J_D; Ansys-SA and Ansys-SST $k-\omega$ CFD validation; laminar-design comparison for the pressure-outlet case

6.2.1 Optimization and Validation Metrics

Throughout the optimization process, the pipe-bend and U-bend benchmarks minimize the average inlet-pressure objective J_p defined in Eq. 5.16. This pressure is averaged over the physical inlet boundary of the fixed non-design extension, where the inlet velocity is prescribed. The outlet pressure is fixed to zero, so reducing J_p directly reduces the pressure level required to drive the prescribed inlet flow through the evolving topology. The optimization problem is still solved on a diffuse Brinkman domain, but once a design is extracted for Ansys Fluent, the Brinkman penalty is removed and the solid-fluid interface becomes a sharp wall.

This choice is made to remain consistent with the Bayat/Alexandersen benchmark objective and reported performance metric. For prescribed-flow internal problems, pressure drop and mechanical dissipation are physically related, so the dissipation objective J_D is also a valid formulation for these bend cases. In the present implementation, however, switching from J_p to J_D is not a pure objective substitution: stable optimization behavior also depends on the Brinkman solid impedance, continuation schedule, and topology-dependent SA and wall-distance penalties. The dissipation formulation is therefore treated as a feasible alternative requiring separate numerical tuning, while the pipe-bend and U-bend results reported here use J_p for consistency with the reference benchmark.

The diffuser benchmark instead minimizes the dissipation objective J_D defined in Eq. 5.15. This matches the objective used by Yoon [63]: physical viscous dissipation plus Brinkman drag is integrated over the design domain and minimized subject to the volume constraint. The original Yoon-style setup prescribes parabolic velocity profiles at both inlet and outlet. Because that outlet Dirichlet condition can influence the optimized outlet geometry, this chapter also introduces a pressure-outlet diffuser variant in which the inlet profile and objective are retained but the outlet velocity profile is replaced by a fixed static pressure. The pressure-outlet variant is used as the primary diffuser result, while the velocity-outlet case is retained as a boundary-condition sensitivity baseline.

All three benchmark cases use the power-law SA sink, the power-law reciprocal wall-distance penalty, and the frozen-turbulence adjoint introduced in Chapter 5.

Bayat et al. [12] report corresponding final average inlet-pressure objectives for their standard k-ε computations in Table 7. Their "conventional" method is the density-based turbulent topology optimization formulation without the proposed implicit wall-functions: solid regions are imposed through Brinkman/body-force penalization, including damping of k and ε toward zero in solid material, on the same diffuse optimization mesh. It is therefore distinct from the body-fitted re-simulations with explicit wall-functions that they use for validation. Since the present thesis uses a wall-resolved SA model instead of k-ε, the conventional values from Bayat et al. are used only as scale comparisons in the individual Bayat benchmark discussions below. For the diffuser benchmark, the reference comparison is qualitative because Yoon [63] reports the laminar and turbulent diffuser layouts and post-processing behavior rather than a directly matching J_D table for the diffuser.

To bridge this gap, the primary validation metric is the physical viscous dissipation integrated over the full exported CFD domain, including the non-design inlet, outlet, and fixed regions that are retained in the final geometry,

$$\Phi_D = \int_\Omega \frac{1}{2}\mu_{\text{eff}} \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right) : \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right) \, d\Omega, \tag{6.1}$$

where Ω denotes the complete sharp-geometry flow domain exported to Ansys and $\mathbf{S} = \frac{1}{2}\left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right)$ is the symmetric strain-rate tensor. The integrand in Eq. 6.1 can therefore also be written as $2\mu_{\text{eff}}\mathbf{S} : \mathbf{S}$. In the Ansys Fluent post-processing, the scalar strain-rate magnitude is $\dot{\gamma} = \sqrt{2\mathbf{S} : \mathbf{S}}$, so the custom field function $\mu_{\text{eff}}\dot{\gamma}^2$ is equivalent to the integrand in Eq. 6.1. The effective viscosity is $\mu_{\text{eff}} = \mu + \mu_t$ for turbulent evaluations and $\mu_{\text{eff}} = \mu$ for laminar reference solves.

The second global validation metric is the static pressure drop between the physical inlet and outlet sections,

$$\Delta p = \bar{p}_{\text{in}} - \bar{p}_{\text{out}}, \qquad \bar{p}_\Gamma = \frac{1}{|\Gamma|} \int_\Gamma p_s \, d\Gamma, \tag{6.2}$$

where p_s is the static pressure and $|\Gamma| = \int_\Gamma 1 \, d\Gamma$ is the boundary length in the two-dimensional model. In Ansys Fluent, p_s is taken directly from the area-weighted average static pressure on the inlet and outlet surfaces. For FEniCS results that use kinematic pressure, the corresponding static pressure is obtained as $p_s = \rho p_k$ before evaluating Eq. 6.2. Pressure drop is retained throughout Chapter 6 because it is the most direct engineering interpretation of the same mechanical losses measured by Φ_D. All two-dimensional dissipation quantities are evaluated per unit depth.

6.3 Pipe Bend Benchmark

The pipe-bend benchmark introduces a direct comparison with the turbulent topology optimization results of Bayat et al. [12]. While the reference study employs a standard k-ε RANS model with wall functions, the present framework tackles the same problem using a wall-resolved SA model under the frozen-turbulence assumption. This comparison tests whether the fundamental topological solution, a streamlined and separation-avoiding bend, remains consistent across different turbulence closures.

6.3.1 Geometry and Boundary Conditions

The boundary conditions and geometric constraints replicate the setup shown in Fig. 10 of Bayat et al. [12], specifically targeting the $V_f = 0.25$ volume-fraction case. The inlet is placed on the left side of the non-design extension, the outlet is placed along the lower boundary, and the outlet pressure is fixed to zero. The optimization minimizes the average pressure on this prescribed inlet boundary required to drive the flow through the evolving bend. The local benchmark geometry used for the present implementation is shown in Figure 6.1.

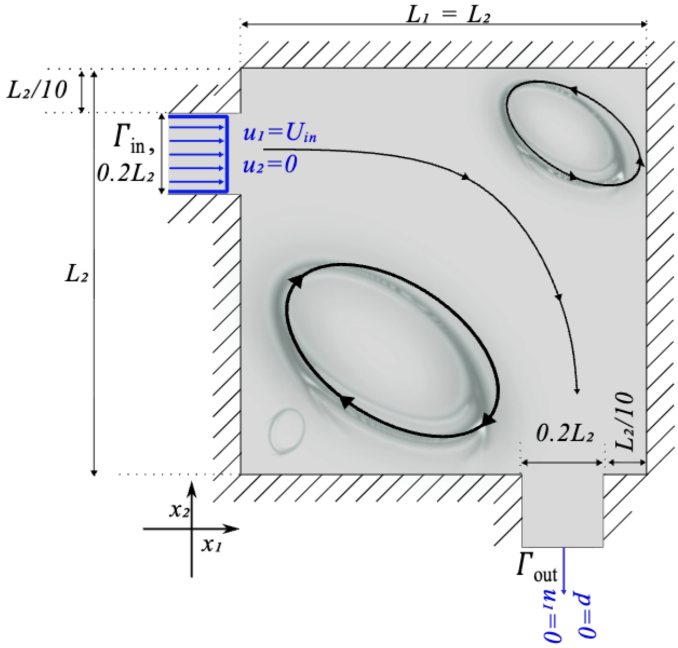
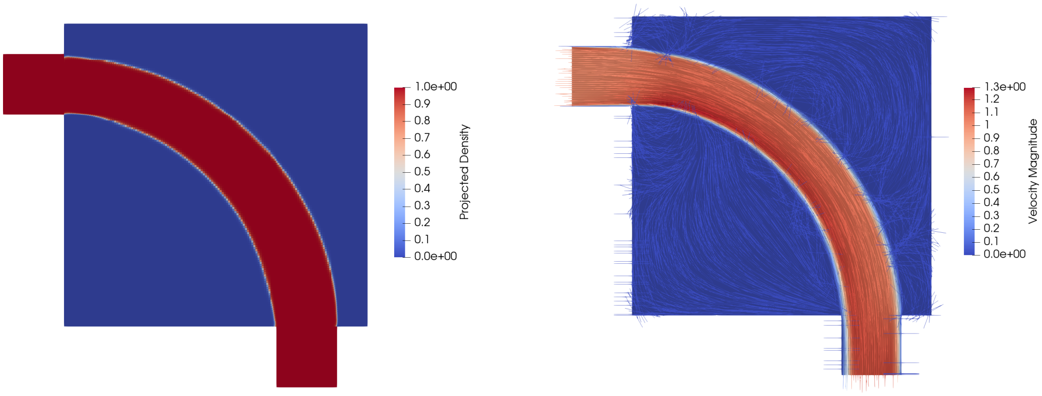


FIGURE 6.1: Geometry and boundary conditions for the pipe-bend benchmark from Bayat et al. [12]. The present evaluation targets the $V_f = 0.25$ case using the SA model.

6.3.2 Optimization Results

Figure 6.2 illustrates the projected density field $\bar{\gamma}$ and the corresponding velocity magnitude. The supporting coordinate finite-difference and Taylor checks for the frozen adjoint are reported in Appendix C. Despite the differing turbulence models, the FEniCS-SA topology is expected to follow the same broad routing strategy as the k-ε reference: the primary high-speed path should bypass the sharpest corners to minimize momentum loss, while boundary-layer treatment can still affect the exact gray-scale density distribution near the walls.



(A) Projected density field $\bar{\gamma}$.

(B) Velocity magnitude and vector field.

Figure 6.2: FEniCS-SA optimization result for the pipe-bend benchmark at $V_f = 0.25$. Final objective value $J_p = [TBD]$ Pa.

For the same $V_f = 0.25$, $Re = 5{,}000$ operating point, Bayat et al. [12] report $J_p = 0.4023$ Pa for their conventional k-ε topology optimization. This value is used only as a reference scale, not as a direct validation target, because the turbulence closure and near-wall treatment differ from the present wall-resolved SA formulation.

The corresponding FEniCS convergence history is reported in Figure C.1 of Appendix C. The objective decreases rapidly during the first part of the optimization, while later continuation updates mainly rebalance the pressure objective and active volume constraint.

6.3.3 CFD Validation

The structural robustness of the extracted pipe-bend topology is verified in Table 6.2. Because the SST $k-\omega$ model is notably more sensitive to adverse pressure gradients than SA, evaluating the extracted geometry under both closures serves as a rigorous stress test for the optimized shape.

Using the Ansys-SA result as the reference, the SST $k-\omega$ evaluation increases the pressure drop by approximately 4.5% and the viscous dissipation by approximately 8.5%. The SST $k-\omega$ model therefore predicts slightly higher losses, as expected for a closure that is more sensitive to adverse pressure gradients, but the changes remain

TABLE 6.2: CFD validation metrics for the extracted pipe-bend topology.

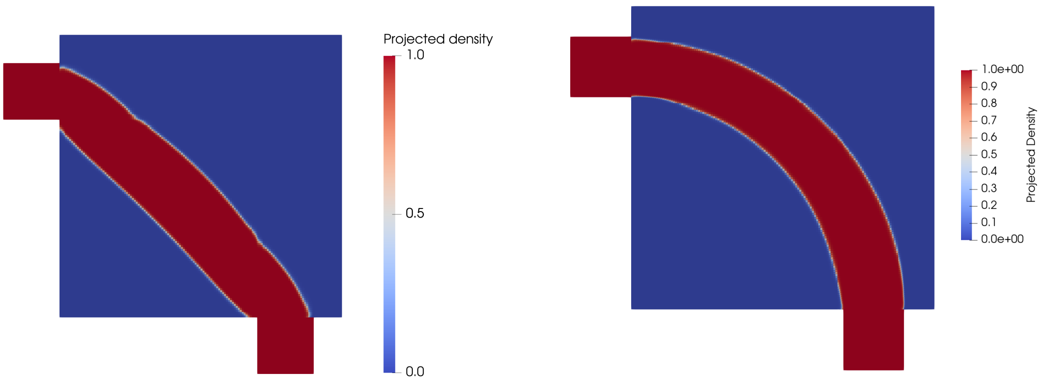
Solver	**Model**	Φ_D **[W/m]**	Δp **[Pa]**	**Comment**
FEniCS diffuse design	SA	[TBD]	[TBD]	Optimization state
Ansys sharp geometry	SA	0.0153	0.1457	Cross-code comparison
Ansys sharp geometry	SST $k-\omega$	0.0166	0.1522	Turbulence-model sensitivity

modest. This suggests that the optimized pipe-bend geometry is robust under a change of turbulence closure, rather than being tuned only to the SA model.

6.3.4 Comparison with Laminar Design

The laminar reference uses the same pipe-bend mesh, design-domain restrictions, inlet and outlet treatment, objective type, and volume fraction, but replaces the SA turbulent state equations by the laminar flow formulation and is optimized at $Re = 1$. The expected distinction is that a laminar bend can minimize wall length and viscous shear in the creeping-flow limit, but may produce a sharper or less separation-aware turn when evaluated at the target turbulent Reynolds number. The comparison is therefore interpreted under identical turbulent operating conditions, rather than by comparing the laminar and turbulent optimization objectives directly.

Figure 6.3 compares the laminar and turbulent optimized topologies before sharp-geometry extraction. The laminar-design mesh-independence study supporting the SST $k-\omega$ result is reported in Table C.8 of Appendix C.

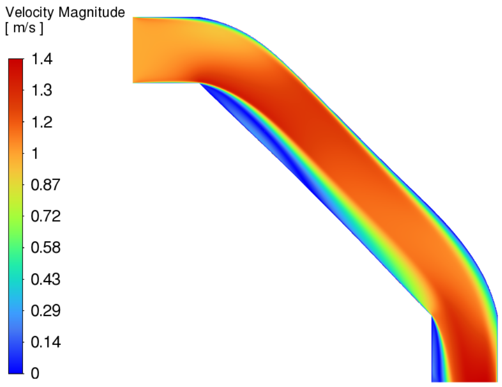


(A) Projected density field $\bar{\gamma}$: laminar-optimized design ($Re = 1$). Final objective value $J_p = 101.742$ Pa.

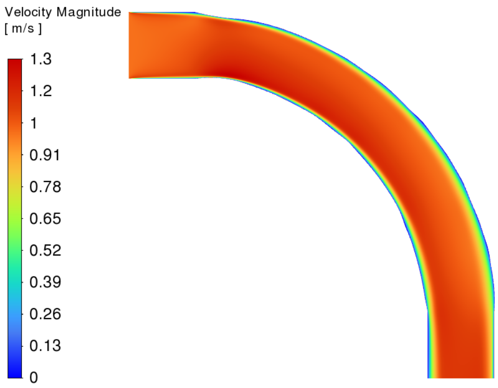
(B) Projected density field $\bar{\gamma}$: turbulent-optimized design ($Re = 5{,}000$). Final objective value $J_p = [TBD]$ Pa.

FIGURE 6.3: Laminar and turbulent topology comparison for the pipe-bend benchmark.

Figure 6.4 shows the corresponding SST $k-\omega$ velocity contours through the extracted laminar and turbulent pipe-bend designs. These contours provide the flow-field evidence behind the scalar metrics: sharper turns and corners in the laminar design are expected to promote stronger local separation and momentum loss when the geometry is operated at the turbulent Reynolds number.

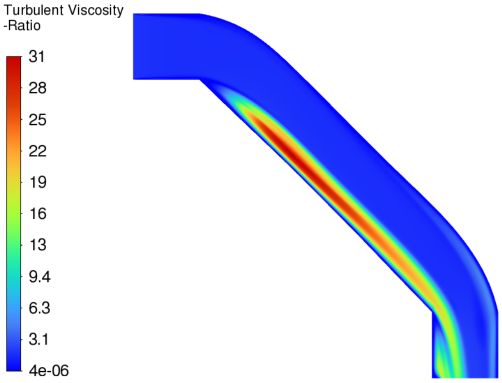


(A) Velocity magnitude field u: laminar-optimized design.

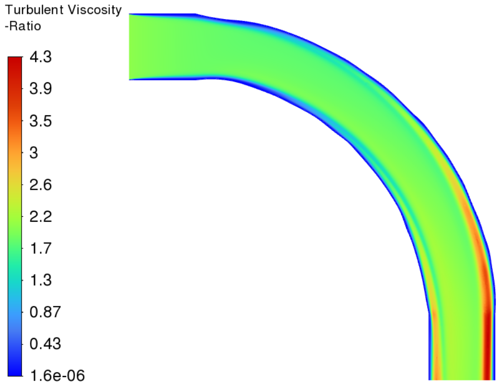


(B) Velocity magnitude field u: turbulent-optimized design.

Figure 6.4: Velocity-contour comparison for the extracted pipe-bend designs under identical turbulent operating conditions (SST $k - \omega$ model).



(A) Turbulent viscosity ratio field μ_t/μ: laminar-optimized design.



(B) Turbulent viscosity ratio field μ_t/μ: turbulent-optimized design.

Figure 6.5: Turbulent viscosity ratio comparison for the pipe-bend designs, again with an identical turbulent flow (SST $k - \omega$ model).

The final SST $k - \omega$ comparison in Table 6.3 evaluates the extracted laminar and turbulent pipe-bend designs under identical turbulent operating conditions. A lower pressure drop and lower viscous dissipation for the turbulent design would support the need for a turbulent topology optimization framework rather than a laminar surrogate.

Table 6.3: Performance comparison of the extracted pipe-bend topologies under identical turbulent operating conditions.

Design origin	Model	Φ_D [W/m]	Δp [Pa]	Relative performance
Laminar TO, $Re = 1$	SST $k - \omega$	0.0313	0.5444	Baseline
Turbulent TO, SA	SST $k - \omega$	0.0166	0.1522	47.0% lower Φ_D; 72.0% lower Δp

6.4 U-Bend Benchmark

The U-bend is the companion benchmark to the pipe bend. It uses the same frozen-SA topology optimization framework, but the geometry forces a 180° return path between two ports on the same side of the domain. This makes the case useful for checking whether the optimizer can form a smooth turbulent return duct without relying on the shorter 90° routing available in the pipe-bend case.

6.4.1 Geometry and Boundary Conditions

The geometry and boundary conditions follow Fig. 11 and Table 4 of Bayat et al. [12]. The square design region has side length L, with a non-design lead duct of length $0.2L$ on the left. The inlet and outlet ports both have height $0.2L$; the top port acts as the inlet and the lower port acts as the pressure outlet. A fixed internal separator bar of thickness $0.10L$ extends from the lead duct into the design domain, forcing the flow to turn around its rounded tip. The target volume fraction is $V_f = 0.27$, and the Reynolds number is $Re = 5{,}000$ based on the port height and maximum inlet velocity. Figure 6.6 shows the corresponding local geometry and boundary-condition layout used for this benchmark.

6.4.2 Optimization Results

Figure 6.7 reports the final projected density and velocity-magnitude field for the U-bend. The main feature to assess is whether the design forms a rounded return passage around the separator tip, rather than a short path with tight local curvature. Because the objective is average inlet pressure, a successful layout reduces the pressure required to maintain the imposed inlet flow while respecting the prescribed volume fraction.

For the matching $V_f = 0.27$, $Re = 5{,}000$ U-bend case, the conventional k-ε result reported by Bayat et al. [12] is $J_p = 0.8327$. As for the pipe bend, this value provides a reference scale only; the comparison remains qualitative because the present solver uses a different turbulence model and near-wall treatment.

The U-bend convergence history is reserved for Figure C.3 of Appendix C, using the same combined objective and volume-residual format as the pipe-bend diagnostic.

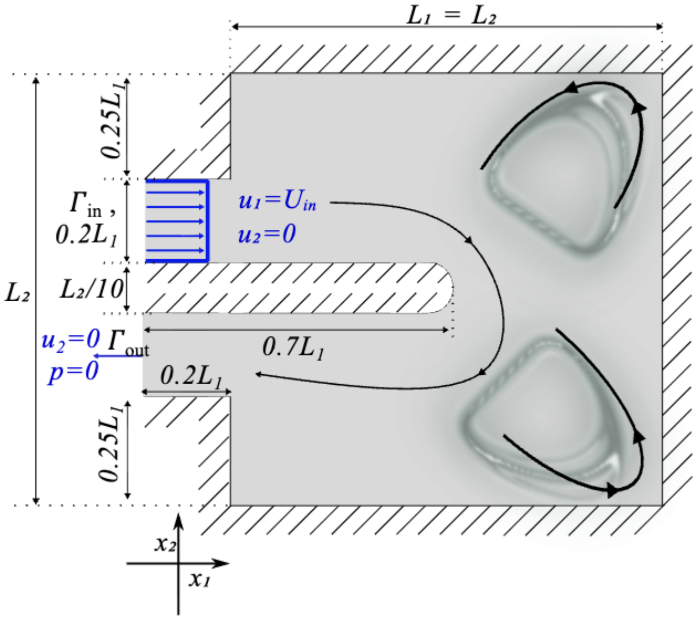


Figure 6.6: Geometry and boundary conditions for the U-bend benchmark from Bayat et al. [12].

Placeholder Projected density field $\bar{\gamma}$, $V_f = 0.27$	**Placeholder** Velocity-magnitude contours
(a) Projected density field $\bar{\gamma}$.	(b) Velocity magnitude and vector field.

Figure 6.7: FEniCS-SA optimization result for the U-bend benchmark at $V_f = 0.27$. Final objective value $J_p = TBD$ Pa.

6.4.3 CFD Validation

The extracted U-bend topology is validated with the same sharp-geometry workflow as the pipe bend. The Ansys SST $k - \omega$ solve tests whether the design remains robust under a two-equation model that is more sensitive to separation and adverse pressure gradients, while the Ansys-SA solve checks cross-code consistency for the same turbulence closure family. The resulting validation metrics are summarized in Table 6.4.

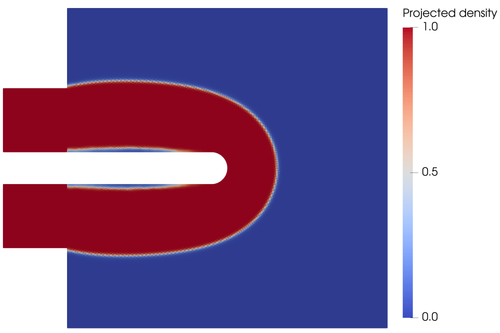
Table 6.4: CFD validation metrics for the extracted U-bend topology.

Solver	Model	Φ_D [W/m]	Δp [Pa]	Comment
FEniCS diffuse design	SA	[TBD]	[TBD]	Optimization state
Ansys sharp geometry	SA	[TBD]	[TBD]	Cross-code comparison
Ansys sharp geometry	SST $k-\omega$	[TBD]	[TBD]	Turbulence-model sensitivity

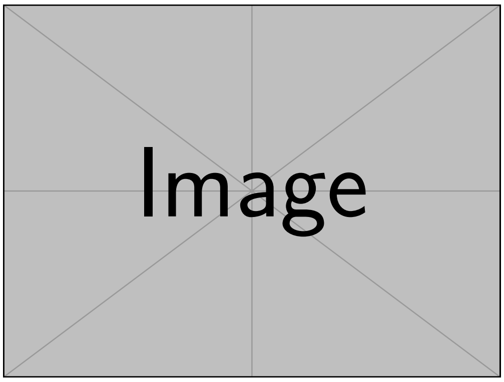
6.4.4 Comparison with Laminar Design

The laminar reference uses the same U-bend mesh, inlet and outlet ports, separator geometry, objective type, and volume fraction, but it is optimized at $Re = 1$ without the SA turbulence model. The extracted laminar and turbulent layouts are then evaluated under the same target turbulent operating conditions. This comparison is expected to show whether the laminar optimizer creates an overly tight return path around the separator tip, while the turbulent optimizer allocates fluid volume to reduce separation and momentum loss through the 180° turn.

Figure 6.8 compares the laminar and turbulent optimized U-bend topologies. The laminar-design mesh-independence study supporting the SST $k-\omega$ result is reported in Table C.13 of Appendix C.



(A) Projected density field $\bar{\gamma}$: laminar-optimized design ($Re = 1$). Final objective value $J_p = 143.254$ Pa.

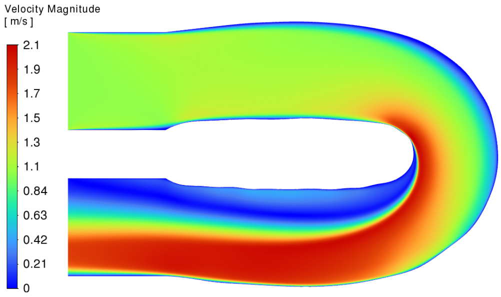


(B) Projected density field $\bar{\gamma}$: turbulent-optimized design ($Re = 5{,}000$). Final objective value $J_p = [TBD]$ Pa.

Figure 6.8: Laminar and turbulent topology comparison for the U-bend benchmark.

Figure 6.9 shows the SST $k-\omega$ velocity contours for the extracted U-bend designs. In this case the comparison is especially useful near the separator tip, where a laminar-optimized shortcut can create a tighter return path, stronger adverse-pressure-gradient effects, and larger separated or low-momentum regions than the turbulent design.

Table 6.5 compares the extracted laminar and turbulent U-bend topologies under identical turbulent operating conditions. If the turbulent design lowers Δp and Φ_D under the SST $k-\omega$ evaluation, the result supports the thesis claim that high-



(A) Velocity magnitude field u: laminar-optimized design.

Placeholder

SST $k-\omega$ velocity contour through turbulent U-bend design

(B) Velocity magnitude field u: turbulent-optimized design.

Figure 6.9: SST $k-\omega$ velocity-contour comparison for the extracted U-bend designs under identical turbulent operating conditions.

Reynolds topology optimization should include turbulence directly in the design loop.

Table 6.5: Performance comparison of the extracted U-bend topologies under identical turbulent operating conditions.

Design origin	**Model**	Φ_D **[W/m]**	Δp **[Pa]**	**Relative performance**
Laminar TO, $Re = 1$	SST $k-\omega$	0.107	1.150	Baseline
Turbulent TO, SA	SST $k-\omega$	[TBD]	[TBD]	[TBD]% vs. laminar

6.5 Diffuser Benchmark

The diffuser benchmark follows the third numerical example of Yoon [63], but it is presented in two stages. The first stage keeps the original velocity-outlet formulation, where both inlet and outlet are prescribed by parabolic velocity profiles. This case is useful as a reference because it follows the published benchmark, but it also exposes a boundary-condition sensitivity: prescribing the outflow profile can encourage the optimizer to widen the geometry near the outlet so that the imposed profile is easier to satisfy. The second stage therefore replaces the outlet velocity profile by a pressure outlet while retaining the same inlet profile, Reynolds number, volume fraction, SA topology-optimization machinery, and dissipation objective J_D. This pressure-outlet variant is treated as the primary diffuser result because the outlet velocity distribution is then a consequence of the optimized geometry rather than a Dirichlet constraint.

6.5.1 Velocity-Outlet Boundary-Condition Sensitivity

The diffuser was first run with the velocity-outlet formulation used by Yoon [63], where parabolic velocity profiles are prescribed at both the full-height inlet and the center-third outlet. That baseline produces a more strongly diverging channel near the outlet. Because the outlet profile is prescribed, this widening is interpreted as partly boundary-condition-driven rather than purely geometry-optimal. The detailed velocity-outlet geometry, FEniCS topology, velocity field, and Ansys diagnostic contour are therefore reported in Appendix C. Replacing the outlet Dirichlet velocity profile by a pressure outlet partly removes this outlet widening, so the pressure-outlet variant is used as the primary diffuser result below.

6.5.2 Pressure-Outlet Setup and Optimization Results

The pressure-outlet variant changes only the downstream boundary treatment relative to the velocity-outlet baseline documented in Appendix C. Its local geometry and boundary-condition layout are shown in Figure 6.10. The inlet remains the shifted parabolic velocity profile with $U_{\text{max,in}} = 3$ m/s and the same $Re = 3{,}000$ definition, while the center outlet is assigned a fixed static pressure $p = 0$ Pa. No outlet velocity profile is prescribed. This makes the outlet mass flux and velocity distribution part of the solution and removes the direct Dirichlet forcing that affected the velocity-outlet topology.

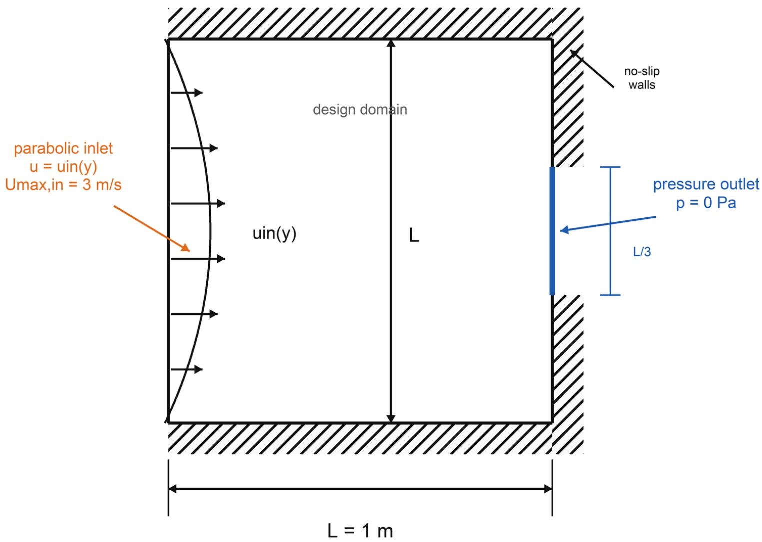
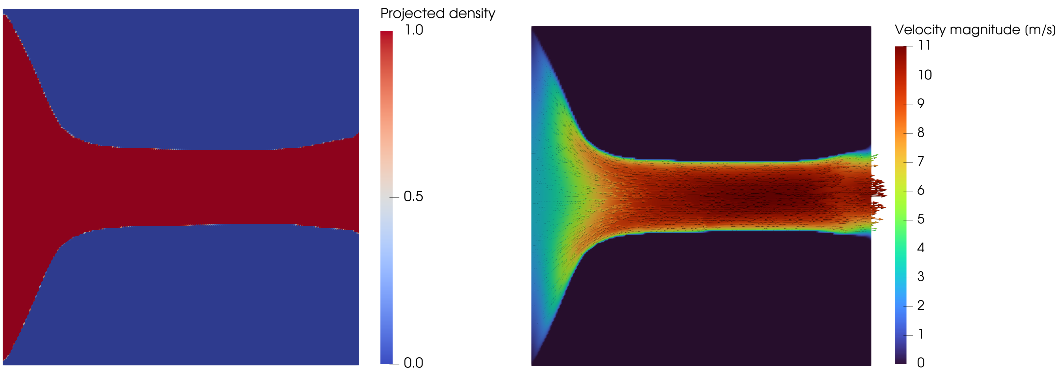


FIGURE 6.10: Geometry and boundary conditions for the pressure-outlet diffuser variant. The full-height inlet retains the shifted parabolic velocity profile, while the center-third outlet is assigned a fixed static pressure $p = 0$ Pa instead of a prescribed velocity profile.

Figure 6.11 reports the pressure-outlet FEniCS-SA topology and velocity field. Compared with Figure C.6, the outlet region is less strongly widened. The pressure-outlet result therefore supports the interpretation that the outlet expansion in the velocity-outlet baseline was at least partly caused by the imposed outflow profile rather than by the diffuser geometry alone. The optimized pressure-outlet objective is also lower than the velocity-outlet objective, decreasing from $J_D = 3.1164 \times 10^5$ W/m to $J_D = 2.7664 \times 10^5$ W/m, or by about 11.2%. This reduction is not interpreted as a direct performance ranking between two equivalent designs, because the boundary-value problem has changed. Both cases were run for 1150 MMA iterations, so the comparison uses the same optimization budget, but the changed outlet condition also changes the state solution, adjoint sensitivities, and local optimization path. The objective difference may therefore contain both boundary-condition effects and local-path effects. The main conclusion is that removing the outlet Dirichlet velocity constraint gives the optimizer a less restrictive outlet treatment: the flow no longer has to be redistributed to match a prescribed parabolic outlet profile, so less loss is required to satisfy the downstream boundary condition.



(A) Projected density field $\bar{\gamma}$. (B) Velocity magnitude and vector field.

Figure 6.11: FEniCS-SA optimization result for the pressure-outlet diffuser at $V_f = 0.30$. Final objective value $J_D = 2.7664 \times 10^5$ W/m.

The corresponding pressure-outlet convergence history through this common comparison point is reported in Figure C.8 of Appendix C.

6.5.3 Pressure-Outlet CFD Validation

The extracted pressure-outlet topology is validated with the same sharp-geometry workflow as the pipe bend and U-bend. The Ansys-SA solve checks the transfer from the diffuse FEniCS design to a sharp-wall geometry under the same closure family, while the Ansys SST $k - \omega$ solve tests whether the ranking is robust under an independent two-equation model. Because the optimization objective is J_D, the validation emphasizes both physical viscous dissipation and static pressure drop.

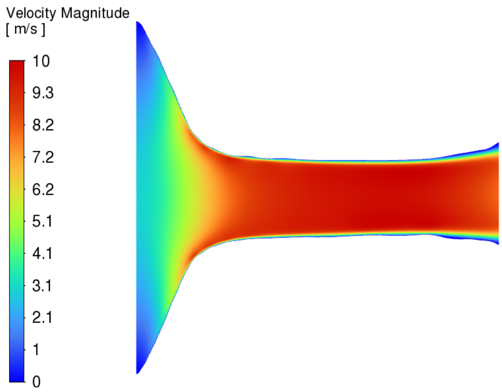
TABLE 6.6: CFD validation metrics for the extracted pressure-outlet diffuser topology.

Evaluation	Φ_D [W/m]	Δp_s [Pa]	Interpretation
FEniCS-SA diffuse topology	2.6307×10^5	2.6022×10^5	Diffuse-domain diagnostic; total objective $J_D = 2.7664 \times 10^5$ W/m
Ansys-SA sharp geometry	1.1885×10^5	1.1541×10^5	Sharp-wall re-analysis with the optimization closure family
Ansys-SST $k - \omega$ sharp geometry	1.2834×10^4	5.0006×10^4	Independent closure check; much lower modeled dissipation

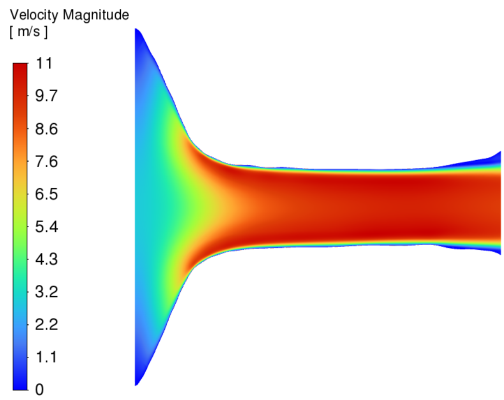
The FEniCS-SA values in Table 6.6 are approximately 2.21 times larger in Φ_D and 2.25 times larger in Δp_s than the Ansys-SA sharp-geometry values. This discrepancy is too large to claim quantitative cross-code agreement. It is interpreted as a transfer limitation of this diffuser case: the FEniCS result is a diffuse Brinkman-domain optimization diagnostic, whereas the Ansys result is a sharp-wall re-analysis of an extracted boundary. The mismatch is amplified by the pressure-outlet diffuser's separated or recovering downstream flow. The final hydraulic performance is therefore judged from the Ansys re-analyses, while the FEniCS values document the optimization objective reached by the design loop.

The Ansys-SA loss is also substantially larger than the SST $k - \omega$ loss. For the turbulent pressure-outlet design, the SA pressure drop is 2.31 times the SST $k - \omega$ value, and the SA dissipation is 9.26 times the SST $k - \omega$ value. Figure 6.12 explains this split. The velocity-magnitude fields are qualitatively similar, but the turbulent-viscosity-ratio fields are not: SA predicts a much larger effective-viscosity level than SST $k - \omega$ in this diffuser. Since the validation dissipation integrand is $2\mu_{\text{eff}}\mathbf{S} : \mathbf{S}$ through Eq. 6.1, this directly increases both the integrated dissipation and the pressure loss. The result should therefore be read as closure sensitivity, not as evidence that one closure is categorically unreliable.

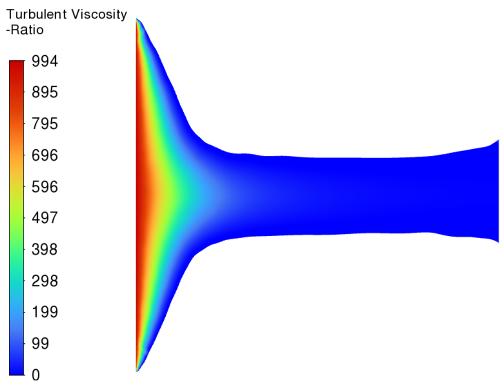
Local dissipation-density contours were also inspected, but they are not retained as a main figure because the field is dominated by the inlet contraction edges and does not clearly identify the downstream separation or recovery behavior. This is physically consistent with the form of the dissipation density, $2\mu_{\text{eff}}\mathbf{S} : \mathbf{S}$: it peaks where strain rates are largest, which in these diffuser designs occurs mainly where the full-height inlet flow is accelerated into the narrow channel. The widened or recovering downstream region can still be important for outlet sensitivity and turbulence-model dependence, but a separated low-speed region does not necessarily create a large local dissipation-density peak if the local strain rate is modest. For this diffuser, the more useful visual diagnostics are therefore the topology, velocity field, and turbulent-viscosity ratio, while the integrated dissipation remains a scalar validation metric.



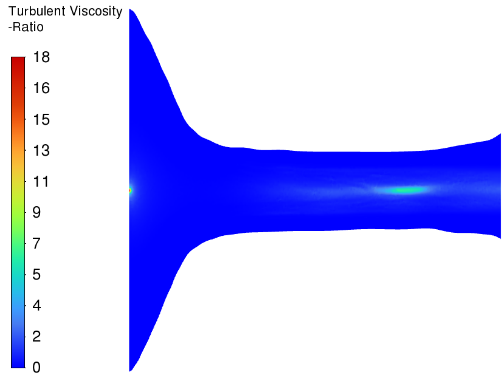
(A) Ansys-SA velocity magnitude.



(B) Ansys-SST $k-\omega$ velocity magnitude.



(C) Ansys-SA turbulent-viscosity ratio.



(D) Ansys-SST $k-\omega$ turbulent-viscosity ratio.

Figure 6.12: Closure diagnostics for the pressure-outlet turbulent diffuser. The velocity contours are similar enough that repeated full-domain velocity plots are not the main evidence; the turbulent-viscosity-ratio contours show the mechanism behind the much larger SA loss prediction.

6.5.4 Pressure-Outlet Laminar-Design Comparison

The laminar reference for the final diffuser comparison uses the same pressure-outlet boundary treatment, objective type, and volume fraction, but it is optimized at $Re = 1$ without the SA turbulence model. The extracted laminar and turbulent pressure-outlet layouts are then evaluated under the same target turbulent operating conditions. This comparison isolates whether the SA-optimized pressure-outlet design remains beneficial when the final sharp geometry is re-solved with the same closure and with SST $k-\omega$.

Table 6.7 compares the extracted laminar and turbulent pressure-outlet diffuser topologies under identical turbulent operating conditions. The topology and velocity fields were inspected, but they do not by themselves reveal the performance ranking clearly enough to justify a separate figure. Because the diffuser is optimized with J_D, lower Φ_D is the primary performance indicator, while the static pressure drop remains the most direct hydraulic interpretation of the extracted sharp geometry. Both SA and SST $k-\omega$ re-analyses are included because the diffuser benchmark is expected to remain turbulence-model-sensitive when separation is present.

TABLE 6.7: Performance comparison of the extracted pressure-outlet diffuser topologies under identical turbulent operating conditions.

Evaluation	Φ_D **[W/m]**	Δp_s **[Pa]**	**Relative performance**
Laminar TO, Ansys-SA	1.3495×10^5	1.3979×10^5	Baseline under SA
Turbulent TO, Ansys-SA	1.1885×10^5	1.1541×10^5	11.9% lower Φ_D; 17.4% lower Δp_s
Laminar TO, Ansys-SST $k-\omega$	1.1289×10^4	3.8332×10^4	Baseline under SST $k-\omega$
Turbulent TO, Ansys-SST $k-\omega$	1.2834×10^4	5.0006×10^4	13.7% higher Φ_D; 30.5% higher Δp_s

The SA re-analysis gives the positive turbulent-TO result sought in this benchmark: the SA-optimized turbulent topology reduces the pressure drop by 17.4% and the viscous dissipation by 11.9% relative to the pressure-outlet laminar topology. Under SST $k-\omega$, however, the ranking reverses. The turbulent topology gives a 30.5% higher pressure drop and a 13.7% higher dissipation than the laminar topology. This is not reported as a failure of the SA topology optimizer, because the optimizer did improve the design under the closure used in the design loop. It is instead reported as a closure-sensitive diffuser result: in this separated pressure-outlet case, the design ranking depends on whether the final sharp geometry is evaluated with SA or SST $k-\omega$.

6.6 Conclusion

This chapter answers the third research question by evaluating whether the turbulent topology optimization framework produces designs that remain credible after independent CFD re-analysis. The pipe-bend result provides the strongest completed evidence: the Appendix C frozen-sensitivity checks confirm the derivative supplied to MMA for J_p, and the extracted design remains robust when re-solved in Ansys Fluent with both SA and SST $k-\omega$.

For the pipe bend, changing the Ansys closure from SA to SST $k-\omega$ increases the validated pressure drop by only about 4.5% and the viscous dissipation by about 8.5%, indicating that the optimized geometry is not narrowly tuned to the SA model. Under identical SST $k-\omega$ operating conditions, the turbulent design also reduces the pressure drop and viscous dissipation relative to the laminar-optimized design, supporting the claim that high-Reynolds turbulence physics should be included directly in the design loop. The diffuser is treated more cautiously. The velocity-outlet baseline shows that a prescribed outlet profile can influence the optimized outlet geometry, so the pressure-outlet variant is used as the primary diffuser result. For this pressure-outlet diffuser, the turbulent SA design improves on the laminar design under Ansys-SA, reducing Δp_s by 17.4% and Φ_D by 11.9%. The same two geometries evaluated with SST $k-\omega$ give the opposite ranking, with the turbulent design increasing Δp_s by 30.5% and Φ_D by 13.7%. The diffuser therefore supports the value of turbulence-aware TO within the SA design model, but it does not support a closure-independent performance claim. It is retained as a closure-sensitivity result for separated diffuser flow. The U-bend remains the main incomplete high-Reynolds benchmark and should be judged by the same validation standard once its remaining metrics are replaced.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

The objective of this thesis was to build and assess a flow-only turbulent topology optimization framework for internal configurations that are relevant to energy and thermal-management systems. The word flow-only is essential. The framework developed here does not solve the energy equation and does not yet optimize conjugate heat transfer. Instead, it addresses the hydraulic prerequisite for those applications: generating low-loss internal flow paths under turbulent operating conditions. This narrower scope is deliberate, because turbulent momentum transport, wall-distance treatment, Brinkman penalization, and adjoint sensitivity assembly already form a difficult and only partially mature optimization problem.

The thesis structure follows that scope. Chapter 2 establishes the turbulence-model background and the expected limits of the Spalart-Allmaras (SA) model. Chapter 3 presents the FEniCS implementation of the SA forward solver, and Chapter 4 verifies it against Ansys Fluent before it is used for topology optimization. Chapter 5 then introduces the density-based turbulent optimization framework, including Brinkman penalization, topology-dependent SA and wall-distance penalties, filtering, projection, MMA, and the frozen-turbulence adjoint. Finally, Chapter 6 evaluates the resulting designs through independent sharp-geometry CFD re-analysis.

Research question 1: How accurately does the Spalart-Allmaras turbulence model predict curved and separated internal turbulent flows, and where do its physical limitations appear? The answer is conditional. SA is a practical baseline model for the present optimization framework because its one-equation, wall-resolved formulation is cheaper and simpler than two-equation closures while still representing near-wall eddy-viscosity effects. The pipe-bend validation supports this use: the optimized bend produces a smooth attached flow path, and Ansys-SA and Ansys-SST $k-\omega$ predict similar pressure-drop and dissipation levels for the extracted turbulent design. However, the diffuser case exposes the important limitation. There, separated low-velocity regions appear near the diverging outlet part, and the Ansys-SA and SST $k-\omega$ re-analyses give substantially different dissipation and pressure-drop values. This behavior is consistent with the known weakness of eddy-viscosity closures, and especially one-equation closures, in separated, anisotropic, adverse-pressure-gradient flow. SA is therefore suitable as a bounded engineering optimization closure, but not as a model whose diffuser predictions can be accepted without independent validation and flow-field diagnosis.

Research question 2: Can a custom FEniCS implementation of the Spalart-Allmaras turbulence model achieve quantitative agreement with Ansys Fluent for turbulent internal-flow benchmarks? Within the two-dimensional and wall-resolved scope of this thesis, the answer is yes. The implemented solver reproduces the governing ingredients needed by the optimizer: incompressible flow coupling, SA transport, eddy-viscosity reconstruction, wall-distance evaluation, and near-wall resolution. The periodic channel case verifies the near-wall turbulence balance in a canonical geometry, while the U-bend verification tests the same solver under curvature and stronger convection. These comparisons do not prove that SA is universally accurate, but they do show that the FEniCS implementation is consistent enough with Ansys-SA to serve as the forward model for the topology optimization studies.

Research question 3: To what extent do SA-driven topology optimizations produce final designs whose predicted performance remains consistent under independent Ansys re-analysis with SA and SST k–ω, and how do they compare with laminar-optimized designs under identical turbulent operating conditions? The strongest completed evidence is the pipe-bend benchmark. The frozen derivative used by MMA was checked directly through coordinate finite differences and a directional Taylor test, giving a maximum coordinate relative discrepancy of 5.92×10^{-5} and approximately second-order Taylor behavior. The extracted pipe-bend topology also remains credible under independent Ansys re-analysis: switching from Ansys-SA to SST $k-\omega$ changes the validated losses only modestly, and the turbulent design substantially outperforms the laminar-optimized reference under identical turbulent operating conditions. The U-bend remains the most important remaining benchmark because it combines high Reynolds number operation with strong flow reversal; any final claim for that case should only be made after the Re = 5,000 turbulent optimization and sharp-geometry validation placeholders are replaced. The diffuser already provides a useful negative result: under SA validation the turbulent design improves on the laminar design, but under SST $k - \omega$ the ranking reverses. That split does not invalidate the framework, but it shows that separated diffuser flows require model-sensitive interpretation rather than a single scalar performance claim.

Overall, the thesis demonstrates that a frozen-SA density-based topology optimization framework can produce credible low-loss turbulent internal-flow designs when the flow remains reasonably attached and when the final geometry is independently re-analyzed. It also shows where the framework is not yet sufficient: strongly separated diffuser behavior, objective oscillations during some MMA continuation stages, and incomplete high-Reynolds U-bend optimization still require additional computation and diagnosis. The most defensible conclusion is therefore not that the method is a finished turbulent thermal-design tool, but that it is a verified flow-only stepping stone toward one.

7.2 Limitations

An objective evaluation of this methodology reveals several important limitations that must be acknowledged:

1. **The Frozen Adjoint Approximation:** By ignoring the derivative of the eddy viscosity with respect to the design field, the optimizer cannot mathematically anticipate future turbulence production. In flows dominated by massive separation, this can blind the optimizer to the downstream hydrodynamic consequences of upstream geometric changes, potentially leading to sub-optimal local minima.

2. **Geometry Extraction Uncertainties:** The transition from a continuous, Brinkman-penalized density field ($\gamma \in [0, 1]$) to a Boolean CAD geometry introduces inherent ambiguity. The specific iso-value threshold chosen to extract the fluid volume dictates the final wall placement. Consequently, the performance discrepancies observed between FEniCS and Ansys are inextricably linked to this manual geometry extraction process, making it difficult to isolate pure solver physics from thresholding effects.

3. **Two-Dimensional Simplification:** The benchmark geometries evaluated in this thesis were restricted to 2D planar domains. Because turbulent phenomena—particularly curvature-induced secondary flows like Dean vortices and turbulent energy cascades—are inherently three-dimensional, the 2D assumption artificially suppresses complex mixing mechanisms that would inevitably occur in physical prototypes.

4. **Lack of Experimental Validation:** While independent cross-code verification against commercial finite-volume solvers strongly corroborates the numerical implementation, it is not a substitute for physical validation. The true operational performance of these algorithmically generated turbulent manifolds requires future validation against experimental pressure-drop and flow-field measurements.

5. **Sensitivity Verification Scope:** The finite-difference and Taylor checks verify the frozen derivative used by MMA, not the derivative of a fully coupled turbulence, wall-distance, and geometry-extraction pipeline. This distinction is important when interpreting the sensitivity evidence: the optimizer is internally consistent with its chosen approximation, but the approximation itself remains a modeling decision.

6. **Optimization Robustness:** Some continuation histories still show objective oscillations, especially when pressure-based objectives are used as proxies for energy loss. Although pressure drop and dissipation are physically related for

steady internal flows, a domain-integrated dissipation objective can provide a less localized and more stable optimization signal than an average inlet-pressure objective. The final benchmark conclusions should therefore distinguish completed, validated runs from remaining runs that still require stricter continuation, smaller MMA move limits, or longer fixed-parameter polishing stages.

7.3 Future Work

The verified turbulent fluid mechanics framework developed in this thesis establishes a mathematically stable foundation for several advanced avenues of future research:

7.3.1 Immediate Benchmark Completion and Diagnostics

Before extending the framework physically, the remaining two-dimensional benchmark evidence should be completed. The highest-priority computation is the $Re =$ 5,000 turbulent U-bend topology optimization, preferably with the dissipation objective J_D rather than the average inlet-pressure proxy J_p. If the full high-Reynolds run remains unstable, an intermediate Reynolds number should be used only as supporting evidence for the onset of asymmetric high-inertia topology formation, not as a replacement for the target benchmark.

The pipe-bend case should be repeated with the same J_D objective and stricter MMA damping to test whether the current objective oscillations are caused primarily by the localized pressure objective. If oscillations persist, a long fixed-continuation polishing run at the most stable q, β, and move-limit setting should be used, and the remaining oscillatory behavior should be reported as an optimization-robustness limitation rather than hidden.

For the diffuser, the main unresolved question is not simply whether the turbulent design has a lower scalar loss, but where the turbulence-model discrepancy originates. The separated low-velocity region near the diverging outlet should be isolated by re-running the extracted geometry after trimming the diverging part, by testing a pressure-outlet topology-optimization variant, and by plotting cellwise dissipation density, turbulent viscosity, and the turbulent component of dissipation for both SA and SST $k - \omega$ re-analyses. These plots would directly test whether the large SA dissipation originates in the separated outlet region and whether it is driven by over-predicted eddy viscosity.

7.3.2 Conjugate Heat Transfer (CHT) and Thermal Management

The ultimate engineering application for high-Reynolds topology optimization is not merely fluid routing, but active thermal management. Historically, the immense computational cost of resolving Conjugate Heat Transfer (CHT) has forced optimization frameworks to rely on laminar approximations (e.g., natural convection as demonstrated by Alexandersen et al. [3]) or simplified flow proxies such as Darcy models [65]. Even when high-resolution, density-based solvers are successfully deployed for microelectronics cooling, they typically circumvent the complexities of turbulent momentum transport [11].

However, high-performance heat sinks still require models that capture turbulent convection. Recent work has introduced turbulence into thermal topology optimization using extended finite element (XFEM) level-set approaches with the SA model [40], computationally reduced multi-layer thermofluid approximations [64], and standard density-based frameworks coupled to two-equation $k-\epsilon$ closures [54].

With the turbulent momentum and SA transport equations implemented and verified within the current FEniCS architecture, the framework provides a practical starting point for turbulent CHT optimization. Extending the current methodology to CHT will require implementing a turbulent Prandtl number (Pr_t) approach to model the turbulent heat flux vector, alongside deriving the corresponding thermal adjoint sensitivities to minimize peak temperatures or maximize overall heat transfer coefficients. Furthermore, the capacity for optimizing two-fluid heat exchangers—currently dominated by laminar assumptions [30]—could be improved by adapting this turbulent solver.

7.3.3 Three-Dimensional Topologies

As noted in the limitations, turbulent secondary flows are inherently 3D phenomena. To fully evaluate the geometric limits of the turbulence models and design physically realizable manifolds, the current 2D FEniCS implementation must be extended to three dimensions. The mathematical structure of the governing equations and adjoint derivations remains similar, but the computational cost changes by orders of magnitude because wall-resolved turbulent optimization requires fine near-wall resolution over a three-dimensional design domain. Large-scale topology optimization studies have demonstrated that parallel finite-element optimization at extreme design-variable counts is possible when the formulation is paired with efficient domain decomposition and scalable solvers [1]. A 3D turbulent extension of the present work should therefore prioritize parallel mesh generation, robust algebraic multigrid preconditioning, and automated sharp-geometry extraction.

7.3.4 Adjoint Model Hierarchies

This thesis deliberately uses the frozen-turbulence adjoint as the robust baseline. Future frameworks should investigate stabilized ways of reintroducing selected turbulence-production sensitivities, especially in separated flows where the constant-eddy-viscosity assumption is least accurate. Existing turbulent topology optimization studies already span several levels of adjoint complexity, from frozen or constant-eddy-viscosity approximations [20, 67] to more complete RANS sensitivity formulations and implementation studies [37, 62]. A practical next step is therefore not necessarily to jump directly to a fully coupled turbulence adjoint, but to build an adjoint hierarchy: frozen SA sensitivities as the baseline, selected SA transport sensitivities as an intermediate model, and two-equation closures such as SST $k - \omega$ for cases where curvature, separation, and anisotropic shear dominate the design response.

Appendices

Appendix A

Chapter 2 Ansys Fluent Benchmark Support

This appendix supports the turbulence-modeling discussion in Chapter 2. It contains the common Ansys Fluent convergence-monitor definitions for the Chapter 2 benchmark runs and the backward-facing-step material that is useful as separated-flow model-sensitivity evidence. The backward-facing-step case is secondary to the main Chapter 2 argument based on the Driver–Seegmiller/NASA literature comparison and is not used as a FEniCS solver-verification benchmark.

Unless stated otherwise, all grid-sensitivity relative differences in this appendix are reported as positive percentage magnitudes. For a mesh-dependent metric ϕ, the coarse–medium difference is computed as $|\phi_{\text{coarse}} - \phi_{\text{medium}}|/|\phi_{\text{coarse}}| \times 100\%$, while the medium–fine difference is computed as $|\phi_{\text{medium}} - \phi_{\text{fine}}|/|\phi_{\text{fine}}| \times 100\%$.

A.1 Fluent Benchmark Convergence Monitors

For the Fluent benchmark runs in Section 2.7, convergence was assessed using Fluent report monitors. Three physical monitors were tracked in addition to the solver residuals: the inlet-to-outlet static pressure drop, the domain-integrated viscous dissipation, and the relative mass imbalance. The pressure-drop monitor was defined from area-weighted static pressures,

$$\Delta p = \bar{p}_{\text{in}} - \bar{p}_{\text{out}}, \tag{A.1}$$

while the viscous-dissipation monitor used a Fluent custom field function,

$$\Phi_D = \int_\Omega \mu_{\text{eff}} \dot{\gamma}^2 \, d\Omega, \tag{A.2}$$

where the custom field function `viscous-dissipation-density` was defined as $\mu_{\text{eff}}\dot{\gamma}^2$ using Fluent's `effective-viscosity` and `strain-rate-magnitude` variables. Here $\mu_{\text{eff}} = \mu + \mu_t$ for turbulent runs and $\dot{\gamma}$ is Fluent's scalar strain-rate magnitude. For a monitored scalar q, the final-window change over N iterations was evaluated as

$$\delta_N(q) = \frac{\left|q^{(k)} - q^{(k-N)}\right|}{\left|q^{(k)}\right|}. \tag{A.3}$$

The relative mass imbalance was computed from the signed inlet and outlet mass-flow-rate reports,

$$\epsilon_m = \frac{|\dot{m}_{\text{in}} + \dot{m}_{\text{out}}|}{|\dot{m}_{\text{out}}|}, \tag{A.4}$$

where $\dot{m}_{\text{out}} < 0$ under Fluent's boundary-normal sign convention for these cases.

TABLE A.1: Convergence metrics used for the Ansys Fluent benchmark runs in Section 2.7. The residual criterion refers to all Fluent-scaled residuals, including continuity, momentum, and turbulence-transport equations.

Benchmark	**Window**	**Pressure-drop change** $\delta_N(\Delta p)$	**Dissipation change** $\delta_N(\Phi_D)$	**Mass imbalance** ϵ_m	**Scaled residuals**
VMFL030 90° bend, 3D	500 iters	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	10^{-6}
VMFL048 U-bend, 3D	500 iters	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	10^{-6}
Backward-facing step, 2D	250 iters	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	10^{-6}

A.2 Ansys SA–SST $k-\omega$ Backward-Facing-Step Benchmark

The third benchmark evaluates the backward-facing-step geometry introduced in Section 2.6. This case is included to expose turbulence-model sensitivity in separated internal flow, not to replace the experimental validation evidence from the Driver-Seegmiller/NASA comparisons discussed above. The geometry and boundary conditions are based on the backward-facing-step configuration from Marcibál's master thesis and accompanying codebase [36, 31]. They are shown in Figure A.1. The flow enters a channel of height $H_{\text{in}} = 2.0$ m with a uniform inlet velocity of $U_{\text{in}} = 25$ m/s. At $x = 0$, the lower wall drops by the step height $h = 0.25$ m, giving an outlet height of $H_{\text{out}} = 2.25$ m. The benchmark Reynolds number is defined using the step height,

$$Re_h = \frac{U_{\text{in}} h}{\nu} = 3.44 \times 10^4, \tag{A.5}$$

because backward-facing-step results are conventionally compared using the step height and the reattachment length x_r/h. For reference, using the inlet-channel height would give $Re_{H,\text{in}} = 2.75 \times 10^5$, but this is not the primary scaling for the separated region.

The Fluent setup mirrors the wall-resolved SA assumptions used throughout this thesis. The inlet velocity is prescribed as $\mathbf{u} = (25, 0)$ m/s, the outlet pressure is fixed at $p = 0$ Pa, and the no-slip walls enforce $\tilde{\nu} = 0$. The inlet modified turbulent viscosity is set to $\tilde{\nu} = 0.18449$ m^2/s, corresponding to the eddy-viscosity level used in the FEniCS backstep configuration. The important comparison quantity is the reattachment length x_r/h, obtained from the downstream lower-wall shear stress

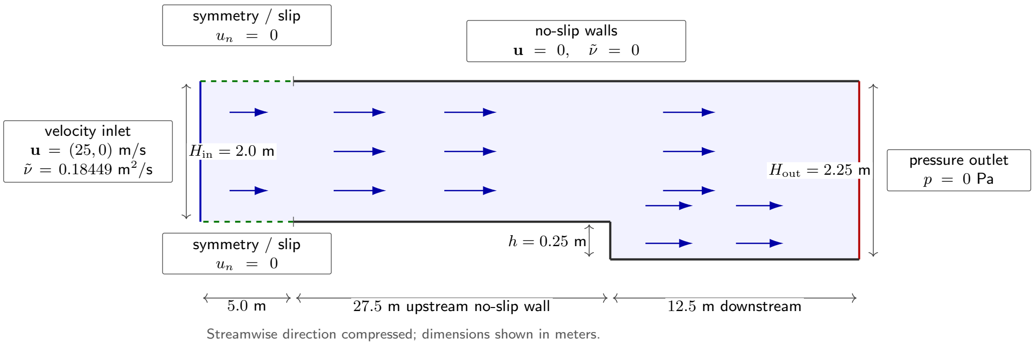


Figure A.1: Backward-facing-step benchmark geometry and boundary conditions, based on the configuration used by Marcibál and the associated code implementation [36, 31]. The first 5.0 m of the upper and lower boundaries are treated as symmetry/slip segments, while all remaining solid boundaries are no-slip walls. The streamwise dimension is compressed for readability.

changing sign from negative to positive. Secondary diagnostics are the size of the recirculation zone, the pressure recovery, and the near-wall skin-friction distribution.

A.2.1 Wall-Resolved Ansys Grid-Sensitivity Study

The grid-sensitivity study uses the same backward-facing-step geometry and named selections as the benchmark definition above. The no-slip wall boundaries are inflated with a fixed first-layer height of $\Delta y_1 = 1.30 \times 10^{-4}$ m. This first-layer height is kept constant on all three grids because the Reynolds number and target wall resolution are unchanged. The mesh refinement is therefore applied through the number of inflation layers, inflation growth rate, and streamwise element sizing. Unlike the attached channel and bend verification cases, this separated-flow benchmark is treated as a grid-sensitivity record rather than a strict verification case, while the SA–SST $k-\omega$ comparison below uses the finest converged mesh for each closure.

The primary grid-sensitivity metric is the normalized reattachment length x_r/h, where x_r is obtained from the downstream zero-crossing of the streamwise wall shear stress on the lower wall associated with the end of the primary recirculation region. The SA reattachment length decreases from $x_r/h = 5.92$ on the coarse grid to 5.83 on the medium grid and 5.69 on the fine grid; the resulting medium–fine change remains above a strict one-percent mesh-independence criterion for this separated-flow metric. The total pressure loss Δp_t is retained as a secondary metric and is computed from the mass-flow-weighted inlet-to-outlet total-pressure difference. Velocity profiles are additionally extracted at $x/h = 1$, 4, 6, and 10, corresponding to $x = 0.25$, 1.00, 1.50, and 2.50 m downstream of the step. These stations cover the initial separated shear layer, the recirculation region, the expected reattachment region, and the downstream recovery region; the SA–SST $k-\omega$ comparison is made using the finest converged mesh available for each turbulence closure.

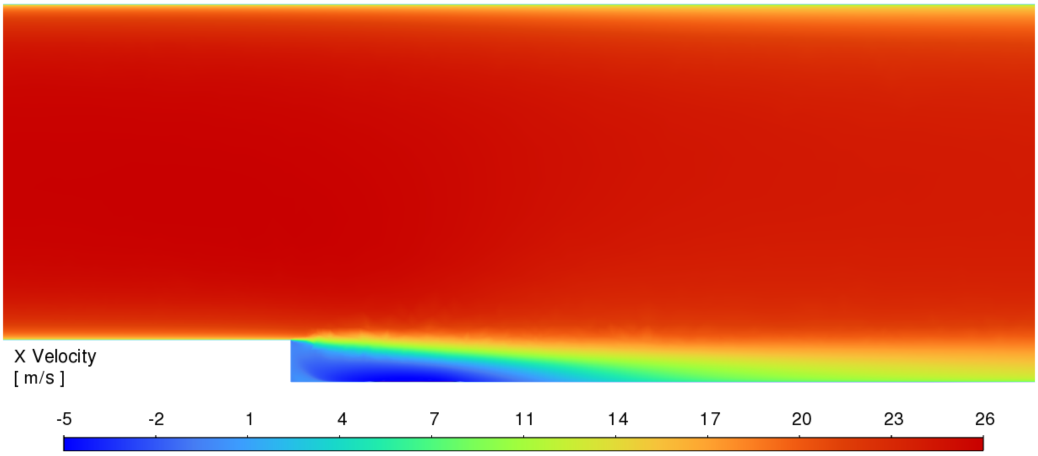
TABLE A.2: Wall-resolved backward-facing-step mesh hierarchy for the Ansys Fluent grid-sensitivity study. The first-layer height is fixed to maintain $y^+ \approx 1$, while the boundary-layer thickness and streamwise resolution are refined.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Mesh method	Quad-dom.	Quad-dom.	Quad-dom.
Element order	Linear	Linear	Linear
Inflation option	First layer	First layer	First layer
Inflated boundaries	No-slip walls	No-slip walls	No-slip walls
Δy_1 [m]	1.30×10^{-4}	1.30×10^{-4}	1.30×10^{-4}
Inflation layers	28	36	44
Growth rate	1.18	1.15	1.12
Near-step wall size [m]	2.50×10^{-2}	1.25×10^{-2}	6.25×10^{-3}
Top-wall size [m]	5.00×10^{-2}	2.50×10^{-2}	1.25×10^{-2}
Inlet/outlet/symmetry size [m]	1.00×10^{-1}	5.00×10^{-2}	2.50×10^{-2}
Nodes	77,725	193,276	457,175
Elements	86,371	192,131	454,431
Grid metrics: Spalart–Allmaras			
x_r/h	5.92	5.44	5.69
$\epsilon_{\text{rel}}(x_r/h)$ [%]	–	8.11	4.39
Total pressure loss Δp_t [Pa]	46.743	46.795	46.531
$\epsilon_{\text{rel}}(\Delta p_t)$ [%]	–	0.11	0.57
Grid metrics: SST $k-\omega$			
x_r/h	6.09	6.13	6.08
$\epsilon_{\text{rel}}(x_r/h)$ [%]	–	0.66	0.82
Total pressure loss Δp_t [Pa]	44.741	45.139	45.142
$\epsilon_{\text{rel}}(\Delta p_t)$ [%]	–	0.89	0.01

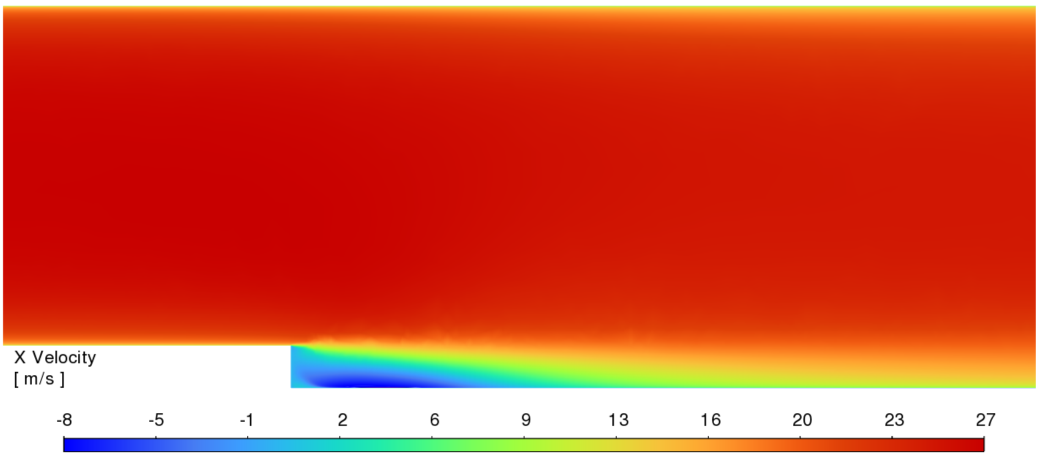
Note. Quad-dom. = quad-dominant. The near-step wall size is applied to `upstream_lower_wall`, `step_wall`, and `downstream_lower_wall`. The inflated no-slip walls are `top_wall`, `upstream_lower_wall`, `step_wall`, and `downstream_lower_wall`; the inlet, outlet, and symmetry boundaries are not inflated. For the ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change. The SA reattachment length is reported as grid-sensitivity evidence rather than proof of strict mesh independence.

The contour plots below should therefore be read as a model-form comparison on the same local backward-facing-step geometry, not as a new validation against the NASA Driver-Seegmiller geometry. The streamwise-velocity fields identify the separated low-speed region and the downstream recovery, while the turbulent-viscosity ratio shows how each closure supplies the modeled momentum diffusion that controls this recovery. The velocity contours use the exported range of each Fluent solution, whereas the turbulent-viscosity plots use a common color scale.

Figure A.2 shows that the two closures give the same qualitative flow topology: separation at the step, a recirculating near-wall region, and redevelopment downstream. The difference is in the local recovery. The SA solution gives a smoother transition from reversed flow to positive streamwise velocity, whereas the SST $k-\omega$ solution retains a more structured low-speed region near the lower wall and a sharper separated shear layer. This is the expected sensitivity for a backward-facing step: reattachment is governed by shear-layer growth and turbulent transport, so small differences in the modeled eddy viscosity can change the local velocity field even



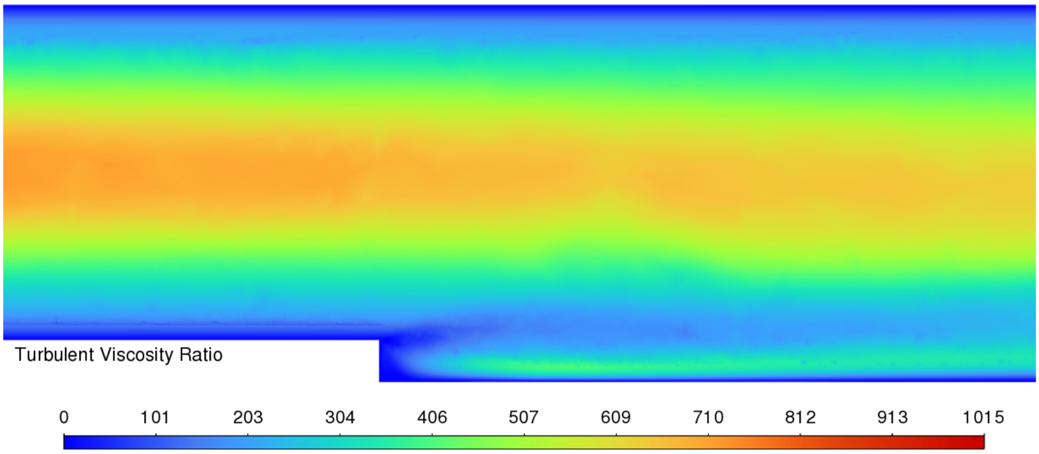
(A) SST $k - \omega$ contour for u_x.



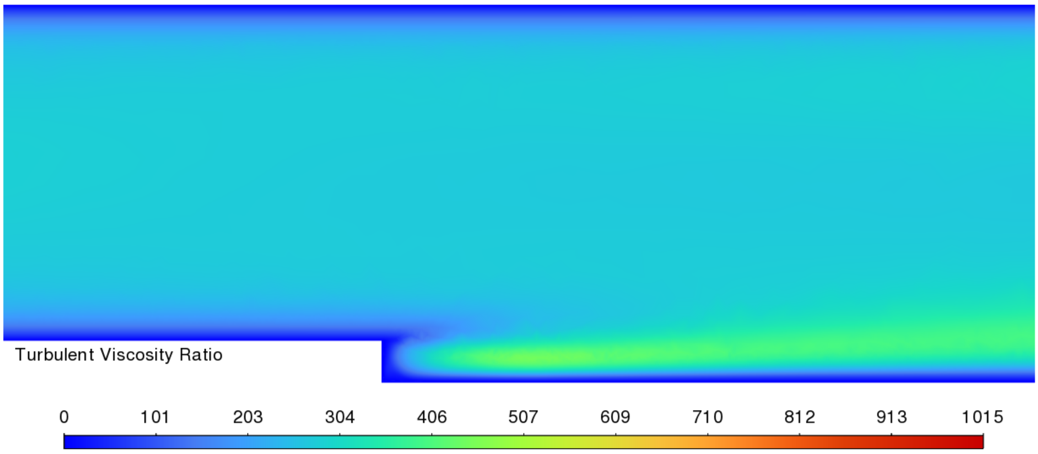
(B) SA contour for u_x.

Figure A.2: Streamwise-velocity contours for the backward-facing-step benchmark. Both closures predict a low-speed or reversed-flow region immediately downstream of the step and gradual recovery farther downstream, but the size and diffusion of this region are visibly model dependent.

when the global recirculation pattern remains plausible.



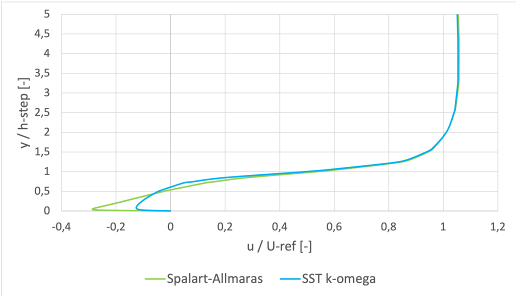
(A) SST $k-\omega$ contour for μ_t/μ.



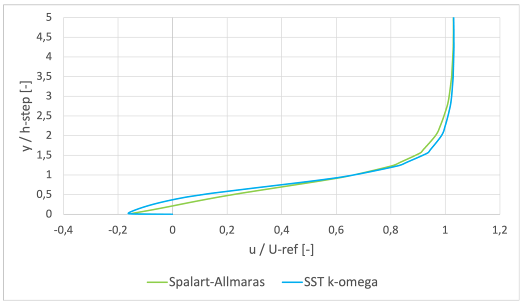
(B) SA contour for μ_t/μ.

FIGURE A.3: Turbulent-viscosity-ratio contours for the backward-facing-step benchmark, plotted with a common color scale. SA produces a smoother, more moderate eddy-viscosity field, while SST $k-\omega$ gives a more stratified distribution with stronger variation between the near-wall/reverse-flow region and the outer shear-dominated flow.

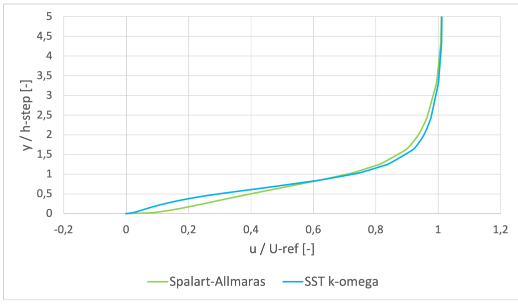
The turbulent-viscosity contours in Figure A.3 explain the velocity differences. SA is a one-equation model: it transports a modified eddy-viscosity variable and reconstructs μ_t mainly from local vorticity and wall-distance effects. Its μ_t/μ field is therefore relatively smooth in this separated channel. SST $k-\omega$ solves separate equations for turbulent kinetic energy and specific dissipation rate, with near-wall $k-\omega$ behavior and outer-flow blending. In this case it preserves lower eddy viscosity in the immediate wall-bounded/reversed-flow region while carrying a stronger non-uniform eddy-viscosity distribution through the outer part of the channel. The resulting difference is not a numerical detail; it is the model-form sensitivity that makes backward-facing-step flow a useful warning case for SA-based topology optimization.



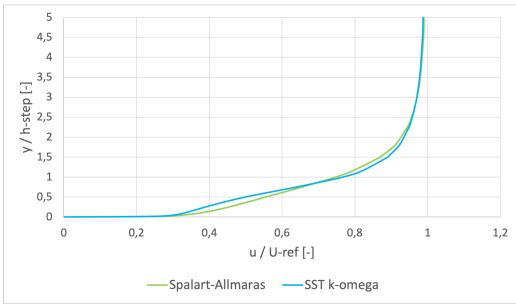
(a) $x/h = 1$



(b) $x/h = 4$



(c) $x/h = 6$



(d) $x/h = 10$

Figure A.4: Normalized streamwise velocity profiles for the backward-facing-step benchmark at four downstream stations. The comparison isolates model-form sensitivity by running SA and SST $k-\omega$ on the same local backward-facing-step geometry and boundary conditions.

Figures A.2 and A.3 provide the full-field contour comparison, while Figure A.4 shows why this benchmark was included even though it is not a direct experimental validation. The purpose is to compare SA and SST $k-\omega$ on the same separated-flow geometry and determine how strongly the modeled mean recirculation and recovery depend on the selected RANS closure. Near the step, at $x/h = 1$ and $x/h = 4$, both models produce a low-speed or reversed-flow region close to the lower wall, which is consistent with a modeled recirculation bubble generated by the sudden expansion. However, the velocity profiles in this separated region are visibly model dependent, especially near the lower wall and in the developing shear layer. Farther downstream, at $x/h = 6$ and especially $x/h = 10$, the near-wall streamwise velocity becomes positive and the two profiles move closer together, consistent with modeled reattachment and redevelopment of the boundary layer.

This case is therefore interpreted as a model-sensitivity study rather than as a validation case. Without a direct comparison to the Driver-Seegmiller experimental data for the exact mesh, inlet condition, and domain definition used here, the Fluent results cannot determine which of the two closures is more accurate for this specific setup. The experimental comparison in Figure 2.5 already shows the broader trend: in separated backward-facing-step flow, SST $k-\omega$ gives better local velocity-profile agreement than SA, even though both models may predict a reasonable reattachment length. The present Fluent comparison extends that lesson to the geometry used later in the thesis. It shows that the separated region and recovery are sensitive to the turbulence model, so an SA-based optimization result containing separation cannot be interpreted as quantitatively reliable on its own. This limitation is fundamental to linear eddy-viscosity RANS closures: the Boussinesq approximation represents turbulent stresses through a scalar eddy viscosity and cannot fully represent stress anisotropy, non-local turbulence transport, or unsteady coherent structures in separated shear layers. The conclusion is therefore deliberately limited and more conservative than “SA captures separation”: SA can generate a plausible RANS mean recirculation pattern, but the separated-flow prediction is model dependent and must be checked with an independent closure such as SST $k-\omega$, and ideally with experimental or higher-fidelity validation.

Appendix B

Chapter 4 Spalart-Allmaras Solver Verification Details

This appendix supports Chapter 4. It contains the full FEniCS configuration tables and the FEniCS/Ansys mesh-independence studies for the channel and U-bend verification cases. The backward-facing-step material is placed in Appendix A, because it supports the Chapter 2 turbulence-model discussion rather than the Chapter 4 forward-solver verification.

Unless stated otherwise, all grid-independence relative differences in this appendix are reported as positive percentage magnitudes. For a mesh-dependent metric ϕ, the coarse–medium difference is computed as $|\phi_{\text{coarse}} - \phi_{\text{medium}}|/|\phi_{\text{coarse}}| \times 100\%$, while the medium–fine difference is computed as $|\phi_{\text{medium}} - \phi_{\text{fine}}|/|\phi_{\text{fine}}| \times 100\%$.

B.1 Verification Benchmark 1: Channel Flow

B.1.1 Physical and Geometric Parameters

The first verification benchmark consists of a pressure-driven periodic 2D channel. To align with fundamental turbulence research conventions, the characteristic length is defined as the full channel height (H), and the Reynolds number (Re_H) is computed strictly based on this geometric parameter rather than the hydraulic diameter. The complete physical setup is detailed in Table B.1.

B.1.2 Numerical Simulation Parameters

The channel flow was resolved using an Incremental Pressure Correction Scheme (IPCS) embedded within a Picard fixed-point iteration loop. The exact numerical solver configurations, convergence tolerances, and relaxation factors utilized in FEniCS are summarized in Table B.2.

TABLE B.1: Geometric, physical, and boundary-condition parameters for the periodic channel flow benchmark.

Parameter	**Channel flow setting**
Benchmark type	Pressure-driven, streamwise-periodic 2D channel
Domain dimensions	Length $L = 2.0$ m; full channel height $H = 2.0$ m
Characteristic length	Full channel height $H = 2.0$ m
Reference velocity	$U_{\mathrm{ref}} = 18.5$ m/s, utilized to define the target Reynolds number
Kinematic viscosity	$\nu = 1.81818 \times 10^{-3}$ m^2/s
Reynolds number	$Re_H \approx 2.03 \times 10^4$
Body force	$\mathbf{f} = (1.0, 0)$ m/s^2 in the kinematic-pressure convention
Driving condition	Streamwise body-force mode, equivalent to a pressure drop of 2 Pa over the periodic length $L = 2$ m
Velocity boundary conditions	Periodic velocity between inlet/outlet planes; no-slip on top and bottom walls
Pressure boundary conditions	Periodic pressure field with no inlet/outlet pressure Dirichlet condition; pressure mean normalized during IPCS
SA boundary conditions	Periodic $\tilde{\nu}$ between inlet/outlet planes; $\tilde{\nu} = 0$ at walls
Initial fields	$\mathbf{u}_0 = (18.5, 0)$ m/s, $p_0 = 0$ Pa, $\tilde{\nu}_0 = 5.0 \times 10^{-3}$ m^2/s
Verification mesh	`Fine_FEniCS`; first layer $\Delta y_1 = 1.709 \times 10^{-3}$ m

B.1.3 Spatial Verification and Grid Independence

To guarantee that the numerical results were independent of the spatial discretization, a formal grid independence study was conducted. Three distinct boundary-layer inflated meshes (Coarse, Medium, and Fine) were evaluated. The geometric properties, node counts, and resolution parameters of this mesh hierarchy are explicitly detailed in Table B.3. Crucially, the first-layer height was kept constant across all grids to strictly enforce the $y^+ \approx 1$ requirement, while the bulk and inflation-layer resolutions were systematically refined.

Visual confirmation of this spatial convergence is provided by the wall-normal velocity profiles plotted in Figure B.1. In each case, the profile is extracted at the streamwise mid-plane, $x = 0.5L$, with the wall-normal coordinate y varied across the full channel height. The `Fine_FEniCS` grid was consequently selected for the final verification against the Ansys Fluent baseline, while the coarser grids demonstrate that the wall-normal velocity profile is already asymptotically converged.

Furthermore, to ensure a rigorous cross-code validation, an equivalent grid independence study was performed for the commercial Ansys Fluent baseline. The Ansys domain was discretized using a comparable mapped mesh hierarchy, maintaining a strict $y^+ \approx 1$ resolution at the walls. In addition to the residual monitors, the bulk velocity was monitored and required to change by less than 1.0×10^{-3} over 250 iterations before accepting a converged solution. The continuity residual criterion was set to 1.0×10^{-5}, while the momentum and SA working-variable residual criteria were set to 1.0×10^{-6}. Mass conservation was also checked from the inlet and outlet mass-flow reports. The properties of these meshes are documented in Table B.4. The wall-normal velocity profiles extracted from the Ansys solver at the

Table B.2: Numerical simulation parameters for the FEniCS steady Spalart-Allmaras channel benchmark.

Parameter	**Symbol**	**Setting**
Velocity space	$\mathbf{u}$	Vector CG2, periodic in streamwise direction
Pressure space	p	Scalar CG1
SA working-variable space	$\tilde{\nu}$	Scalar CG1, periodic in streamwise direction
Quadrature degree	–	2
Maximum outer Picard steps	$N_{\text{Picard,max}}$	400
SA solves per Picard step	–	1
Outer velocity tolerance	$\epsilon_{\mathbf{u}}$	1.0×10^{-5}
Outer pressure tolerance	ϵ_p	1.0×10^{-6}
Outer SA tolerance	$\epsilon_{\tilde{\nu}}$	1.0×10^{-6}
Outer diagnostic convergence	–	Midline bulk velocity window of 100 Picard steps with relative tolerance 1.0×10^{-3}
IPCS pseudo-time step	Δt	5.0×10^{-3} s
Maximum IPCS steps per Picard step	–	500
IPCS velocity tolerance	–	1.0×10^{-6}
IPCS pressure tolerance	–	1.0×10^{-6}; pressure convergence is diagnostic in body-force mode
IPCS pressure normalization	–	Enabled; the pressure correction is mean-normalized
Velocity relaxation factor	α_u	0.3
Pressure relaxation factor	α_p	0.3
SA relaxation factor	$\alpha_{\tilde{\nu}}$	0.3
SA lower bound	$\tilde{\nu}_{\min}$	1.0×10^{-12}
Wall-distance equation	–	Relaxed Eikonal, $\sigma_w = 0.1$, $G_0 = 20$
Wall-distance Newton settings	–	Relative tolerance 1.0×10^{-8}; absolute tolerance 1.0×10^{-10}; maximum 80 iterations; relaxation 0.5
Linear solver strategy	–	IPCS blocks: direct MUMPS; SA transport: default solver

same mid-channel location, illustrated in Figure B.2, exhibited identical asymptotic convergence, confirming that the baseline reference data is mathematically robust and completely independent of spatial discretization errors.

TABLE B.3: Wall-resolved channel mesh hierarchy used for the FEniCS grid-independence study. The first-layer height is kept fixed to maintain $y^+ \approx 1$, while the streamwise resolution, wall-normal resolution, and inflation-layer resolution are refined.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Streamwise cells	10	20	40
Wall-normal cells	42	51	68
Δy_1 [m]	1.709×10^{-3}	1.709×10^{-3}	1.709×10^{-3}
$\Delta y_{\min}$ [m]	1.709×10^{-3}	1.709×10^{-3}	1.709×10^{-3}
$\Delta y_{\max}$ [m]	7.707×10^{-2}	8.454×10^{-2}	7.589×10^{-2}
Inflation layers per wall	10	14	18
Growth ratio	1.45	1.35	1.25
Vertices	473	1,092	2,829
Elements	840	2,040	5,440
Grid metric			
U_{bulk} [m/s]	16.923	18.489	18.472
$\epsilon_{\text{rel}}(U_{\text{bulk}})$ [%]	–	9.25	0.09

Note. For the ϵ_{rel} row, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

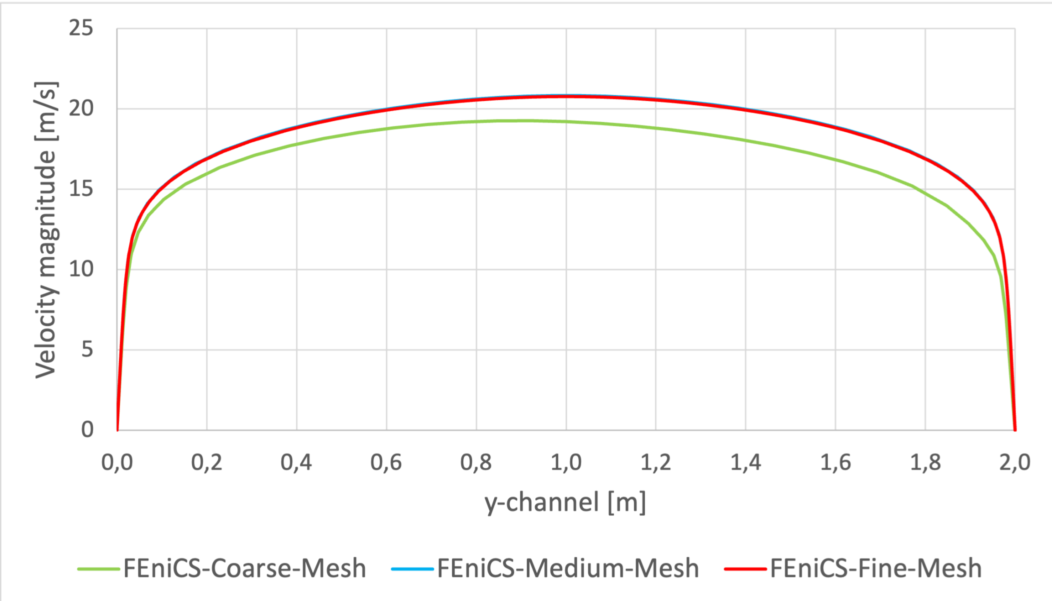


FIGURE B.1: FEniCS wall-normal velocity profiles extracted at $x = 0.5L$ for the Coarse, Medium, and Fine grids in the periodic channel flow benchmark. The overlapping profiles demonstrate asymptotic spatial convergence.

Table B.4: Wall-resolved channel mesh hierarchy used for the Ansys grid-independence study.

Parameter	Coarse	Medium	Fine
Mesh settings			
Mesh type	MultiZone	MultiZone	MultiZone
Face type	All quad	All quad	All quad
N_x	64	160	220
N_y	80	144	180
Bias type	Two-sided	Two-sided	Two-sided
Bias factor	59	28	21
Approx. Δy_1 [m]	1.700×10^{-3}	1.700×10^{-3}	1.700×10^{-3}
Nodes	7,740	26,448	43,010
Elements	7,565	26,123	42,594
Grid metric			
U_{bulk} [m/s]	18.305	18.114	17.977
$\epsilon_{\text{rel}}(U_{\text{bulk}})$ [%]	–	1.04	0.76

Note. N_x and N_y are the streamwise and wall-normal edge divisions. For the ϵ_{rel} row, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

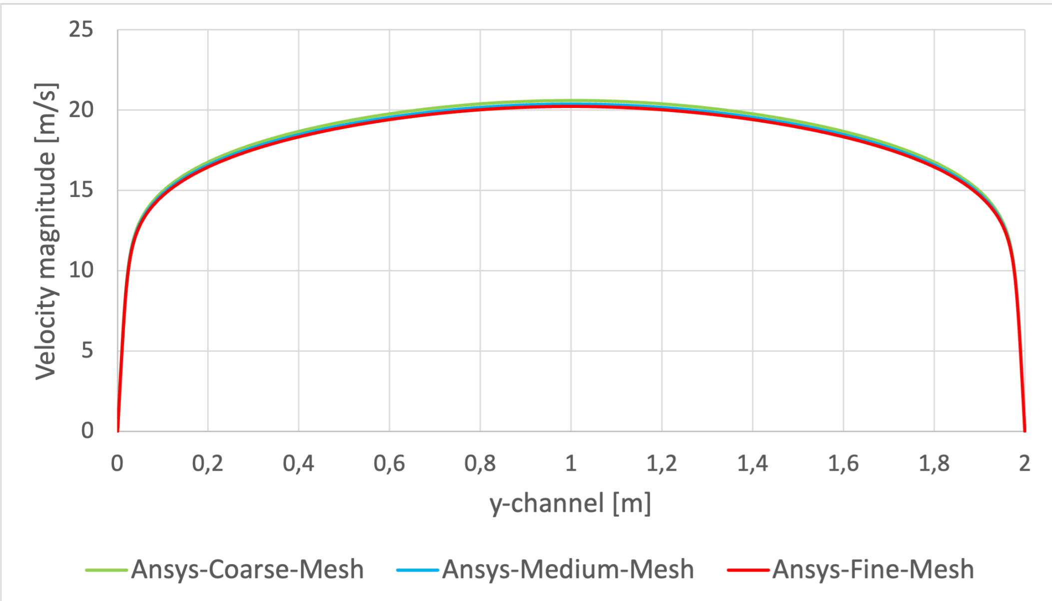


Figure B.2: Wall-normal velocity profiles extracted at $x = 0.5L$ for the Coarse, Medium, and Fine grids in the Ansys periodic channel flow benchmark, demonstrating spatial convergence of the baseline data.

B.2 Verification Benchmark 2: U-Bend Geometry

B.2.1 Physical and Geometric Parameters

The second verification benchmark evaluates a highly convective U-bend. Because this FEniCS domain is a 2D planar approximation of the symmetry mid-plane of the 3D Ansys VMFL048 case, the 3D physical pipe diameter (D) is explicitly retained as the characteristic length scale. This ensures the Reynolds (Re_D) and Dean (De) numbers remain non-dimensionally consistent with the commercial baseline. The straight inlet and outlet legs are shorter than in the original VMFL048 Ansys mesh to reduce meshing and computational effort; both the FEniCS and Ansys runs use this same short-leg U-bend geometry. The configuration is detailed in Table B.5.

TABLE B.5: Geometric, physical, and boundary-condition parameters for the U-bend benchmark.

Parameter	**U-bend setting**
Benchmark type	Short-leg 2D symmetry-plane U-bend based on Ansys VMFL048
Domain dimensions	Pipe diameter $D = 0.028$ m; centreline bend radius $R_c = 0.125$ m; straight-leg length $30D = 0.840$ m
Characteristic length	Physical pipe diameter $D = 0.028$ m
Reference / inlet velocity	Uniform prescribed inlet velocity $\mathbf{u}_{in} = (0, -1.42)$ m/s
Kinematic viscosity	$\nu = 8.9 \times 10^{-7}$ m^2/s
Reynolds number	$Re_D \approx 4.47 \times 10^4$
Dean number	$De \approx 1.50 \times 10^4$
Body force	$\mathbf{f} = (0, 0)$ m/s^2
Driving condition	Prescribed inlet velocity and zero relative static pressure at the outlet
Velocity boundary conditions	No-slip on all curved and straight pipe walls; natural velocity condition at pressure outlet
Pressure boundary conditions	$p = 0$ Pa at outlet; pressure left free at inlet and walls
SA boundary conditions	$\tilde{\nu}_{in} = 9.34 \times 10^{-5}$ m^2/s at inlet; $\tilde{\nu} = 0$ at walls; natural condition at outlet
Initial fields	$\mathbf{u}_0 = (0, -1.42)$ m/s, $p_0 = 0$ Pa, $\tilde{\nu}_0 = \tilde{\nu}_{in}$
Verification mesh	`Medium_FEniCS`; first layer $\Delta y_1 = 1.20 \times 10^{-5}$ m

B.2.2 Numerical Simulation Parameters

Due to the adverse pressure gradients and secondary flow separation induced by the streamline curvature, the U-bend represents a significantly more demanding numerical challenge than the channel. Consequently, the FEniCS solver required a smaller pseudo-time step, tighter under-relaxation, and advanced iterative preconditioners to maintain stability, as detailed in Table B.6.

Table B.6: Numerical simulation parameters for the highly convective FEniCS U-bend benchmark.

Parameter	Symbol	Setting
Velocity space	$\mathbf{u}$	Vector CG2
Pressure space	p	Scalar CG1
SA working-variable space	$\tilde{\nu}$	Scalar CG1
Quadrature degree	–	4
Maximum outer Picard steps	$N_{\text{Picard,max}}$	180
SA solves per Picard step	–	1
Outer velocity tolerance	$\epsilon_{\mathbf{u}}$	1.0×10^{-4}
Outer pressure tolerance	ϵ_p	1.0×10^{-4}
Outer SA tolerance	$\epsilon_{\tilde{\nu}}$	1.0×10^{-6}
IPCS pseudo-time step	Δt	1.0×10^{-4} s
Maximum IPCS steps per Picard step	–	180
Maximum IPCS steps for final flow	–	600
IPCS velocity tolerance	–	1.0×10^{-4}
IPCS pressure tolerance	–	2.0×10^{-3}
Pressure-drop convergence monitor	–	Final-window length 100 with relative tolerance 5.0×10^{-4}; flow convergence required
Velocity relaxation factor	α_u	0.3
Pressure relaxation factor	α_p	0.1
SA relaxation factor	$\alpha_{\tilde{\nu}}$	0.01
SA pseudo-time stabilization	–	SA transport pseudo-time step 5.0×10^{-2} s
SA lower bound	$\tilde{\nu}_{\text{min}}$	1.0×10^{-12}
Wall-distance equation	–	Relaxed Eikonal, $\sigma_w = 0.1$, $G_0 = 20$
Wall-distance Newton settings	–	Relative tolerance 1.0×10^{-8}; absolute tolerance 1.0×10^{-10}; maximum 80 iterations; relaxation 0.5
Linear solver strategy	–	IPCS blocks: GMRES with ILU velocity/correction preconditioning and Hypre-AMG pressure preconditioning; SA transport: default solver

B.2.3 Spatial Verification and Grid Independence

Managing the highly stretched, non-orthogonal inflation layers around the curved arcs of the U-bend required custom mesh generation via Gmsh. A spatial grid independence study was conducted across three refined unstructured grids. The node counts, element densities, and inflation layer parameters for these generated meshes are explicitly documented in Table B.7. Similar to the channel benchmark, the wall-adjacent element height was strictly maintained at 1.20×10^{-5} m to satisfy the low-Reynolds SA wall constraints, while the internal element sizing was systematically refined.

The reported FEniCS static pressure-drop metric is computed from boundary-measure-weighted mean pressures over the inlet and outlet sections, rather than from local point samples. Let Γ_{in} and Γ_{out} denote the inlet and outlet boundaries, and let p_k be the kinematic pressure solved in FEniCS. The physical static pressure drop reported in pascals is

$$\Delta p_s = \rho\left(\overline{p}_{k,\text{in}} - \overline{p}_{k,\text{out}}\right) = \rho\left[\frac{1}{|\Gamma_{\text{in}}|}\int_{\Gamma_{\text{in}}} p_k \,\mathrm{d}\Gamma - \frac{1}{|\Gamma_{\text{out}}|}\int_{\Gamma_{\text{out}}} p_k \,\mathrm{d}\Gamma\right], \tag{B.1}$$

where $|\Gamma| = \int_\Gamma 1 \,\mathrm{d}\Gamma$ is the FEniCS boundary measure and $\rho = 997$ kg/m^3. In the implementation this corresponds to integrating over the marked inlet and outlet facets with `ds(2)` and `ds(3)`, respectively. The density factor is required because the IPCS formulation stores pressure in kinematic form, while the benchmark table reports the static pressure drop in Pa. Although the outlet pressure is prescribed as zero relative pressure, the outlet term is still evaluated as a boundary average so that the diagnostic follows the same inlet-minus-outlet definition on every grid.

TABLE B.7: Wall-resolved U-bend mesh hierarchy used for the FEniCS grid-independence study. The first-layer height is kept fixed to maintain $y^+ \approx 1$, while the bulk element size and inflation-layer resolution are refined.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
h_{bulk} [m]	2.80×10^{-3}	1.40×10^{-3}	8.00×10^{-4}
Δy_1 [m]	1.20×10^{-5}	1.20×10^{-5}	1.20×10^{-5}
Inflation layers	10	14	18
Growth ratio	1.45	1.30	1.25
t_{infl} [m]	1.07×10^{-3}	1.53×10^{-3}	2.62×10^{-3}
Vertices	23,749	73,978	182,870
Elements	45,994	144,952	360,486
Grid metric			
Static pressure drop Δp_s [Pa]	*No completed three-grid pressure-drop set is checked in*		
$\epsilon_{\text{rel}}(\Delta p_s)$ [%]	–		*Not reported*

Note. For the ϵ_{rel} row, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change. The checked-in U-bend pressure-drop residual files do not contain a completed coarse–medium–fine sequence, so no pressure-drop grid metric is reported for the FEniCS U-bend hierarchy.

The checked-in FEniCS U-bend mesh hierarchy is documented by the mesh parameters in Table B.7. The repository does not currently include extracted Coarse–Medium–Fine bend-exit velocity-profile data for this FEniCS hierarchy, so Figure B.3 records the data-availability status rather than a plotted convergence profile. The selected `Medium_FEniCS` grid is therefore reported as the configured verification mesh, while the missing profile and incomplete pressure-drop sequence are not promoted to quantitative mesh-convergence evidence.

No checked-in FEniCS U-bend Coarse–Medium–Fine bend-exit velocity-profile data are available for this appendix figure.

Figure B.3: Data-availability status for the FEniCS U-bend mesh-independence velocity-profile figure.

Similarly, to guarantee the integrity of the cross-code validation in this highly convective regime, an equivalent mesh independence study was executed for the Ansys Fluent baseline. Using a parallel unstructured meshing strategy with strict $y^+ \approx 1$ boundary layer inflation (documented in Table B.8), the Ansys velocity profiles at the bend exit across the same pipe-diameter line, shown in Figure B.4, were checked for spatial convergence. For each Ansys solve, convergence was accepted only after the scaled continuity, momentum, and Spalart–Allmaras residuals reached 1.0×10^{-6} and the monitored static pressure drop changed by less than a relative tolerance of 5.0×10^{-4} over the final 100 iterations. The inlet and outlet mass-flow rates were also compared for each grid to verify the global mass balance before accepting the pressure-drop values. The `Fine_Ansys` grid is retained as the commercial verification baseline, while the deliberately coarser `Coarse_Ansys` level is selected to bracket the mesh-convergence trend.

TABLE B.8: Wall-resolved U-bend mesh hierarchy used for the Ansys grid-independence study.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Mesh type	Quad-dom.	Quad-dom.	Quad-dom.
h_{face} [m]	4.00×10^{-3}	2.00×10^{-3}	1.25×10^{-3}
Curvature angle	22°	12°	8°
Inflation option	First layer	First layer	First layer
Δy_1 [m]	1.20×10^{-5}	1.20×10^{-5}	1.20×10^{-5}
Inflation layers	10	16	22
Growth rate	1.35	1.22	1.16
t_{infl} [m]	6.55×10^{-4}	1.59×10^{-3}	2.17×10^{-3}
Nodes	14,525	47,707	106,460
Elements	13,984	46,640	104,763
Grid metric			
Static pressure drop Δp_s [Pa]	814.109	796.804	792.995
$\epsilon_{\text{rel}}(\Delta p_s)$ [%]	–	2.13	0.48

Note. Quad-dom. = quad-dominant. For the ϵ_{rel} row, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

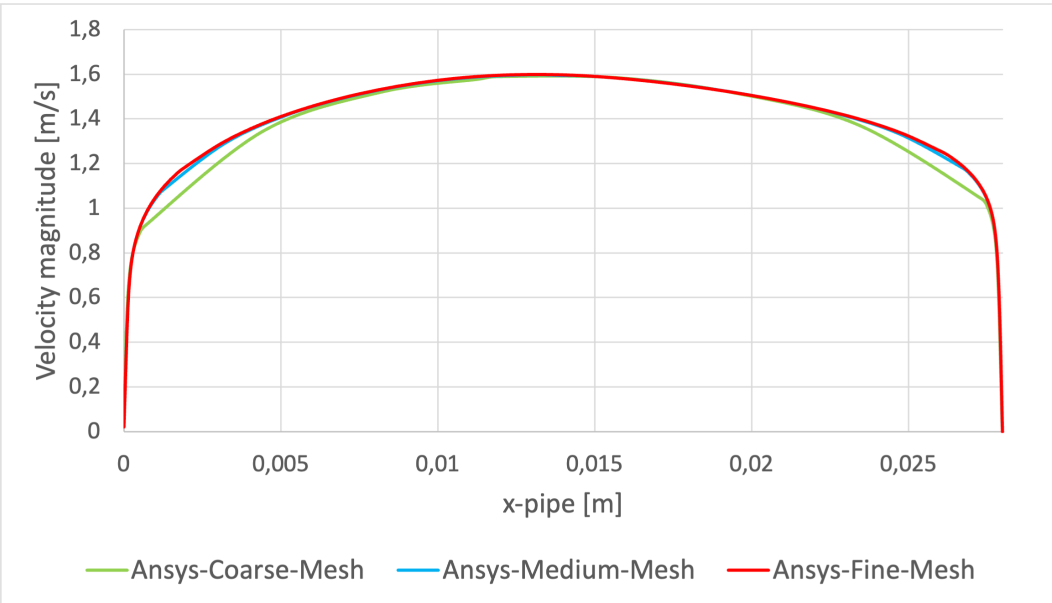


FIGURE B.4: Velocity profiles at the U-bend exit, extracted across the pipe diameter $D = 0.028$ m, for the Coarse, Medium, and Fine Ansys grids, demonstrating spatial convergence of the commercial baseline.

Appendix C

Chapter 6 Topology-Optimization Benchmark Details

This appendix supports Chapter 6. It combines the turbulent and laminar topology-optimization support material in one place and groups it by benchmark case. Within each benchmark, the ordering is: turbulent-TO FEniCS settings and diagnostics, frozen-sensitivity checks where applicable, turbulent-design Ansys mesh independence, laminar-TO FEniCS settings and diagnostics, and laminar-design Ansys mesh independence.

Unless stated otherwise, all mesh-independence relative differences in this appendix are reported as positive percentage magnitudes. For a mesh-dependent metric ϕ, the coarse–medium difference is computed as $|\phi_{\text{coarse}} - \phi_{\text{medium}}|/|\phi_{\text{coarse}}| \times 100\%$, while the medium–fine difference is computed as $|\phi_{\text{medium}} - \phi_{\text{fine}}|/|\phi_{\text{fine}}| \times 100\%$.

C.1 Validation and Meshing Strategy

The FEniCS meshes used during topology optimization are treated as fixed optimization discretizations rather than as a separate mesh-convergence hierarchy. This distinction is important because refining the FEniCS mesh would also change the DG_0 design space, the filter stencil, the number of MMA design variables, and the diffuse Brinkman interface location. The pipe-bend optimization uses a rectangular-coordinate triangular mesh with near-wall clustering based on a smooth-wall $y^+ \approx 1$ estimate at the prescribed inlet Reynolds number. The U-bend optimization uses the corresponding wall-resolved triangular Gmsh mesh, with a near-wall size field based on the same $y^+ \approx 1$ estimate and a far-field target size of $h_{\max} = 0.007L$ used to set the density-filter scale. The diffuser benchmark from Yoon [63] uses a wall-resolved rectangular-coordinate triangular mesh over the square diffuser domain, refined around the full-height inlet, the center-third outlet, and the no-slip walls. In all cases, the mesh is chosen to provide sufficient wall-resolved SA and design resolution for the optimization problem; quantitative grid-convergence evidence is instead reported for the extracted sharp geometries in Ansys Fluent.

For each extracted turbulent topology optimization design, the Ansys validation mesh is refined through coarse, medium, and fine resolutions. Both SA and SST $k-\omega$ validation runs use the same extracted geometry, inlet condition, outlet condition, and fluid properties. The mesh-independence comparison is reported for both turbulence models because near-wall behavior and separated-flow predictions may respond differently to grid refinement.

All Fluent mesh-independence runs in this appendix use the same convergence and acceptance criteria. The static pressure drop must change by less than 5×10^{-4} over the final 100 iterations, while the viscous dissipation must change by less than 10^{-3} over the final 100 iterations. The fine mesh is accepted only when the medium–fine relative difference is below 1% for static pressure drop and below 2% for viscous dissipation for both SA and SST $k-\omega$. The reported refinement differences are computed as $|\phi_{\text{coarse}} - \phi_{\text{medium}}|/|\phi_{\text{coarse}}|$ and $|\phi_{\text{medium}} - \phi_{\text{fine}}|/|\phi_{\text{fine}}|$ for each validation metric.

C.2 Shared Turbulence-Topology Coupling Conventions

All three turbulent topology optimization benchmarks use the same basic coupling idea between the density field, the SA working variable, and the reciprocal wall-distance equation. Topology-created solids act as walls for the SA working-variable penalty and the reciprocal wall-distance solve. The wall-density source is therefore `design`, the wall-distance mode is `reciprocal_penalized`, and both the SA and wall-distance penalties use the power-law solid indicator defined in Chapter 5. The same frozen-turbulence adjoint convention is also used: the forward loop updates $\tilde{\nu}$, ν_t, and G, while the derivative freezes $\tilde{\nu}$ and G during gradient assembly.

The pipe-bend and U-bend benchmarks then share a Bayat-style pressure-loss setup, while the diffuser follows the Yoon dissipation benchmark. The common mechanisms and the benchmark-specific parameter values are summarized in Table C.1.

The pipe-bend finite-difference sensitivity verification in Section C.3.1 uses a dedicated sensitivity configuration. It inherits the archived `Jp_Old` pipe-bend settings and stores its own finite-difference output tables, so it should be read as a frozen-adjoint implementation check rather than as the current production pipe-bend continuation table.

Table C.1: Shared turbulence-topology coupling conventions and benchmark-specific parameter values.

Parameter	Pipe bend and U-bend	Diffuser
SA wall-distance mode	`reciprocal_penalized`	`reciprocal_penalized`
SA wall-density source	`design`	`design`
SA penalty interpolation	Power-law solid indicator	Power-law solid indicator
Wall-distance penalty interpolation	Power-law solid indicator	Power-law solid indicator
Wall-distance recovery	Reciprocal recovery from the relaxed Eikonal variable G	Reciprocal recovery from the relaxed Eikonal variable G
Adjoint treatment	Frozen-turbulence adjoint; $\tilde{\nu}$ and G frozen during gradient assembly	Frozen-turbulence adjoint; $\tilde{\nu}$ and G frozen during gradient assembly
Solid threshold for wall source	$\bar{\gamma} = 0.50$	No separate threshold parameter in the diffuser configuration
SA working-variable penalty	Power-law solid indicator with amplitude continuation ending at 10^5 and exponent $n = 3$	Power-law solid indicator with amplitude 10^5 and exponent $n = 3$
Wall-distance penalty	Power-law solid indicator with amplitude continuation ending at 10^5 and exponent $n = 3$	Power-law solid indicator with amplitude 10^5 and exponent $n = 3$
Wall-distance parameters	$\sigma_w = 0.1$, $G_0 = 20$	$\sigma_w = 0.1$, $G_0 = 20$
SA inlet treatment	Turbulence-intensity estimate: $I = 7.5\%$, $\ell/H = 0.05$, eddy-viscosity ratio capped at 50	Prescribed $\tilde{\nu} = 1$ at inlet and outlet
Optimization objective	Average inlet pressure with fixed outlet pressure $p = 0$	Dissipation objective J_D; the velocity-outlet baseline prescribes inlet and outlet profiles, while the primary pressure-outlet case prescribes the inlet profile and fixes outlet static pressure

C.3 Pipe Bend Benchmark

The pipe-bend case follows the setup of Bayat et al. [12]. The reference paper uses a standard k-ε model with wall functions; the present thesis uses the wall-resolved SA implementation and evaluates the $V_f = 0.25$ design.

TABLE C.2: Geometric, physical, and boundary-condition parameters for the pipe-bend benchmark.

Parameter	**Setting**
Reference case	Pipe-bend benchmark from Bayat et al. [12]
Reference figure for setup	Bayat et al. Fig. 10
Reference velocity figure	Bayat et al. Fig. 15
Reference turbulence model	Standard k-ε with wall functions
Turbulence model in this thesis	Spalart-Allmaras
Design-domain dimensions	Square design region $L \times L$ with $L = 1$
Full computational domain	$x/L \in [-0.2, 1]$, $y/L \in [-0.2, 1]$
Non-design inlet duct	Length $0.2L$, height $0.2L$, inlet over $y/L \in [0.7, 0.9]$
Non-design outlet duct	Width $0.2L$, length $0.2L$, outlet over $x/L \in [0.7, 0.9]$ on the lower boundary
Volume fraction	$V_f = 0.25$
Reynolds number	$Re = 5{,}000$, based on $U_{\text{in}} H_{\text{in}} \rho/\mu$ with $H_{\text{in}} = 0.2L$
Density and viscosity	$\rho = 1.0$, $\mu = 4.0 \times 10^{-5}$
Inlet condition	Uniform inlet velocity $\mathbf{u} = (1, 0)$; SA inlet estimated from the shared turbulence-intensity settings in Table C.1
Outlet condition	Pressure outlet $p = 0$ on the lower outlet, with the normal/-tangential component constraint used by the FEniCS configuration
Wall conditions	No-slip velocity and $\tilde{\nu} = 0$ on external walls; topology-created solids enter through the design-density penalties
Validation metrics	Average inlet pressure, static pressure drop, Φ_D, and relative changes between FEniCS-SA, Ansys-SA, and Ansys-SST $k-\omega$

C.3.1 Frozen Sensitivity Verification

The frozen adjoint is checked directly on the pipe-bend sensitivity configuration in `Config_PipeBendAlexandersen_Frozen_Sensitivity.py`. That file inherits the archived `Config_PipeBendAlexandersen_Frozen_Jp_Old.py` path, not the current production pipe-bend continuation table above. It therefore uses the same inlet-pressure objective and frozen-turbulence adjoint algebra, but with the old single-stage J_p setup: $\alpha_{\text{solid}} = 100$, $q = 1/150$, $\beta = 4$, move limit 0.010, one frozen-SA Picard update with SA relaxation 0.20, wall-distance parameter $\sigma_w = 0.01$, and power-law SA/wall-distance penalty exponent $n = 4$. No full turbulence-adjoint or wall-distance-adjoint comparison is made; the purpose is to compare the frozen derivative supplied to MMA with objective changes obtained by re-solving the perturbed velocity-pressure state. If a current five-stage production pipe-bend or U-

Table C.3: Numerical FEniCS solver, topology-optimization, and sensitivity-verification parameters for the pipe-bend benchmark.

Parameter	**Symbol / option**	**Setting**
Flow finite elements	$(\mathbf{u}, p)$	Taylor–Hood P_2/P_1 continuous Galerkin elements
SA working-variable space	$\tilde{\nu}$	P_1 continuous Galerkin
Design-density space	γ	DG_0 cellwise constant density
FEniCS optimization mesh	–	`mesh_yplus1.xdmf`; 39,128 vertices, 77,334 triangles, $h_{\max} = 0.007L$
Objective	J_p	Average inlet pressure
Initial design density	γ_0	$V_f = 0.25$, matched to the filtered volume
Brinkman penalty	$\alpha(\bar{\gamma})$	RAMP form with $\alpha_{\text{fluid}} = 0$, $\alpha_{\text{solid}} = 10^3$ for the active J_p path
RAMP continuation	q	[0.02, 0.04, 0.08, 0.12, 0.18]
Projection continuation	β	[1.5, 3, 5, 8, 12] with threshold $\eta = 0.50$
MMA move limits	–	[0.012, 0.010, 0.007, 0.005, 0.003]
Maximum optimization iterations	–	Five continuation stages with [60, 70, 90, 110, 140] MMA iterations; stop if objective change is below 10^{-5} for eight iterations
Filter radius	r_f	$4h_{\max} = 0.028L$
SA and wall-distance penalty continuation	–	Shared J_p amplitude schedule $[10^3,\ 3 \times 10^3,\ 10^4,\ 3 \times 10^4,\ 10^5,\ 10^5,\ 10^5]$; the first five entries are used by the active five-stage continuation
Forward flow solver	–	IPCS flow solves for frozen-SA Picard updates, followed by a warm-started SNES Newton line-search final flow solve
Frozen NS-SA coupling	–	Five Picard updates per design iteration with SA relaxation factor 0.04 for the active J_p path
Linear solver	–	MUMPS for SNES/state and adjoint solves; BiCGSTAB with ILU preconditioning for IPCS warm-start solves
Adjoint treatment	–	Frozen-turbulence adjoint in $(\mathbf{u}^*, p^*)$; $\tilde{\nu}$ and G are frozen during gradient assembly
Sensitivity verification samples	–	Five active DG0 cells, seed 13
Sensitivity verification step	Δh	10^{-6}, central finite difference
Sensitivity verification fields held fixed	–	SA working variable $\tilde{\nu}^0$ and reciprocal wall-distance field G^0

bend campaign is used as the final design source, the same diagnostic should be rerun on that exact configuration.

The checked-in coordinate finite-difference result is performed at the initial pipe-bend design state, with base objective $J_0 = 1.8343209502$. In every perturbation, the SA working variable $\tilde{\nu}^0$ and reciprocal wall-distance field G^0 are held fixed. This matches the frozen-turbulence assumption used by the adjoint: the flow state is re-solved, but the turbulence and wall-distance updates are not differentiated.

The sensitivity configuration evaluates five active DG_0 samples with seed 13. For sampled cell i, the raw density vector $\boldsymbol{\gamma}$ is perturbed by $\pm\Delta h\,\mathbf{e}_i$, where $\mathbf{e}_i$ is the unit vector for that active design variable and $\Delta h = 10^{-6}$. The central finite-difference derivative is

$$S_i^{\mathrm{FD}} = \frac{J_p\big(\boldsymbol{\gamma} + \Delta h\,\mathbf{e}_i; \mathbf{u}^+, p^+, \tilde{\nu}^0, G^0\big) - J_p\big(\boldsymbol{\gamma} - \Delta h\,\mathbf{e}_i; \mathbf{u}^-, p^-, \tilde{\nu}^0, G^0\big)}{2\Delta h}. \qquad \text{(C.1)}$$

The frozen-adjoint value for the same cell is denoted by S_i^{fr}; the superscripts FD and fr denote finite-difference and frozen-adjoint quantities, respectively. The absolute and relative discrepancies are computed as

$$\Delta_i^{\mathrm{abs}} = \left|S_i^{\mathrm{fr}} - S_i^{\mathrm{FD}}\right|, \qquad \varepsilon_i^{\mathrm{fr}} = \frac{\Delta_i^{\mathrm{abs}}}{\max\!\big(\left|S_i^{\mathrm{fr}}\right|, \left|S_i^{\mathrm{FD}}\right|, 10^{-30}\big)}. \qquad \text{(C.2)}$$

Table C.4: Coordinate sensitivity check for the frozen pipe-bend configuration. Values are taken from `SensitivityVerificationTable.tsv`.

Cell	**Finite difference** S_i^{FD}	**Frozen adjoint** S_i^{fr}	**Absolute error** Δ_i^{abs}	**Relative error** $\varepsilon_i^{\mathrm{fr}}$
CV_1	-2.7011×10^{-5}	-2.7010×10^{-5}	1.5875×10^{-10}	5.8774×10^{-6}
CV_2	-2.1146×10^{-6}	-2.1148×10^{-6}	1.2513×10^{-10}	5.9171×10^{-5}
CV_3	-1.6664×10^{-4}	-1.6664×10^{-4}	4.2394×10^{-10}	2.5440×10^{-6}
CV_4	-1.7209×10^{-5}	-1.7209×10^{-5}	4.3775×10^{-10}	2.5437×10^{-5}
CV_5	2.3641×10^{-5}	2.3641×10^{-5}	2.0497×10^{-10}	8.6700×10^{-6}

The coordinate checks show close agreement between the frozen-adjoint derivative and the re-solved central finite differences across all five samples. The maximum absolute discrepancy is 4.38×10^{-10}, and the maximum relative discrepancy is 5.92×10^{-5}. The sign of the derivative also agrees in all five samples.

The same sampled active cells are also used for a directional Taylor check with steps $(10^{-3}, 3 \times 10^{-4}, 10^{-4}, 3 \times 10^{-5})$ and seed 29. The directional derivative is $9.6668564775 \times 10^{-5}$, and the central finite differences remain within 8.66×10^{-8} relative error over the tested step range. The symmetric Taylor remainder decays with approximately second-order behavior, as reported in Table C.5.

This check verifies the implemented frozen-turbulence derivative for the archived pipe-bend sensitivity setup that generated the stored TSV file. The conclusion is deliberately limited to that derivative: the SA working variable and reciprocal wall-distance field are held fixed during the checks, matching the optimization adjoint; consequently, the result should not be interpreted as validation of a fully coupled

Table C.5: Directional Taylor check for the frozen pipe-bend sensitivity configuration. Values are taken from `TaylorCheckLog.tsv`.

Step ϵ	Frozen directional derivative	Central finite difference	Directional relative error	Symmetric Taylor order
1.0×10^{-3}	$9.6668564775 \times 10^{-5}$	$9.6668565153 \times 10^{-5}$	3.9055×10^{-9}	–
3.0×10^{-4}	$9.6668564775 \times 10^{-5}$	$9.6668563524 \times 10^{-5}$	1.2939×10^{-8}	2.0006
1.0×10^{-4}	$9.6668564775 \times 10^{-5}$	$9.6668562044 \times 10^{-5}$	2.8252×10^{-8}	2.0077
3.0×10^{-5}	$9.6668564775 \times 10^{-5}$	$9.6668573146 \times 10^{-5}$	8.6596×10^{-8}	2.1339

turbulence sensitivity or as a sensitivity check of the newer five-stage production bend continuation. Repeating the same diagnostic on the current pipe-bend and U-bend configurations would test the exact continuation, penalty, and IPCS settings used for those production runs.

C.3.2 Turbulent FEniCS Optimization Diagnostics

Figure C.1 records the checked-in convergence-history status for the turbulent pipe-bend topology. The pressure objective is normalized by its initial value J_0, while the volume constraint is defined as the relative residual

$$r_V^{(k)} = 100 \frac{v_f^{(k)} - V_f}{V_f}, \tag{C.3}$$

where $v_f^{(k)}$ is the realized fluid volume fraction at MMA iteration k. With this sign convention, the dashed line at $r_V = 0$ represents the active volume constraint. Values below this line are feasible and correspond to a design using slightly less fluid volume than the prescribed limit.

Checked-in turbulent pipe-bend production traces contain only global MMA iterations 0–3. No completed five-stage convergence history is available for this appendix figure.

Figure C.1: Data-availability status for the turbulent pipe-bend optimization convergence history. A complete plot would show the normalized pressure objective J/J_0 and the relative volume-fraction residual r_V from Eq. (C.3).

The checked-in `ObjectiveFunction_Convergence.txt` and `VolumeConstraint_Convergence.txt` files for the active turbulent pipe-bend J_p results contain a header and four data rows, corresponding to global MMA iterations 0–3. This is not a completed five-stage production history, so no late-stage continuation oscillations or final objective-reduction trend are reported here.

The extracted turbulent pipe-bend geometry is meshed in Ansys Fluent with a patch-conforming triangular bulk mesh and wall-resolved inflation on all no-slip walls. The first-layer height is kept fixed from the smooth-wall $y^+ \approx 1$ estimate, while the mesh hierarchy refines the bulk face size, curvature angle, and inflation distribution.

TABLE C.6: Ansys Fluent mesh-independence table for the extracted turbulent pipe-bend benchmark geometry.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Mesh type	Tri.	Tri.	Tri.
h_{bulk} [m]	1.10×10^{-2}	5.00×10^{-3}	3.25×10^{-3}
Curvature angle	15°	7°	4.5°
Near-wall	WR infl.	WR infl.	WR infl.
Δy_1 [m]	6.00×10^{-4}	6.00×10^{-4}	6.00×10^{-4}
Inflation layers	8	14	18
Growth rate	1.15	1.10	1.08
t_{infl} [m]	8.24×10^{-3}	1.68×10^{-2}	2.25×10^{-2}
Nodes	5,147	21,343	44,042
Elements	7,551	32,780	68,743
Validation metrics			
Φ_D^{SA} [W/m]	0.0143	0.0151	0.0153
$\epsilon_{\mathrm{rel}}(\Phi_D^{\mathrm{SA}})$ [%]	–	5.38	1.32
Φ_D^{SST} [W/m]	0.0150	0.0164	0.0166
$\epsilon_{\mathrm{rel}}(\Phi_D^{\mathrm{SST}})$ [%]	–	9.04	1.22
Static pressure drop Δp_s^{SA} [Pa]	0.1423	0.1449	0.1457
$\epsilon_{\mathrm{rel}}(\Delta p_s^{\mathrm{SA}})$ [%]	–	1.84	0.55
Static pressure drop $\Delta p_s^{\mathrm{SST}}$ [Pa]	0.1481	0.1516	0.1522
$\epsilon_{\mathrm{rel}}(\Delta p_s^{\mathrm{SST}})$ [%]	–	2.36	0.40
Selected validation mesh		Fine_Ansys	

Note. WR = wall-resolved; Φ_D = viscous dissipation; Δp_s = static pressure drop; SST denotes SST k–ω. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

C.3.3 Laminar FEniCS Configuration Differences

The laminar pipe-bend reference is a completed FEniCS topology-optimization run from `Config_PipeBendAlexandersen_Laminar.py`. It reuses the turbulent pipe-bend setup in Tables C.2 and C.3 wherever not stated otherwise: the geometry, wall-resolved mesh, non-design inlet and outlet ducts, uniform inlet velocity, pressure-outlet treatment, $V_f = 0.25$ volume constraint, average inlet-pressure objective J_p, filter radius, and projection threshold are unchanged. The Brinkman resistance and continuation controls are laminar-specific and are listed in Table C.7.

Table C.7: Laminar-specific differences from the turbulent pipe-bend FEniCS setup above.

Parameter	Laminar setting
Flow model	Laminar incompressible Navier–Stokes; the SA working variable, reciprocal wall-distance equation, topology-dependent SA penalties, and frozen-turbulence coupling are omitted
Reynolds number	$Re = 1$, based on $U_{\text{in}} H_{\text{in}} \rho/\mu$ instead of the turbulent $Re = 5{,}000$ setup
Density and viscosity	$\rho = 1.0$ and $\mu = 0.2$, with the same $U_{\text{in}} = 1$ and $H_{\text{in}} = 0.2L$ as the turbulent case
Brinkman penalty	RAMP form with $\alpha_{\text{fluid}} = 0$ and $\alpha_{\text{solid}} = 10^5$
Flow finite elements	Taylor–Hood P_2/P_1 for $(\mathbf{u}, p)$; no separate P_1 SA working-variable space is used
Forward and adjoint solves	SNES/MUMPS laminar Newton forward solves with relative tolerance 10^{-6}, absolute tolerance 10^{-8}, and maximum forward iteration count 220; the adjoint is solved as a linear system with MUMPS
Continuation and optimizer controls	Seven-stage schedule $q = \{0.01, 0.02, 0.04, 0.08, 0.08, 0.12, 0.20\}$, $\beta = \{1, 2, 4, 8, 8, 12, 16\}$, move limits $\{0.010, 0.008, 0.006, 0.004, 0.003, 0.002, 0.0015\}$, and inner-iteration budgets $\{12, 25, 45, 80, 120, 160, 220\}$; stop if objective change is below 10^{-6} for ten iterations

C.3.4 Laminar FEniCS Optimization Diagnostics

Figure C.2 records the completed laminar pipe-bend optimization history. The normalized inlet-pressure objective drops rapidly in the first continuation stage and then remains nearly flat, while the volume-fraction residual repeatedly returns to the active constraint after continuation updates.

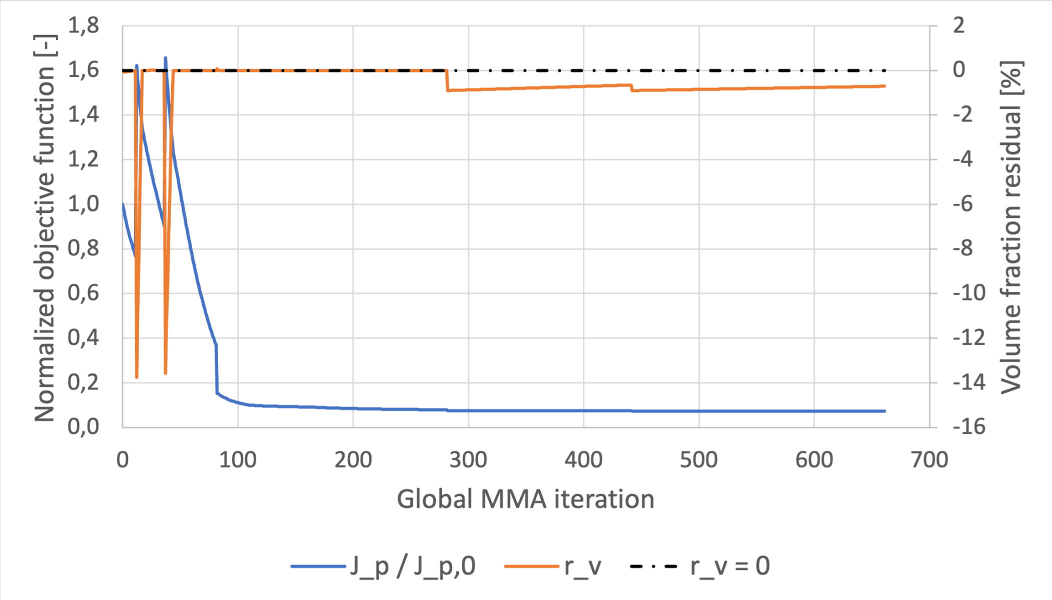


FIGURE C.2: FEniCS convergence history for the laminar pipe-bend topology-optimization run. The blue curve is the normalized inlet-pressure objective $J_p/J_{p,0}$, the orange curve is the relative volume-fraction residual, and the dashed black line marks the active volume constraint.

C.3.5 Laminar-Design Mesh Independence

Table C.8 reports the Ansys mesh-independence study for the extracted laminar pipe-bend design. This table is separate from the turbulent-design mesh study in Table C.6 of Section C.3 because the laminar and turbulent topologies generally produce different wall shapes and near-wall mesh requirements.

Table C.8: Ansys mesh-independence table for the extracted laminar pipe-bend design.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Mesh type	Tri.	Tri.	Tri.
h_{bulk} [m]	4.50×10^{-3}	2.60×10^{-3}	2.00×10^{-3}
Curvature angle	8°	4°	3°
Δy_1 [m]	1.20×10^{-3}	1.20×10^{-3}	1.20×10^{-3}
Inflation layers	12	15	16
Growth rate	1.06	1.045	1.04
t_{infl} [m]	2.02×10^{-2}	2.49×10^{-2}	2.62×10^{-2}
Nodes	22,494	56,468	89,804
Elements	35,948	93,828	153,163
Validation metrics			
Φ_D^{SST} [W/m]	0.0293	0.0310	0.0313
$\epsilon_{\text{rel}}(\Phi_D^{\text{SST}})$ [%]	–	5.82	0.95
Static pressure drop Δp_s^{SST} [Pa]	0.5229	0.5420	0.5444
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SST}})$ [%]	–	3.65	0.45
Selected comparison mesh		Fine_Ansys	

Note. Φ_D = viscous dissipation; Δp_s = static pressure drop; SST denotes SST k–ω. The checked-in repository does not contain a completed turbulent U-bend Ansys validation hierarchy, so no mesh-independence metrics are reported for this design.

C.4 U-Bend Benchmark

The U-bend case follows Fig. 11 and Table 4 of Bayat et al. [12]. It uses the same turbulent topology optimization framework as the pipe bend, but forces a 180° return path around a fixed internal separator.

C.4.1 Turbulent FEniCS Optimization Diagnostics

Figure C.3 records the checked-in convergence-history status for the turbulent U-bend topology. No frozen U-bend result directory or convergence-history files are currently checked in under `FluidTO/Results_Frozen`, so the normalized inlet-pressure objective and relative volume-fraction residual are not reported here.

TABLE C.9: Geometric, physical, and boundary-condition parameters for the U-bend benchmark.

Parameter	**Setting**
Reference case	U-bend benchmark from Bayat et al. [12]
Reference figure/table for setup	Bayat et al. Fig. 11 and Table 4
Reference turbulence model	Standard k-ε with wall functions
Turbulence model in this thesis	Spalart-Allmaras
Design-domain dimensions	Square design region $L \times L$ with $L = 1$
Full computational domain	$x/L \in [-0.2, 1]$, $y/L \in [0, 1]$
Port geometry	Two left-side ports of height $0.2L$; inlet over $y/L \in [0.55, 0.75]$, outlet over $y/L \in [0.25, 0.45]$
Fixed separator	Bar thickness $0.10L$, rounded-tip radius $0.05L$, extending from $x/L = -0.2$ to tip at $x/L = 0.5$
Volume fraction	$V_f = 0.27$
Reynolds number	$Re = 5{,}000$, based on $U_{\text{in}} H_{\text{port}} \rho/\mu$ with $H_{\text{port}} = 0.2L$
Density and viscosity	$\rho = 1.0$, $\mu = 4.0 \times 10^{-5}$
Inlet condition	Uniform inlet velocity $\mathbf{u} = (1, 0)$; SA inlet estimated from the shared turbulence-intensity settings in Table C.1
Outlet condition	Pressure outlet $p = 0$ on the lower left port, with the component constraint used by the FEniCS configuration
Wall conditions	No-slip velocity and $\tilde{\nu} = 0$ on external walls and the fixed separator; topology-created solids enter through the design-density penalties
Validation metrics	Average inlet pressure, static pressure drop, Φ_D, and relative changes between FEniCS-SA, Ansys-SA, and Ansys-SST k−ω

No checked-in turbulent U-bend frozen-optimization convergence-history files are available for this appendix figure.

FIGURE C.3: Data-availability status for the turbulent U-bend optimization convergence history.

Table C.10: Numerical FEniCS solver and topology-optimization parameters for the U-bend benchmark.

Parameter	Symbol / option	Setting
Flow finite elements	$(\mathbf{u}, p)$	Taylor–Hood P_2/P_1 continuous Galerkin elements
SA working-variable space	$\tilde{\nu}$	P_1 continuous Galerkin
Design-density space	γ	DG_0 cellwise constant density
Reference FEniCS mesh	–	`mesh_yplus1.xdmf`; wall-resolved triangular Gmsh mesh with 45,280 vertices, 85,780 triangles, near-wall refinement based on $y^+ \approx 1$, and far-field $h_{\max} = 0.007L$
Objective	J_p	Average inlet pressure
Initial design density	γ_0	$V_f = 0.27$, matched to the filtered volume
Brinkman penalty	$\alpha(\bar{\gamma})$	RAMP form with $\alpha_{\text{fluid}} = 0$, $\alpha_{\text{solid}} = 10^3$ for the active J_p path
RAMP continuation	q	[0.02, 0.04, 0.08, 0.12, 0.18]
Projection continuation	β	[1.5, 3, 5, 8, 12] with threshold $\eta = 0.50$
MMA move limits	–	[0.012, 0.010, 0.007, 0.005, 0.003]
Maximum optimization iterations	–	Five continuation stages with [60, 70, 90, 110, 140] MMA iterations; stop if objective change is below 10^{-5} for ten iterations
Filter radius	r_f	$4h_{\max} = 0.028L$
SA and wall-distance penalty continuation	–	Shared amplitude schedule [10^3, 3×10^3, 10^4, 3×10^4, 10^5, 10^5, 10^5]; the first five entries are used by the active five-stage continuation
Forward flow solver	–	IPCS flow solves for frozen-SA Picard updates, followed by a warm-started SNES Newton line-search final flow solve
Frozen NS-SA coupling	–	Five Picard updates per design iteration with SA relaxation factor 0.04 for the active J_p path
Linear solver	–	MUMPS for SNES/state and adjoint solves; BiCGSTAB with ILU preconditioning for IPCS warm-start solves
Adjoint treatment	–	Frozen-turbulence adjoint in $(\mathbf{u}^*, p^*)$; $\tilde{\nu}$ and G are frozen during gradient assembly

TABLE C.11: Ansys Fluent mesh-independence table for the extracted turbulent U-bend benchmark geometry.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Mesh-study status	*No extracted turbulent U-bend Ansys validation mesh hierarchy is checked in*		
Validation metrics			
Φ_D^{SA} [W/m]		*Not reported*	
Φ_D^{SST} [W/m]		*Not reported*	
Static pressure drop Δp_s^{SA} [Pa]		*Not reported*	
Static pressure drop Δp_s^{SST} [Pa]		*Not reported*	
Selected validation mesh		*Not reported*	

Note. Φ_D = viscous dissipation; Δp_s = static pressure drop; SST denotes SST k–ω. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

C.4.2 Laminar FEniCS Configuration Differences

The laminar U-bend reference is a completed FEniCS topology-optimization run from `Config_UBendAlexandersen_Laminar.py`. It reuses the turbulent U-bend setup in Tables C.9 and C.10 wherever not stated otherwise: the geometry, fixed separator, wall-resolved mesh, non-design ports, uniform inlet velocity, pressure-outlet treatment, $V_f = 0.27$ volume constraint, average inlet-pressure objective J_p, filter radius, and projection threshold are unchanged. The Brinkman resistance and continuation controls are laminar-specific and are listed in Table C.12.

C.4.3 Laminar FEniCS Optimization Diagnostics

Figure C.4 records the completed laminar U-bend optimization history. As in the pipe-bend case, the objective reduction occurs mainly at the start of the run, after which the optimizer preserves a nearly active volume constraint while only small continuation-related corrections remain.

Table C.12: Laminar-specific differences from the turbulent U-bend FEniCS setup above.

Parameter	Laminar setting
Flow model	Laminar incompressible Navier–Stokes; the SA working variable, reciprocal wall-distance equation, topology-dependent SA penalties, and frozen-turbulence coupling are omitted
Reynolds number	$Re = 1$, based on $U_{\text{in}} H_{\text{port}} \rho/\mu$ instead of the turbulent $Re = 5{,}000$ setup
Density and viscosity	$\rho = 1.0$ and $\mu = 0.2$, with the same $U_{\text{in}} = 1$ and $H_{\text{port}} = 0.2L$ as the turbulent case
Brinkman penalty	RAMP form with $\alpha_{\text{fluid}} = 0$ and $\alpha_{\text{solid}} = 10^5$
Flow finite elements	Taylor–Hood P_2/P_1 for $(\mathbf{u}, p)$; no separate P_1 SA working-variable space is used
Forward and adjoint solves	SNES/MUMPS laminar Newton forward solves with relative tolerance 10^{-6}, absolute tolerance 10^{-8}, and maximum forward iteration count 260; the adjoint is solved as a linear system with MUMPS
Continuation and optimizer controls	Seven-stage schedule $q = \{0.01, 0.02, 0.04, 0.08, 0.08, 0.12, 0.20\}$, $\beta = \{1, 2, 4, 8, 8, 12, 16\}$, move limits $\{0.010, 0.008, 0.006, 0.004, 0.003, 0.002, 0.0015\}$, and inner-iteration budgets $\{12, 25, 45, 80, 120, 160, 220\}$; stop if objective change is below 10^{-6} for ten iterations

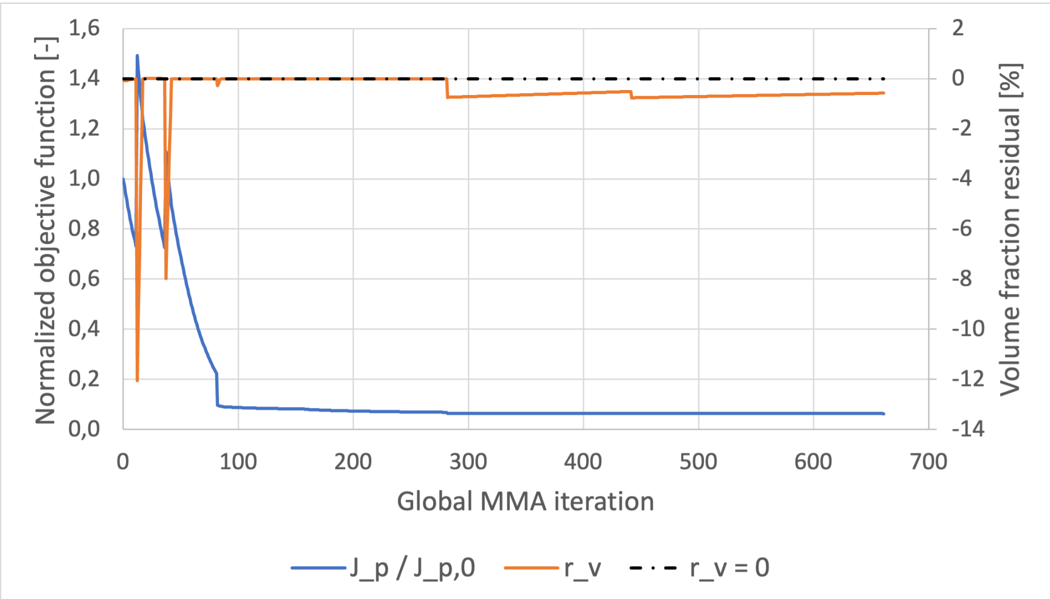


Figure C.4: FEniCS convergence history for the laminar U-bend topology-optimization run. The blue curve is the normalized inlet-pressure objective $J_p/J_{p,0}$, the orange curve is the relative volume-fraction residual, and the dashed black line marks the active volume constraint.

C.4.4 Laminar-Design Mesh Independence

Table C.13 reports the Ansys mesh-independence study for the extracted laminar U-bend design. The turbulent U-bend mesh study is reported separately in Table C.11 of Section C.4. For the $Re = 5{,}000$ U-bend operating point, the inlet-port height is $0.2L$ and $U_{\text{in}} = 1\ \text{m}\,\text{s}^{-1}$, giving $\nu = 4.0 \times 10^{-5}\ \text{m}^2\,\text{s}^{-1}$. Using the same smooth-wall estimate as the FEniCS mesh generator gives $u_\tau \approx 6.59 \times 10^{-2}\ \text{m}\,\text{s}^{-1}$ and a wall distance of $y_+ = 1$ at 6.07×10^{-4} m. The Ansys mesh hierarchy therefore keeps the first layer fixed at 6.00×10^{-4} m and refines the bulk triangular mesh, curvature resolution, and inflation-layer distribution around the separator tip and 180° return wall.

TABLE C.13: Ansys mesh-independence table for the extracted laminar U-bend design.

Parameter	Coarse	Medium	Fine
Mesh settings			
Mesh type	Tri.	Tri.	Tri.
h_{bulk} [m]	6.00×10^{-3}	4.00×10^{-3}	2.75×10^{-3}
Curvature angle	8°	6°	4°
Δy_1 [m]	6.00×10^{-4}	6.00×10^{-4}	6.00×10^{-4}
Inflation layers	12	16	20
Growth rate	1.10	1.08	1.06
t_{infl} [m]	1.28×10^{-2}	1.82×10^{-2}	2.21×10^{-2}
Nodes	16,106	32,580	62,622
Elements	24,460	50,067	98,084
Validation metrics			
Φ_D^{SST} [W/m]	0.101	0.105	0.107
$\epsilon_{\text{rel}}(\Phi_D^{\text{SST}})$ [%]	–	3.96	1.90
Static pressure drop Δp_s^{SST} [Pa]	1.125	1.141	1.150
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SST}})$ [%]	–	1.42	0.79
Selected comparison mesh		Fine_Ansys	

Note. Φ_D = viscous dissipation; Δp_s = static pressure drop; SST denotes SST k–ω. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

C.5 Diffuser Benchmark

The diffuser case follows Fig. 20–22 of Yoon [63]. The original velocity-outlet formulation prescribes parabolic velocity profiles at both the inlet and the outlet and uses the dissipation objective J_D. Chapter 6 uses that case as a boundary-condition sensitivity baseline, then treats the pressure-outlet variant as the primary diffuser result. Both variants use the shared `design`, `reciprocal_penalized`, power-law, and frozen-adjoint conventions summarized in Table C.1.

For the body-fitted Fluent evaluations, the inlet parabolic profile is shifted to the extracted inlet segment. In the velocity-outlet baseline, the outlet profile is also shifted to the extracted outlet segment and its maximum velocity is adjusted

Table C.14: Geometric, physical, and boundary-condition parameters for the diffuser benchmark.

Parameter	**Setting**
Reference case	Diffuser benchmark from Yoon [63]
Reference figures for setup and topology	Yoon Fig. 20–22
Reference turbulence model	Spalart-Allmaras
Turbulence model in this thesis	Spalart-Allmaras with frozen-turbulence adjoint
Design-domain dimensions	Square design region $L \times L$ with $L = 1$ m
Full computational domain	$x/L \in [0, 1]$, $y/L \in [0, 1]$
Inlet geometry	Exported Fluent inlet segment from $(1.16 \times 10^{-3}, 1.38 \times 10^{-2})$ to $(1.16 \times 10^{-3}, 0.979)$, with $H_{\text{in}} = 0.96558$ m
Outlet geometry	Exported Fluent outlet segment from $(0.999, 0.336)$ to $(0.999, 0.670)$, with $H_{\text{out}} = 0.33333$ m
Volume fraction	$V_f = 0.30$ fluid, corresponding to Yoon's 70% solid mass usage
Reynolds number	$Re = 3{,}000$, based on $U_{\text{max,in}} L\rho/\mu$
Density and viscosity	$\rho = 1000$ kg/m^3, $\mu = 1.0$ Pa s
Inlet condition	Shifted parabolic velocity profile with $U_{\text{max,in}} = 3$ m/s; $\tilde{\nu} = 1$ for the FEniCS-SA solve
Outlet condition	Velocity-outlet baseline: shifted parabolic outflow profile with maximum velocity set by incompressible flow-rate conservation and pressure pinned at $(0, 0)$ for gauge fixing. Pressure-outlet primary case: fixed static pressure $p = 0$ Pa on the center-third outlet with no prescribed outlet velocity.
Wall conditions	No-slip velocity and $\tilde{\nu} = 0$ on the top and bottom walls and on the right boundary outside the outlet
Brinkman solid resistance	$\alpha_{\text{solid}} = 10^9$, $\alpha_{\text{fluid}} = 0$
Optimization objective	Dissipation objective J_D, including physical viscous dissipation and Brinkman drag over the design domain
Validation metrics	J_D, static pressure drop, Φ_D, and relative changes between FEniCS-SA, Ansys-SA, and Ansys-SST $k-\omega$

by incompressible flow-rate conservation. In the pressure-outlet variant, the outlet velocity distribution is solved rather than prescribed.

C.5.1 Velocity-Outlet Boundary-Condition Baseline

The geometry and boundary conditions follow Fig. 20 of Yoon [63]. The design domain is a square of side length $L = 1$ m. A parabolic inflow enters through the full left boundary with maximum speed $U_{\max,\mathrm{in}} = 3$ m/s, while a parabolic outflow is prescribed on the center third of the right boundary with maximum speed $U_{\max,\mathrm{out}} = 9$ m/s. The remaining external boundaries are treated as no-slip walls. The Reynolds number is $Re = 3{,}000$ based on $U_{\max,\mathrm{in}} L\rho/\mu$, and the present density convention converts Yoon's 70% solid mass usage into a maximum fluid volume fraction of $V_f = 0.30$. Figure C.5 shows the corresponding local geometry and boundary-condition layout used for the velocity-outlet baseline.

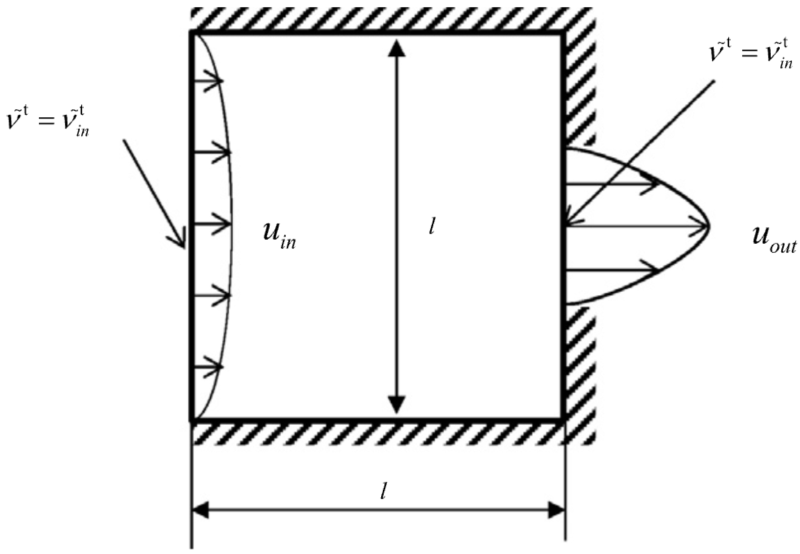


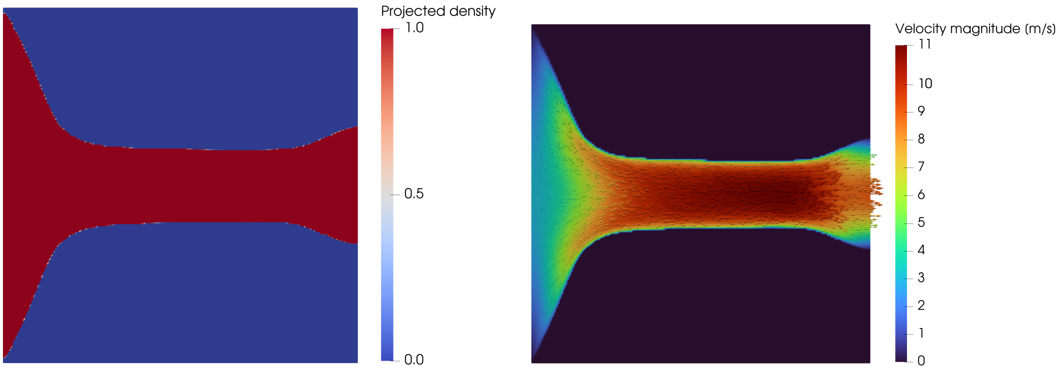
FIGURE C.5: Geometry and boundary conditions for the velocity-outlet diffuser baseline following Yoon [63]. Both inlet and outlet are prescribed by shifted parabolic velocity profiles.

Figure C.6 reports the FEniCS-SA topology and velocity field for the velocity-outlet baseline. The outlet region widens before the prescribed outlet segment. This feature should not be interpreted as a purely geometry-driven diffuser optimum: the optimizer must deliver the imposed parabolic outflow over the fixed outlet boundary, and opening the channel near that boundary reduces the shear and redistribution needed to satisfy the Dirichlet profile.

Figure C.7 shows the corresponding Ansys-SA velocity contour for the extracted velocity-outlet design. The contour is used here only as a diagnostic of the baseline geometry: the widened outlet region contains low-speed separated or recovering flow,

Table C.15: Numerical FEniCS solver and topology-optimization parameters for the diffuser benchmark.

Parameter	Symbol / option	Setting
Flow finite elements	$(\mathbf{u}, p)$	Taylor–Hood P_2/P_1 continuous Galerkin elements
SA working-variable space	$\tilde{\nu}$	P_1 continuous Galerkin
Design-density space	γ	DG_0 cellwise constant density
Reference FEniCS mesh	–	`mesh_yoon_yplus1.xdmf`; wall-resolved triangular mesh with 38,745 vertices, 76,704 triangles, far-field $h_{\max} = L/180 = 5.56 \times 10^{-3}L$, and first-cell width $1.82 \times 10^{-3}L$
Objective	J_D	Dissipation objective over the design domain
Initial design density	γ_0	$V_f = 0.30$, matched to the filtered volume
Brinkman penalty	$\alpha(\bar{\gamma})$	RAMP form with $\alpha_{\text{fluid}} = 0$, $\alpha_{\text{solid}} = 10^9$
RAMP continuation	q	8-stage schedule: 0.01, 0.02, 0.04, 0.08, 0.08, 0.12, 0.20, 0.35
Projection continuation	β	8-stage schedule: 1, 2, 4, 8, 8, 12, 16, 32, with threshold $\eta = 0.50$
MMA move limits	–	8-stage schedule: 0.05, 0.04, 0.03, 0.02, 0.012, 0.010, 0.0075, 0.005
Maximum optimization iterations	–	8-stage schedule: 50, 60, 80, 100, 200, 180, 220, 260 MMA iterations; stop if objective change is below 10^{-6} for ten iterations
Filter radius	r_f	$4h_{\max} = 2.22 \times 10^{-2}L$
SA working-variable penalty	–	Power-law solid indicator with amplitude 10^5 and exponent $n = 3$
Wall-distance penalty	–	Power-law solid indicator with amplitude 10^5 and exponent $n = 3$
Wall-distance equation	–	Reciprocal penalized relaxed Eikonal equation with $\sigma_w = 0.1$ and $G_0 = 20$
Forward flow solver	–	IPCS flow solves for frozen-SA Picard updates, followed by an SNES Newton line-search final flow solve
Frozen NS-SA coupling	–	Three Picard updates per design iteration with SA relaxation factor 0.15
Linear solver	–	MUMPS for SNES/state and adjoint solves; BiCGSTAB with ILU preconditioning for IPCS warm-start solves
Adjoint treatment	–	Frozen-turbulence adjoint in $(\mathbf{u}^*, p^*)$; $\tilde{\nu}$ and G are frozen during gradient assembly



(A) Projected density field $\bar{\gamma}$. (B) Velocity magnitude and vector field.

FIGURE C.6: FEniCS-SA result for the velocity-outlet diffuser baseline at $V_f = 0.30$. Final objective value $J_D = 3.1164 \times 10^5$ W/m.

which makes the original diffuser result strongly sensitive to the outflow treatment and turbulence closure.

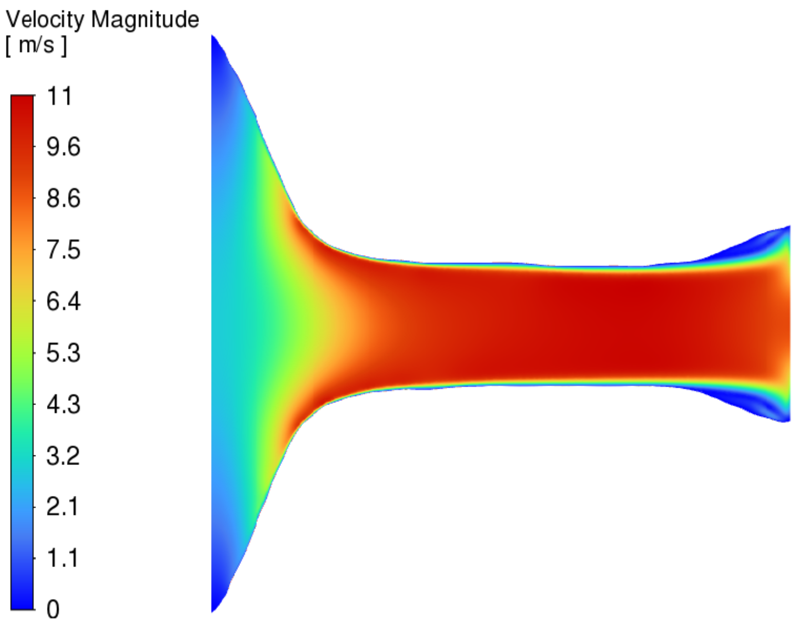


FIGURE C.7: Ansys-SA velocity-magnitude contour for the extracted velocity-outlet diffuser baseline. The low-speed region near the widened outlet motivates the pressure-outlet sensitivity study.

C.5.2 Pressure-Outlet Turbulent FEniCS Optimization Diagnostics

Figure C.8 gives the pressure-outlet diffuser optimization history through the common 1150-iteration comparison point used in Chapter 6. The dissipation objective is normalized by its initial value, while the volume trace is plotted as the relative residual $100(v_f^{(k)} - V_f)/V_f$ so that constraint activity can be read on the same iteration axis.

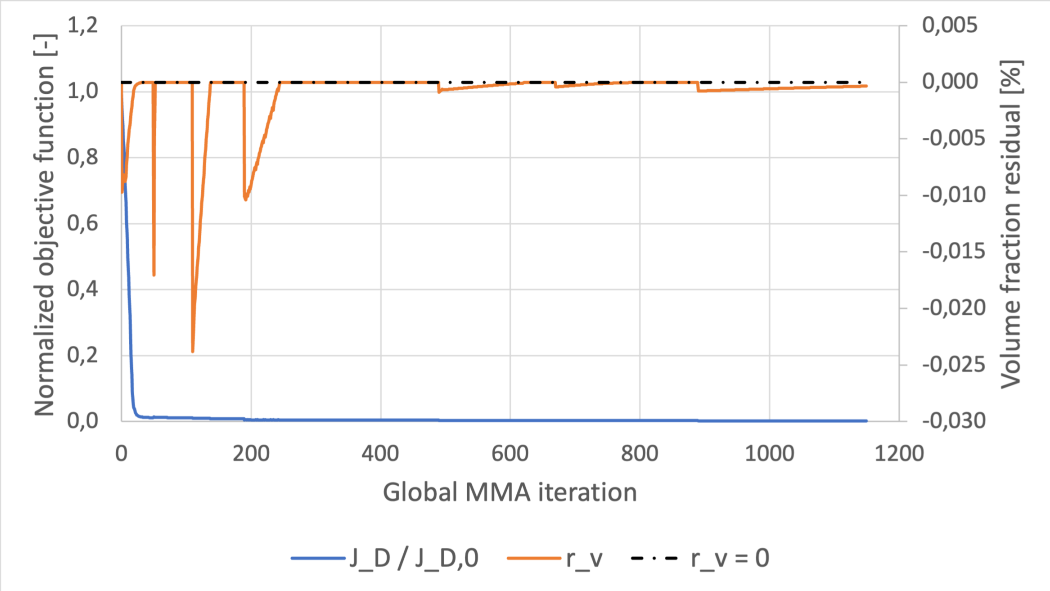


Figure C.8: Optimization convergence history for the turbulent pressure-outlet diffuser benchmark. The blue curve shows the normalized dissipation objective $J_D/J_{D,0}$, the orange curve shows the relative volume-fraction residual, and the dashed black line marks the active volume constraint.

The extracted turbulent diffuser geometry is meshed with the same wall-resolved triangular strategy as the laminar diffuser comparison in Table C.18, but with a distinct coarse–medium–fine hierarchy. The first-layer height is kept fixed at $\Delta y_1 = 1.50 \times 10^{-3}$ m to match the laminar-design diffuser meshes, while the bulk size, curvature angle, and inflation-layer distribution are refined independently for the turbulent topology.

TABLE C.16: Ansys Fluent mesh-independence table for the extracted turbulent diffuser benchmark geometry.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Mesh type	Tri.	Tri.	Tri.
Near-wall	WR infl.	WR infl.	WR ST infl.
h_{bulk} [m]	7.50×10^{-3}	5.00×10^{-3}	4.00×10^{-3}
Curvature angle	$9°$	$6°$	$5°$
Δy_1 [m]	1.50×10^{-3}	1.50×10^{-3}	1.50×10^{-3}
Inflation layers	10	14	16
Growth rate	1.11	1.09	1.08
t_{infl} [m]	2.51×10^{-2}	3.90×10^{-2}	4.55×10^{-2}
Nodes	8,104	16,166	21,238
Elements	12,330	24,570	40,268
Validation metrics			
Φ_D^{SA} [W/m]	1.2921×10^{5}	1.2966×10^{5}	1.2995×10^{5}
$\epsilon_{\mathrm{rel}}(\Phi_D^{\mathrm{SA}})$ [%]	–	0.35	0.22
Φ_D^{SST} [W/m]	1.5296×10^{4}	1.5570×10^{4}	1.5725×10^{4}
$\epsilon_{\mathrm{rel}}(\Phi_D^{\mathrm{SST}})$ [%]	–	1.79	0.99
Static pressure drop Δp_s^{SA} [Pa]	1.2460×10^{5}	1.2449×10^{5}	1.2422×10^{5}
$\epsilon_{\mathrm{rel}}(\Delta p_s^{\mathrm{SA}})$ [%]	–	0.09	0.22
Static pressure drop $\Delta p_s^{\mathrm{SST}}$ [Pa]	4.2412×10^{4}	4.2034×10^{4}	4.1833×10^{4}
$\epsilon_{\mathrm{rel}}(\Delta p_s^{\mathrm{SST}})$ [%]	–	0.89	0.48
Selected validation mesh		Fine_Ansys	

Note. WR = wall-resolved; ST = smooth-transition; Φ_D = viscous dissipation; Δp_s = static pressure drop; SST denotes SST k–ω. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

C.5.3 Laminar FEniCS Configuration Differences

The existing laminar diffuser reference is a completed FEniCS topology-optimization run from `Config_DiffuserYoon_Laminar.py`. It reuses the pressure-outlet turbulent diffuser setup in Tables C.14 and C.15 wherever not stated otherwise: the square $L = 1$ m domain, full-height parabolic inlet, center-third pressure outlet with $p = 0$ Pa, no prescribed outlet velocity, $V_f = 0.30$ volume constraint, dissipation objective J_D, Brinkman limits, mesh, filter radius, and projection threshold are unchanged. The laminar Reynolds number, solver settings, and continuation controls are listed in Table C.17.

C.5.4 Laminar FEniCS Optimization Diagnostics

Figure C.9 records the completed pressure-outlet laminar diffuser optimization history. The normalized objective drops rapidly during the early continuation stages and then remains nearly flat, while the volume-fraction residual stays close to the active constraint. This figure is kept in the appendix because it documents the laminar reference run rather than the turbulent performance comparison itself.

Table C.17: Laminar-specific differences from the pressure-outlet turbulent diffuser FEniCS setup above.

Parameter	**Laminar setting**
Flow model	Laminar incompressible Navier–Stokes; the SA working variable, SA wall-distance equation, and frozen-turbulence coupling are omitted
Reynolds number	$Re = 1$, based on $U_{\max,\mathrm{in}} L\rho/\mu$ instead of the turbulent $Re = 3{,}000$ setup
Density and viscosity	$\rho = 1000$ kg/m^3 and $\mu = 3000$ Pa s, with the same $U_{\max,\mathrm{in}} = 3$ m/s as the turbulent pressure-outlet case; no outlet velocity is prescribed
Flow finite elements	Taylor–Hood P_2/P_1 for $(\mathbf{u}, p)$; no separate P_1 SA working-variable space is used
Forward and adjoint solves	SNES/MUMPS laminar Newton forward solves with method **`newtontr`**, relative tolerance 10^{-6}, absolute tolerance 10^{-9}, and maximum forward iteration count 80; the adjoint is solved as a linear system with MUMPS
Continuation and optimizer controls	Seven-stage schedule $q = \{0.01, 0.02, 0.04, 0.08, 0.08, 0.12, 0.20\}$, $\beta = \{1, 2, 4, 8, 8, 12, 16\}$, move limits $\{0.010, 0.008, 0.006, 0.004, 0.003, 0.002, 0.0015\}$, and inner-iteration budgets $\{12, 25, 45, 80, 120, 160, 220\}$; stop if objective change is below 10^{-6} for ten iterations

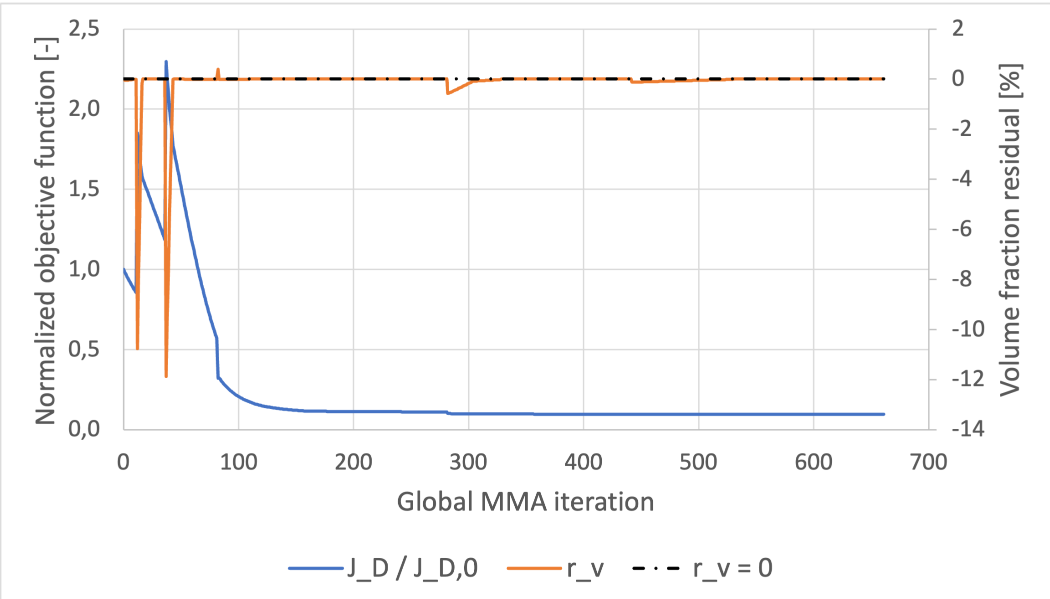


Figure C.9: FEniCS convergence history for the pressure-outlet laminar diffuser topology-optimization run. The blue curve is the normalized dissipation objective $J_D/J_{D,0}$, the orange curve is the relative volume-fraction residual, and the dashed black line marks the active volume constraint.

C.5.5 Laminar-Design Mesh Independence

Table C.18 reports the Ansys mesh-independence study for the extracted pressure-outlet laminar diffuser design. The turbulent diffuser mesh study is reported separately in Table C.16 of Section C.5. The Fluent solves use the same shifted inlet-profile and pressure-outlet convention as the turbulent diffuser validation setup.

TABLE C.18: Ansys mesh-independence table for the extracted pressure-outlet laminar diffuser design.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Mesh type	Tri.	Tri.	Tri.
Near-wall	WR infl.	WR infl.	WR infl.
h_{bulk} [m]	1.20×10^{-2}	5.50×10^{-3}	3.75×10^{-3}
Curvature angle	15°	7°	5°
Δy_1 [m]	1.50×10^{-3}	1.50×10^{-3}	1.50×10^{-3}
Inflation layers	6	12	16
Growth rate	1.15	1.10	1.08
t_{infl} [m]	1.31×10^{-2}	3.19×10^{-2}	4.55×10^{-2}
Nodes	3,317	13,444	25,485
Elements	5,085	20,888	39,582
Validation metrics			
Φ_D^{SST} [W/m]	1.2643×10^4	1.2983×10^4	1.3077×10^4
$\epsilon_{\text{rel}}(\Phi_D^{\text{SST}})$ [%]	–	2.69	0.72
Static pressure drop Δp_s^{SST} [Pa]	3.6628×10^4	3.6693×10^4	3.6464×10^4
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SST}})$ [%]	–	0.18	0.62
Φ_D^{SA} [W/m]	1.4296×10^5	1.4315×10^5	1.4333×10^5
$\epsilon_{\text{rel}}(\Phi_D^{\text{SA}})$ [%]	–	0.13	0.13
Static pressure drop Δp_s^{SA} [Pa]	1.4191×10^5	1.4276×10^5	1.4200×10^5
$\epsilon_{\text{rel}}(\Delta p_s^{\text{SA}})$ [%]	–	0.60	0.53
Selected comparison mesh		Fine_Ansys	

Note. WR = wall-resolved; Φ_D = viscous dissipation; Δp_s = static pressure drop; SST denotes SST k–ω. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change. The SA rows are re-evaluations for turbulence-model sensitivity.

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Code of conduct and transparency statement on the use of GenAI for KU Leuven-students (academic year 2025–2026)

Generative AI (GenAI) assistance tools can be used to generate various types of content, including text, images, code, video, music, or combinations thereof. Common examples of such tools include ChatGPT, Google Gemini, Microsoft Copilot, Midjourney, Claude.ai, Perplexity.ai, and DALL · E, among others.

This code of conduct is a tool that helps students to be transparent about the use of GenAI and fits within the university's principles on academic integrity.

Important guidelines and remarks

- **Sensitive or personal data:** Some GenAI tools protect your input and sensitive or personal data better than others. There is often no transparency on what the owners of the AI applications do with the data entered. Therefore, **do not enter sensitive or personal data in free GenAI tools.** More info about what sensitive and personal data are, is described in the three confidentiality levels of the KU Leuven data classification model (non-confidential, confidential, strictly confidential). For confidential data you should preferably use **M365 Copilot Chat with your KU Leuven account**. If you use another GenAI tool, you must be **absolutely certain** it does **not store or reuse the data you enter. Try to avoid entering these data. Do so only if strictly necessary, and only to the extent required. For personal data, work with anonymous or pseudonymized data.** Additionally, in case of strictly confidential data this should first be discussed with the teaching staff of the course or your thesis supervisor.
- **Copyrighted materials:** For **lawfully obtained copyrighted material** you should preferably use M**365 Copilot Chat with your KU Leuven**

account. If you use another GenAI tool, you must be **absolutely certain** it does n**ot store or reuse the data you enter.**

- GenAI assistance may not be used for data or topics covered by a Non-Disclosure Agreement (NDA). Even in the absence of an NDA, certain information may still need to be treated as confidential—for example, due to regulatory requirements or the risk of significant harm to the university if disclosed. In case of doubt, check with your teaching staff or supervisor.

- If your master's thesis is under embargo, you should first discuss with your supervisor whether the use of GenAI is permitted.

- Before using a GenAI tool, always consider whether its use is responsible, including from a sustainability perspective (e.g. when using GenAI as a search engine, language assistant,. . .).

- **Take a scientific and critical attitude** when interacting with GenAI assistance and interpreting its output, that may not always be correct.

- As a student you are responsible for complying with Article 84 of the Regulations on Education and Examinations: your report or thesis should reflect your own knowledge, understanding and skills. Be aware that plagiarism rules also apply to (work that is the result of) the use of GenAI assistance tools. **Exam Regulations Article 84:** *"Every conduct individual students display with which they (partially) inhibit or attempt to inhibit a correct judgement of their own knowledge, understanding and/or skills or those of other students, is considered an irregularity which may result in a suitable penalty. A special type of irregularity is plagiarism, i.e. copying the work (ideas, texts, structures, designs, images, plans, codes , ...) of others or prior personal work in an exact or slightly modified way without adequately acknowledging the sources. Every possession of prohibited resources during an examination (see article 65) is considered an irregularity."*

- In order to maintain academic integrity and avoid plagiarism, **more information about being transparent on the use of GenAI assistance and about correctly citing and referencing GenAI** can be found on this website for students (Dutch/English).

- **Additional reading: KU Leuven guidelines on responsible use of Generative AI tools, and other information** (Dutch/English)

A few final words

If you are uncertain whether or not you should declare your use of GenAI tools, we suggest that you discuss this with your instructor or supervisor. It is always safer to declare GenAI use, even when it is not strictly required. However, declaring GenAI use does not entail that its use is

allowed; the right column in the table below provides more detailed instructions in this regard (code of conduct).

Moreover, advanced AI tools are evolving rapidly, and their capabilities have expanded significantly in a short period of time. As a result, we do not yet have all the answers about their responsible use. Finally, it is important to follow-up on the most recent evolutions in AI technologies, to have an open mind but also to be a bit cautious, to communicate with instructors, teaching assistants, supervisors and peers, to be as transparent as we can, and to learn together as we move along.

Student name: .... **Student number:** ....

Please indicate with "X" whether it relates to a course assignment or to the Bachelor's or Master's thesis:

O This form is related to a **course assignment.**
Course name: .... **Course number:** ....

This form is related to my O **Bachelor's** or O **Master's thesis.**
Title Bachelor's or Master's thesis: .... **Supervisor:** ....

Daily supervisor: ....

Please indicate with "X":

O I **did not use** any GenAI assistance tool.

O I **did use** GenAI Assistance. In this case **specify which ones** (e.g. ChatGPT, M365 Copilot, ...): ....

GenAI assistance used as/for:	**Name of the GenAI tool(s) used** If helpful, also describe in which way you were using GenAI related to what is specified as code of conduct.	**Code of conduct:** **For each of the categories below, always take into account the important guidelines and remarks mentioned above (e.g. copyrighted data, sensitive or personal data,...).**
As a language assistant for reviewing or improving texts I wrote myself	*add text here*	This use is similar to using spelling and grammar check tools. In general, you do not have to refer to such kind of GenAI use in the text. **However, be careful:** • When using GenAI tools on texts you did not write yourself to improve the text, **you have to refer to the original source or author**, otherwise you are committing plagiarism and thus an irregularity.
As a paraphrasing tool	*add text here*	This use is allowed except when it is prohibited by the teacher or the program of study. You may paraphrase your own text or texts by an author other than yourself and take inspiration from what a GenAI tool or another tool suggests (unless it is not allowed). In general, you do not have to refer to such kind of GenAI use or other paraphrasing tools in the text. **However, be careful**: • If it entails text by an author other than yourself, **you are not allowed to include that paraphrased text without reference to the original source or author**. Without such reference, you would be committing plagiarism and thus an irregularity.

For translation aid to improve texts I wrote myself or to better understand text from others	*add text here*	This use is allowed except when it is prohibited by the teacher or the program of study. It is similar to using translation tools (Google translate, DeepL, ...). In general, you do not have to refer to such kind of GenAI use in the text. **However, be careful:** • **You are not allowed to include that translated text without reference to the original source or author.** Without such reference, you would be committing plagiarism and thus an irregularity. • Always check the translated text for correctness and meaning.
As a search engine to get information on a topic or to search for existing research on the topic	*add text here*	This use is similar to e.g. a Google search or checking Wikipedia. If you write your own text based on this information, you do not have to refer to the use of GenAI in the text. **You only have to refer to the existing research and references you have checked and used** (without such references you would be committing plagiarism and thus committing an irregularity). **However, be careful:** • Be aware that the output of the GenAI tool cannot be guaranteed as a 100% reliable source of information. The output may not be entirely correct and/or be limited due to the databases it uses. Moreover, knowledge evolves and may change over time; therefore, the database of the GenAI tool may not be up to date. Therefore, **verify the information and do not just copy-paste it** as you should understand and critically process everything you are writing.

For literature search	*add text here*	This use is comparable to e.g. a Google Scholar search. You do not have to refer to such kind of GenAI use; **you only have to refer to the literature references you have checked and used** (without such references you would be committing plagiarism and thus committing an irregularity). **However, be careful:** • Be aware that the search output is restricted to the database the GenAI tool is built on. After this initial search, look for scientific sources and conduct your own analysis of the source documents. **Interpret, analyse and process the information you obtained; verify it and do not just copy-paste it** as you should understand and critically process everything you are writing. • Be aware that some GenAI tools may **output no or wrong references.** As a student you are responsible for further checking and verifying the absence or correctness of references; do not just copy-paste it.
For generating programming code	*add text here*	Use of GenAI for coding is allowed except when it is prohibited by the teacher or the program of study. If used for coding, correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparant about the use of GenAI.

For generating new (research) ideas	*add text here*	This use of GenAI is allowed except when it is prohibited by the teacher or the program of study. Correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparent about the use of GenAI. **Be careful:** • Further verify in this case whether the idea is novel or not. It is likely that it is related to existing work. **If so, that existing work should be correctly referenced in the text** (without such reference you would be committing plagiarism and thus committing an irregularity).
For generating synthetic data	*add text here*	Use of GenAI for generating synthetic data is allowed, **provided that it is methodologically and ethically justifiable**, except when it is prohibited by the teacher or the program of study. Always correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparent about the use of GenAI. **Be careful:** • Always **carefully evaluate** the generated synthetic data for quality and possible bias since the output is highly dependent on the quality of the data on which the models are trained.

For generating blocks of text (other than the allowed use without referencing mentioned above)		According to Article 84 of the Regulations on Education and Examinations your text should allow to correctly and properly assess your own knowledge, understanding and skills. **Therefore, inserting blocks of text without quotes and a reference to GenAI assistance in your work is not allowed.** **Be careful:** • If it is really needed to insert a block of text from a GenAI tool, for instance because of the nature of your assignment, mention it as a citation by using quotes and correctly mention the use of GenAI assistance in accordance with the instructions page on being transparent about the use of GenAI. • However, in general, such GenAI use should be kept to an **absolute minimum**; you should always check the original sources.
For generating visuals, video or audio	*add text here*	Use of GenAI for generating visuals, audio or video is allowed except when it is prohibited by the teacher or the program of study. **Be careful:** • If used, refer to GenAI for visuals according to the usual referencing style, following the instructions on the page on referencing GenAI. • If you work with existing visuals, audio or video, **that existing work should be correctly referenced in the text** (without such reference you would be committing plagiarism and thus an irregularity). • Explain the usage in the methods section (if there is one) and optionally attach (or link to) the prompts with the full output (or history).

Other use (specify here; this may also include a combination of types of use mentioned above):	*add text here*	To make sure other use of GenAI is allowed within the course or thesis, and if so, the conditions that may apply, contact the teaching staff of the course or the supervisor of the thesis **beforehand** and explain the intended GenAI use. Also inform the programme director. Motivate how you would comply with Article 84 of the exam regulations. Explain the use and the added value of the AI tool you consider to use and how it is in accordance with the assignment or thesis and the KU Leuven guidelines on responsible use of Generative AI tools. Depending on the kind of GenAI use, it may be needed to properly reference it in the text, in accordance with the instructions page on being transparent about the use of GenAI.